MATH1110 Real Analysis

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0 Introduction

Fundamentally, the importance of mathematics in the modern world comes from our ability to know that mathematical statements are correct. Arguably, one of the mathematical ideas that has had the most impact on the world (and continues to have impact) is that of calculus. Almost every area of science and engineering requires some use of calculus and it has been central to our understanding of the world for at least the last two centuries. However, when we start to delve into calculus, we start to get to some statements which may trip us up (and have tripped up many of the greatest mathematicians of the last two centuries). This can (and has) lead to errors in solutions of equations, mistakes in proofs, entire books that are incorrect... This is where Real Analysis comes in: Real Analysis is the rigorous study of the theory of real numbers and functions on the real numbers. It aims to completely clarify (i.e. prove) the statements that we require as mathematicians.

Here are some statements - which of these is true? (Recall that a continuous function is one that "you can draw without taking your pen off the page")

- $1 = 0.999999999999999999999999999999\ldots$
- If a, b and c are real numbers and $a \leq b$ then $ac \leq bc$.
- Every continuous function is differentiable.
- Every differentiable function is continuous.
- Given a sequence of differentiable functions which get closer and closer together, they will limit to a differentiable function.
- There is no continuous curve in the plane (i.e. a curve which you can draw without taking your pen off the page) which covers every point on the plane.
- Suppose that a function f has $f'(a) > 0$. Then, near f the function is strictly increasing.

In fact, only the first and fourth bullet points are true and the rest are false! When trying to decide which are true and which are false, you may at some point have felt some uncertainty. Indeed understanding which of these is true starts right at the beginning: What does continuous even **really** mean? What does it mean for a function to be differentiable? What does 0.9999999999999999... even really mean?

So far, you may have been given vague "definitions" such as the "$f'(a)$ is the slope of the graph of f at a", or (as above) "f is continuous if you can draw it without taking the pen off the page". However, when it comes to actually doing anything at all with these, they are useless! For example we couldn't either prove or disprove any of the above, which is pretty bad given the importance of calculus.

In this course, we will understand more about the real numbers and functions on the real numbers. Starting from just the real numbers, we will see why calculus works and build up the techniques necessary to prove the statements that we need to solve mathematical problems. Beyond this, we will build up proof techniques that will have application throughout your degree.

1 Numbers and inequalities

1.1 Sets of numbers

We start with various different types of numbers. We express these as sets, which you will be learning about in more detail in Core Mathematics - essentially a set is a collection of objects, and we can think of a set as a "box" containing mathematical objects. Here are five very important sets which occur regularly in mathematics:

$\mathbb{Z}^+$, the set of positive integers: This is the set of "counting" numbers, that is the set of all whole numbers larger than zero, which can be written as

$$\mathbb{Z}^+ = \{1, 2, 3, 4, \ldots\} \ .$$

Here we have used set notation, the start of the set denoted by { and the end of it by } (so everything in between is in the set).[1]

$\mathbb{Z}$, the set of integers: This is the set of all whole numbers, including 0 and negative numbers. That is

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\} \ .$$

$\mathbb{Q}$, the set of rational numbers . This is the set of all numbers that may be written as a fraction of integers. That is

$$\mathbb{Q} = \text{"any number which may be written as } \frac{a}{b} \text{ where } a \in \mathbb{Z} \text{ and } b \in \mathbb{Z}^+\text{"}$$

In the above the $\mathbb{Q}$ stands for quotients.

$\mathbb{R}$, the set of real numbers: This is the set of all real numbers, including algebraic real numbers, such as $\sqrt{2} \in \mathbb{R}$ and transcendental real numbers, for example $e, \pi \in \mathbb{R}$.

$\mathbb{C}$, the set of complex numbers: All complex numbers, including for example $\sqrt{-1} = i \in \mathbb{C}$.

An object that is in the a set is called an element of that set. To say that an object is in a set we use $\in$, so given a set A "$x \in A$" means "x is an element of A". If we want to say an object isn't in the set we use $\notin$. So, for example

$$7 \in \mathbb{Z}^+, \qquad 54392375004 \in \mathbb{Z}^+, \qquad \text{but} \qquad \frac{1}{2} \notin \mathbb{Z}^+, \qquad -10 \notin \mathbb{Z}^+ \ .$$

Note that in the above, every natural number is also an integer, every integer is a rational number, every rational number is a real number and so on. However, there are real numbers that are not rational numbers, and rational numbers that are not integers and so on.

A set A is said to be a **subset** of another set B (or we might say A is contained in B) if every element in set A is also in set B. We write this, $A \subseteq B$. We have that:

$$\mathbb{Z}^+ \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C} \ .$$

In this course we will be interested in understanding the properties of $\mathbb{R}$. Our aim is to derive all of the properties of the real numbers, starting only from the fundamental sets described above.

In this course, we will be particularly interested in the real numbers, $\mathbb{R}$. As well as being a set, $\mathbb{R}$ has further structures which have been familiar to you since primary school:

$\mathbb{R}$ is a field: The real numbers come with standard operations which are familiar to you, namely addition, subtraction, multiplication and division, that is $(+ \ - \ \times \ \div)$, along with the familiar algebraic properties (for example, for any real numbers a, b, c we know that $a \times (b + c) = (a \times b) + (a \times c)$).

$\mathbb{R}$ is an ordered field: As well as the above we have an ordering relation $<$. Specifically, given **any** two real numbers $a, b \in \mathbb{R}$ exactly one of $a = b$, $a < b$ and $a > b$ is true. The ordering $<$ is assumed to have all of the usual properties that you are used to (for example, for any $a, b, c \in \mathbb{R}$, if $a < b$ and $0 < c$ then $ac < bc$).

[1] Some mathematicians call this set $\mathbb{N}$, but others disagree and say that $\mathbb{N}$ should include 0. To avoid confusion we use $\mathbb{Z}^+$.

Our aim is to use the properties the above properties to build up the theorems that we need. For now, we have a further property that we will assume:

FACT: (The Archimidean property of real numbers) Given any real number $x \in \mathbb{R}$ there is a positive integer $n \in \mathbb{Z}^+$ with $x < n$.

We now restrict ourselves to these properties and statements that we may derive from these properties! For example, for now we will avoid using functions that we haven't proven the properties of (or even rigorously defined) functions such as $\sin(x)$, $\cos(x)$, $\exp(x)$, x^p for p not an integer. We need to prove the properties that we know about these before using them!

1.2 Modulus and measurement on the real numbers

We now define an important function which will be extremely important to us in this course is the **modulus function** (or absolute value function). This is written as $|x|$ and is defined by

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

So $|4| = 4$, $|-2| = -(-2) = 2$ and so on. Clearly $|x| \geq 0$ for all x, by definition.

If we think of real numbers as lying on an infinite straight line, then $|x|$ can be interpreted geometrically as the **distance** from x to 0. Furthermore, given any **pair** of real numbers x, y, the absolute value of their difference $|x - y|$ can be thought of as the distance between them. Note that $|x - y| = |y - x|$ for all x and y. For example the distance between 4 and -2 is

$$|4 - (-2)| = |6| = 6 \quad \text{or} \quad |(-2) - 4| = |-6| = -(-6) = 6.$$

We emphasize that this is just a mental model of $\mathbb{R}$ which gives a nice interpretation of $|\cdot|$. We are not **defining** real numbers as points on an infinite line, nor are we **defining** $|x|$ to be the distance from x to 0.

We will often care about inequalities involving $|x|$.

Lemma 1.1 *For any real numbers x and c, $|x| < c$ if and only if $-c < x < c$.*

In the above "if and only if" means that $|x| < c$ is true **exactly** when $-c < x < c$ (and $|x| < c$ is false exactly when $-c < x < c$ is false). As mathematicians are busy people, we often shorten "if and only if" to "iff". The above seems reasonable – we might draw the following helpful picture: However, drawing a graph helps with intuition, but does not constitute a proof!

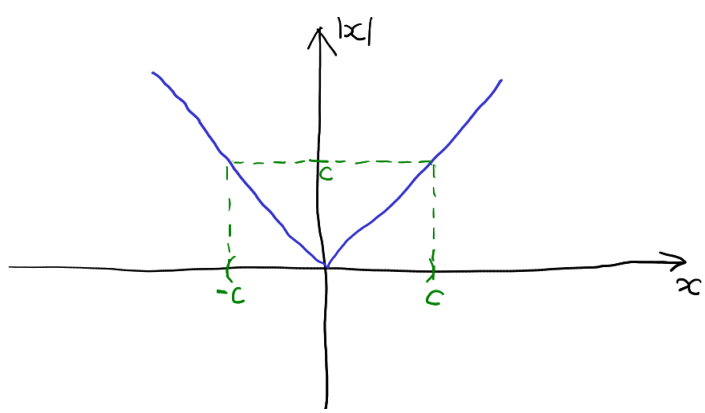


Figure 1: Our intuition for Lemma 1.1.

Proof:

Suppose first that $|x| < c$ holds. Note that in this case, $0 < c$ (as $0 \leq |x| < c$). We have two cases to consider:

- If $x \geq 0$ then this implies that $0 \leq x = |x| < c$ so we have that $-c < x < c$.
- If $x < 0$, we have $-x = |x| < c$. Rearranging we have that $-c < x$. Therefore in this case $-c < x < 0$ which implies $-c < x < c$.

As these are the only two possibilities for x, if $|x| < c$ we always have $-c < x < c$

Now suppose that $-c < x < c$: Again, this implies that $0 \leq c$ (this follows from rearranging $-c < c$). If $x \geq 0$ then $|x| = x < c$. If $x < 0$ then $|x| = -x < c$ (where this inequality comes from rearranging $-c < x$). Therefore we see that if $-c < x < c$ then $|x| < c$. $\square$

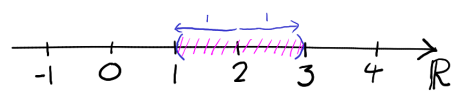
Note that in the above we had to have two steps to prove $|x| < c$ iff $-c < x < c$. We had to prove **both** "if $|x| < c$ is true then $-c < x < c$ is true" **and** "if $-c < x < c$ is true then $|x| < c$". In general, we need to do both! There are lots of true statements of the form "If A then B" is true but "If B then A" is false. For example, "If I am wearing a hat then there is someone in the room that I am in who is wearing a hat" is always true. However, "If there is someone in the room that I am in who is wearing a hat then I am wearing a hat" is false in general.

Remark 1.2 There is a second statement which is the weak inequality version (or "$\leq$ version") of Lemma 1.1:

For any real numbers x and c, $|x| \leq c$ if and only if $-c \leq x \leq c$.

This is also true - the proof is identical to the proof of Lemma 1.1 (with $<$ replaced with $\leq$). I leave it as an exercise to check this!

Example 1.3 We may see that $|x-2| < 1$ is all numbers x that have $-1 < x-2 < 1$ or $1 < x < 3$. We may think of these as all numbers of distance less than 1 from 2.



Lemma 1.4 (Properties of the modulus) *For any $x \in \mathbb{R}$ we have that*

$$|-x| = |x|, \qquad x \leq |x| \qquad -x \leq |x| \ ,$$

and for any $x, y \in \mathbb{R}$ we have that

$$|xy| = |x||y| \ .$$

Proof: These follows from the definition of the modulus by checking the two cases $x \geq 0$ and $x < 0$ (similarly to Lemma 1.1). I leave these as an exercise! $\square$

Finally, we come to an inequality which will be very important throughout the course.

Lemma 1.5 (The triangle inequality) *Suppose that $a, b \in \mathbb{R}$. Then*

$$|a+b| \leq |a| + |b| \ .$$

Proof: From Lemma 1.4, we have that $a \leq |a|$ and $b \leq |b|$. Therefore $a+b \leq |a|+b \leq |a|+|b|$. On the other hand, $-a \leq |a|$, $-b \leq |b|$ so $-a-b \leq |a|+|b|$, or (rearranging) $-|a|-|b| \leq a+b$. Putting these together we have that

$$-|a|-|b| \leq a+b \leq |a|+|b| \ ,$$

so by Lemma 1.1, $|a+b| \leq |a|+|b|$ as claimed. $\square$

Corollary 1.6 (The reverse triangle inequality) *Suppose that $x, y \in \mathbb{R}$. Then*

$$||x| - |y|| \leq |x - y|$$

Proof: We have that from the triangle inequality (setting $a = x - y$, $b = y$)

$$|x| \leq |x - y| + |y| \qquad \Longrightarrow \qquad |x| - |y| \leq |x - y| \ ,$$

and (setting $a = y - x$, $b = x$)

$$|y| \leq |y - x| + |x| \qquad \Longrightarrow \qquad -|x - y| \leq |x| - |y| \ .$$

Putting these together we have that

$$-|c - d| \leq |c| - |d| \leq |c - d|$$

which is equivalent to the statement (by Lemma 1.1 and Remark 1.2) □

Summary

- We have defined several sets of numbers $\mathbb{Z}^+$ (the set of positive integers), $\mathbb{Z}$ (the set of integers), $\mathbb{Q}$ (the set of rational numbers) and $\mathbb{R}$ (the set of real numbers).
- We say set notation "$\in$" means "is an element of", "$\notin$" means "is an not element of" and "$\subseteq$" means "is a subset of". So for example, $1 \in \mathbb{Z}^+$, $-36 \notin \mathbb{Z}^+$ and $\mathbb{Z} \subseteq \mathbb{R}$.
- The modulus function gives the distance of a point to 0.
- $|x| < c$ is equivalent to $-c < x < c$.
- The triangle inequality holds for any real numbers a and b, that is, $|a + b| \leq |a| + |b|$.
- The reverse triangle inequality holds for any real numbers a and b, that is, $||a| - |b|| \leq |a - b|$.

2 Sequences

A **real sequence** is a mapping $a : \mathbb{Z}^+ \to \mathbb{R}$, that is, a rule which assigns to each positive integer n some real number $a(n)$. Actually, for sequences, unlike pretty much all other functions, it's conventional to write the image of n under the mapping a as a_n instead of $a(n)$, and call it the **$\mathbf{n^{th}}$ term** of the sequence, rather than the "image of n." It's also conventional to denote the mapping as $(a_n)_{n \in \mathbb{Z}^+}$, or just (a_n) for short, rather than $a : \mathbb{Z}^+ \to \mathbb{R}$. We will follow these conventions.

Often we will specify a sequence just by giving a formula for its nth term.

Example 2.1 $a_n = \frac{n^2+5}{n^2}$ is a sequence whose first 4 terms are

$$6, \qquad \frac{9}{4}, \qquad \frac{14}{9}, \qquad \frac{21}{16}.$$

As n gets very large, a_n gets very close to 1: for example $a_n = 1$ to 3 decimal places for all $n \geq 101$. So for n this large, we can approximate the sequence by the constant number 1, provided we're happy to accept errors no larger than 5×10^{-4}. What if we're a bit fussier than this, and need to

know a_n to at least 5 decimal places? How big must n be before we can approximate a_n by the constant 1? We need

$$\begin{aligned}
& 0.999995 \leq a_n < 1.000005 \\
\Leftrightarrow\quad & 0.999995 \leq 1 + \frac{5}{n^2} < 1.000005 \\
\Leftrightarrow\quad & -0.000005 \leq \frac{5}{n^2} < 0.000005 \\
\Leftrightarrow\quad & \frac{5}{n^2} < 5 \times 10^{-6} \\
\Leftrightarrow\quad & n^2 > 10^6 \\
\Leftrightarrow\quad & n > 1000.
\end{aligned}$$

So, for all $n \geq 1001$, $a_n = 1$ to 5 decimal places. In fact, however many decimal; places of accuracy we insist on, there is a point in the sequence beyond which, to that accuracy, a_n is just 1.

Example 2.2 Contrast this with $a_n = \sin n$, whose first 4 terms (to 5 decimal places) are

$$0.84147, \quad 0.90930, \quad 0.14112 \quad \text{and} - 0.75680.$$

The behaviour of this sequence as n grows large is not at all uniform; the terms bounce around seemingly at random with $-1 < a_n < 1$.

Sometimes it's more convenient to define a sequence **inductively**. In its simplest form, this is when we specify the first term a_1, and give a rule for constructing a_{n+1} from a_n.

Example 2.3 Let (a_n) be the sequence whose first term is $a_1 = 1$, for which

$$a_{n+1} = \frac{a_n}{1 + a_n^2}.$$

The first 4 terms of this sequence are

$$\begin{aligned}
a_1 &= 1 \\
a_2 &= \frac{1}{1+1^2} = \frac{1}{2} \\
a_3 &= \frac{1/2}{1+(1/4)} = \frac{2}{5} \\
a_4 &= \frac{2/5}{1+(4/25)} = \frac{10}{29}
\end{aligned}$$

What happens to a_n as n gets very large? This is actually quite hard to answer directly, even using a computer package such as Maple, because the terms become very unwieldy as n gets even moderately large. For example

$$a_8 = \frac{26844038679865941898849 0}{1099331737522548368039021},$$

and Maple has trouble computing even a_{30}, because the integers involved get so large. We can convert each a_n to a truncated decimal expansion, using the command `evalf`, which greatly speeds up the calculation, but this gives rounding errors which accumulate as n increases, so by the time we get to a_{1000}, say,it's not obvious whether the answer Maple produces (0.02234293705, for what it's worth) bears any relation to reality.

Another interesting question is whether the large n behaviour of this sequence depends on our choice of initial term, $a_1 = 1$. What if $a_1 = 0$? Or $a_1 = 1000$? We will develop methods which will allow us to show that, whatever a_1 we choose, a_n for large n becomes very close to 0 – despite the fact that we have no idea how to write down a_n in general!

2.1 Convergence of sequences

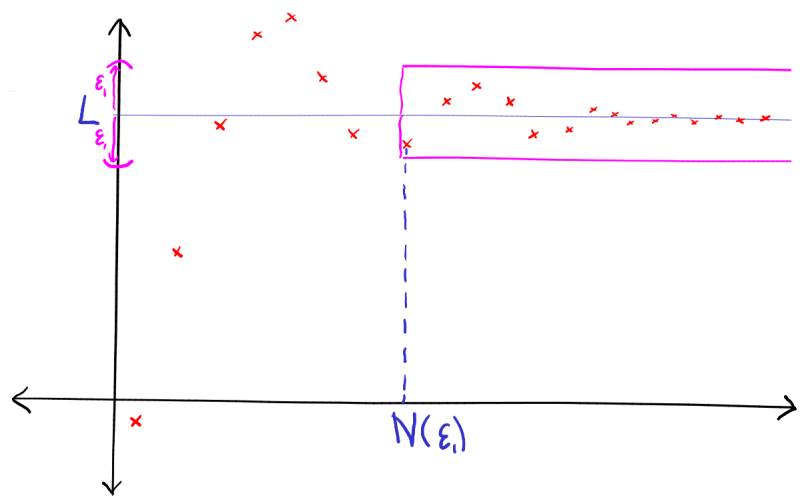
In Example 2.1, the terms of the sequence "tend to" 1 as n gets arbitrarily large, whereas in Example 2.2, the terms bounce around indefinitely, without tending to a particular value. We say that $a_n = (n^2+5)/n^2$ **converges** to 1, while $a_n = \sin n$ does not converge. It is now time to make this concept of convergence precise.

What does this have to do with convergence of sequences? In saying that $a_n = \frac{(n^2+5)}{n^2}$ converges to 1, we are saying, roughly, that the terms a_n are close to 1 for large n. In terms of absolute values, this means that $|a_n - 1|$ is small for large n. More precisely, we make the following definition:

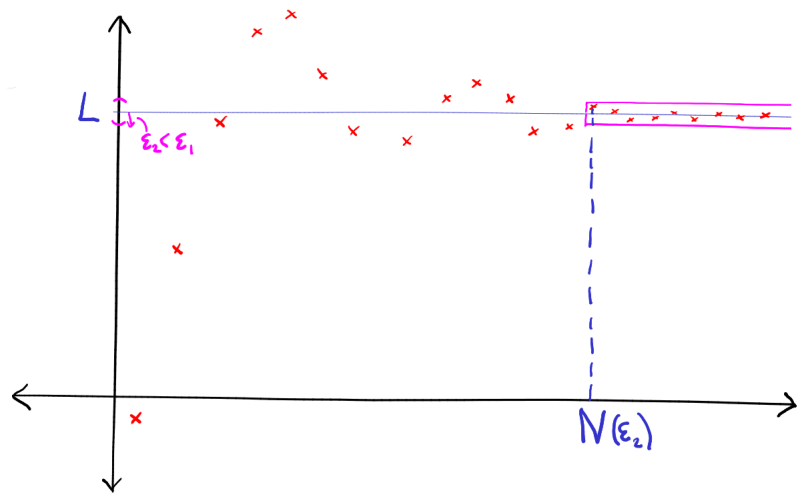
Definition 2.4 A real sequence (a_n) **converges** to a real number L if, for each $\varepsilon > 0$ there exists some positive integer N such that for all $n \geq N$, $|a_n - L| < \varepsilon$. In this case, we will write $a_n \to L$. The number L is called the **limit** of the sequence (a_n), and is often denoted $\lim a_n$ or $\lim_{n\to\infty} a_n$.

Remarks

- This is the most important definition in this course, and it is vitally important that you understand it thoroughly. It is the first rigorous definition of limit we have encountered. Other definitions of limit (e.g. $\lim_{x\to a} f(x)$) will be constructed from it.
- The definition says that given **any** $\varepsilon > 0$, no matter how small, there is a term in the sequence, a_N, after which **all** terms in the sequence lie within distance ε of L.
- Note that $|a_n - L| < \varepsilon$ is equivalent to $L - \varepsilon < a_n < L + \varepsilon$.
- If we imagine plotting the terms of the sequence a_n on a graph (with n along the x axis and a_n along the y axis), we can construct a geometric picture of what the definition means. Given any positive number ε, construct the horizontal strip of height 2ε centred on the line $y = L$. Then there exists a point N on the x axis, to the right of which all points on the graph lie in the strip.



This is true no matter how small we make ε. If we make ε smaller, the strip gets narrower, and the N beyond which all terms lie in the strip (probably) moves to the right.



- So N will, in general, depend on ε: if ε is large, a fairly small N will probably work, whereas if ε is very small, N will probably have to be very large. For our purposes the important thing is to establish that given any $\varepsilon > 0$ there is a corresponding N which works for that particular ε. The relationship between ε and N is not important, nor is it important to know the smallest N which will work for a given ε.

- Note that, directly from the definition, $a_n \to L$ is and only if the sequence $b_n = a_n - L$ satisfies $b_n \to 0$.

If I ask you to **prove from first principles** that a given sequence (a_n) converges to a given limit L, I am asking you to show directly that, given any $\varepsilon > 0$, there is some $N \in \mathbb{Z}^+$ such that whenever $n \geq N$, $|a_n - L| < \varepsilon$. The key to doing this is to "estimate" (that is, rigorously find an upper bound on) the quantity $|a_n - L|$.

Example 2.5 Claim: The sequence $a_n = \frac{1}{n} \to 0$
Proof: We suppose that we are given $\epsilon > 0$ - to prove that $a_n \to 0$ we need to find an $N \in \mathbb{N}$ such that if $n > N$ then $|a_n - 0| = |a_n| < \epsilon$. As $1/\epsilon$ is a real number, by the Archimidean property of the reals, there is an integer $N > \frac{1}{\epsilon}$. Then, for any $n \geq N$,

$$|a_n| = \left|\frac{1}{n}\right| = \frac{1}{n} \leq \frac{1}{N} < \frac{1}{\frac{1}{\epsilon}} = \epsilon$$

where we repeatedly used that if $a > b$ then $\frac{1}{b} > \frac{1}{a}$. $\square$

Example 2.6 Claim: $a_n = \frac{n^2+5}{n^2} \to 1$.
Proof: Let $\varepsilon > 0$ be given. Then $5/\varepsilon$ is a real number, so by the Archimedean Property, there exists a positive integer N such that $N > \frac{5}{\varepsilon}$. Now, for all $n \geq N$,

$$\begin{aligned} |a_n - 1| &= \left|\frac{n^2+5}{n^2} - 1\right| = \left|\frac{5}{n^2}\right| = \frac{5}{n^2} \leq \frac{5}{n} \qquad \text{(since } n \geq 1\text{)} \\ &\leq \frac{5}{N} < \varepsilon. \end{aligned}$$

Hence, given any $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that $|a_n - 1| < \varepsilon$ for all $n \geq N$. Hence $a_n \to 1$. $\square$

Note that Definition 2.4 does **not** say that a_n "gets closer and closer to L as n gets larger", because this isn't necessarily true. For example, the sequence

$$a_n = \begin{cases} \frac{1}{n} & n \text{ odd} \\ \\ \frac{1}{n^2} & n \text{ even} \end{cases} \tag{2.1}$$

does not have this property, because each odd term is further away from 0 than the previous even term. Nonetheless, (a_n) certainly converges to 0. To prove this, we only need a minor variation of the argument used in Example 2.6

Example 2.7 Claim: the sequence defined in equation (2.1) converges to 0.
Proof: Let $\varepsilon > 0$ be given. Then $1/\varepsilon$ is a real number, so by the Archimedean Property, there exists a positive integer N such that $N > \frac{1}{\varepsilon}$. Now, for all $n \geq N$,

$$\begin{aligned} |a_n - 0| &= a_n \qquad \text{(since } a_n > 0 \text{ for all } n\text{)} \\ &= \begin{cases} \frac{1}{n} & n \text{ odd} \\ \frac{1}{n^2} & n \text{ even} \end{cases} \\ &\leq \frac{1}{n} \qquad \text{(since } n \geq 1\text{)} \\ &\leq \frac{1}{N} \\ &< \varepsilon. \end{aligned}$$

Hence, given any $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that $|a_n - 0| < \varepsilon$ for all $n \geq N$. Hence $a_n \to 0$. □

Note that Definition 2.4 does **not** merely say that given any $\varepsilon > 0$ there is some $N \in \mathbb{Z}^+$ such that $|a_N - L| < \varepsilon$. The definition certainly implies this, but is considerably stronger. For example, consider the sequence $a_n = (-1)^n$. Then given any $\varepsilon > 0$, there is an $N \in \mathbb{Z}^+$ such that $|a_n - 1| < \varepsilon$. For example, $N = 2$ will do for any $\varepsilon > 0$, since $|a_2 - 1| = |1 - 1| = 0 < \varepsilon$. But clearly a_n does not converge to 1. In fact, a_n does not converge to anything. Can we prove this? Yes:

Example 2.8 Claim: the sequence $a_n = (-1)^n$ does not converge (to **any** limit).
Proof: Assume, to the contrary, that $a_n \to L$. Then, given any positive number ε, there exists $N \in \mathbb{Z}^+$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. In particular, this must be true in the case $\varepsilon = \frac{1}{2}$: there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $|a_n - L| < \frac{1}{2}$. But then N and $N + 1$ are consecutive integers, so one is odd and the other is even, and hence $|a_{N+1} - a_N| = 2$. But, by the triangle inequality,

$$|a_{N+1} - a_N| = |(a_{N+1} - L) - (a_N - L)| \leq |a_{N+1} - L| + |a_N - L| < \frac{1}{2} + \frac{1}{2} = 1$$

by the definition of N. Hence $2 < 1$, a contradiction. □

We next establish that two rather simple, but important, sequences converge to their expected limits.

Example 2.9 The constant sequence, $a_n = c$ for all n, converges to c.
Proof: Given any $\varepsilon > 0$ we can take $N = 1$, since for all $n \geq 1$, $|a_n - c| = |c - c| = 0 < \varepsilon$. □

Direct proofs of this kind are often called "ε–N proofs". We finish this section by showing how a direct ε–N proof can be made even for a reasonably complicated sequence.

Example 2.10 Claim: $a_n = \dfrac{n^2 - \sin n}{(2n-7)(3n+1)} \to \dfrac{1}{6}$.

Proof: The key is to estimate $|a_n - \frac{1}{6}|$. For all $n \in \mathbb{Z}^+$,

$$\begin{aligned}
\left|a_n - \frac{1}{6}\right| &= \frac{1}{6}\left|\frac{6n^2 - 6\sin n - (2n-7)(3n+1)}{(2n-7)(3n+1)}\right| \\
&= \frac{1}{6}\frac{|6n^2 - 6\sin n - (6n^2 - 19n - 7)|}{|2n-7||3n+1|} \\
&= \frac{1}{6}\frac{|19n - 6\sin n + 7|}{(2n-7)(3n+1)} \qquad (\text{if } n \geq 4) \\
&\leq \frac{1}{6}\frac{19n + 6|\sin n| + 7}{(2n-7)(3n+1)} \qquad (\text{triangle inequality}) \\
&\leq \frac{1}{6}\frac{32n}{(2n-7)(3n+1)} \qquad (|\sin n| \leq 1 \text{ and } 1 \leq n) \\
&< \frac{1}{6}\frac{32n}{(2n-n)3n} \qquad (\text{if } n \geq 7) \\
&< \frac{2}{n}
\end{aligned}$$

So we've shown that for all $n \geq 7$, $|a_n - \frac{1}{6}| < \frac{2}{n}$.

Now, let $\varepsilon > 0$ be given. Then, by the Archimedean Property, there exists $N \in \mathbb{Z}^+$ such that $N \geq \max\{\frac{2}{\varepsilon}, 7\}$. Then for all $n \geq N$, $n \geq 7$ and $n \geq \frac{2}{\varepsilon}$, so by the above estimate

$$\left|a_n - \frac{1}{6}\right| < \frac{2}{n} \leq \varepsilon.$$

□

Notation: In the above question, I used the following notation. For real numbers $a, b \in \mathbb{R}$, we write $\max\{a, b\}$ for the largest number of a and b (or maximum of a and b), that is, in this case,

$$\max\{a, b\} = \begin{cases} a & \text{if } a \geq b \\ b & \text{otherwise} \end{cases}.$$

We will see more about the maximum later in the course.

2.2 Some limit theorems

It's very important that you get plenty of practice in proving directly from first principles that sequences converge (i.e. constructing ε–N proofs) because this is the best way to get to grips with what Definition 2.4 really means. It is also good preparation for understanding the proofs of the following basic limit theorems. Once we've proved these, we can often use them, together with a few standard examples, such as Examples 2.9 and 2.5, to show indirectly that a given sequence converges, without having to give an ε–N argument.

Proposition 2.11 (Uniqueness of Limits) *If a sequence converges, its limit is unique.*

Proof: Assume, to the contrary, that a sequence a_n converges to both L_1 and L_2 with $L_1 \neq L_2$. Let $\varepsilon = \dfrac{1}{2}|L_1 - L_2| > 0$. Since $a_n \to L_1$, there exists $N_1 \in \mathbb{Z}^+$ such that, for all $n \geq N_1$,

$$|a_n - L_1| < \varepsilon.$$

Similarly, since $a_n \to L_2$, there exists $N_2 \in \mathbb{Z}^+$ such that, for all $n \geq N_2$,

$$|a_n - L_2| < \varepsilon.$$

Now let n be any positive integer greater than both N_1 and N_2 (for example $n = N_1 + N_2$). Then both the above inequalities hold, and so,

$$\begin{aligned} 2\varepsilon = |L_2 - L_1| &= |(a_n - L_1) - (a_n - L_2)| \\ &\leq |a_n - L_1| + |a_n - L_2| \qquad \text{(by the Triangle Inequality)} \\ &< \varepsilon + \varepsilon \qquad \text{(since } n > N_1 \text{ and } n > N_2\text{)} \end{aligned}$$

that is, $2\varepsilon < 2\varepsilon$, a contradiction. Hence, the original assumption (that (a_n) has two different limits) must be false. □

So we are justified in speaking of **the** limit of a convergent sequence.

Definition 2.12 We say that a real sequence (a_n) is **bounded above** if its rangeis bounded above, that is, if there exists $M \in \mathbb{R}$ such that for all $n \in \mathbb{Z}^+$, $a_n \leq M$. Similarly, (a_n) is **bounded below** if there is a $L \in \mathbb{R}$ such that for all $n \in \mathbb{Z}^+$, $L \leq a_n$. We will say that (a_n) is **bounded** if it is bounded both above and below.

Proposition 2.13 *If a sequence converges then it is bounded.*

Proof: Assume $a_n \to L$. Then given any $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $|a_n - L| < \varepsilon$. This is true, in particular, in the case $\varepsilon = 1$. Hence, there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $|a_n - L| < 1$. Let

$$K = \max\{|a_1 - L|, |a_2 - L|, \ldots, |a_{N-1} - L|, 1\},$$

which exists since the set is finite. Then for all $n \in \mathbb{Z}^+$, $|a_n - L| \leq K$, and hence, $L - K \leq a_n \leq L + K$. Hence, the sequence (a_n) is bounded (above by $L + K$ and below by $L - K$). □

Of course, the converse of this Proposition is false: not every bounded sequence is convergent, as we have already seen (for example $a_n = (-1)^n$ is bounded, but not convergent). **If** a sequence is convergent, it's bounded above and below, and it's not hard to show that its limit must lie between the upper and lower bounds:

Proposition 2.14 *If $K \leq a_n \leq M$ for all $n \in \mathbb{Z}^+$ and (a_n) converges to L, then $K \leq L \leq M$.*

Proof: Exercise. The obvious strategy is a proof by contradiction. That is, assume $a_n \to L$ but $L \notin [K, M]$. Then either (i) $L > M$ or (ii) $L < K$. Now in each of these cases, show that a contradiction arises. □

Note that a sequence (a_n) is bounded if and only if $|a_n|$ is bounded above.

Theorem 2.15 (Algebra of Limits) *If $a_n \to A$ and $b_n \to B$ then*

(i) $a_n + b_n \to A + B$,

(ii) $a_n b_n \to AB$.

If, in addition, $a_n \neq 0$ for all n and $A \neq 0$, then

(iii) $\dfrac{1}{a_n} \to \dfrac{1}{A}$.

Proof:

(i) Let $\varepsilon > 0$ be given. Then $\frac{\varepsilon}{2}$ is also a positive number, and $a_n \to A$, so there exists $N_1 \in \mathbb{Z}^+$ such that, for all $n \geq N_1$, $|a_n - A| < \frac{\varepsilon}{2}$. Similarly, $b_n \to B$, so there exists $N_2 \in \mathbb{Z}^+$ such that, for all $n \geq N_2$, $|b_n - B| < \frac{\varepsilon}{2}$. Let $N = \max\{N_1, N_2\}$. Then, for all $n \geq N$,

$$\begin{aligned} |(a_n + b_n) - (A + B)| &= |(a_n - A) + (b_n - B)| \\ &\leq |a_n - A| + |b_n - B| \qquad \text{(Triangle Inequality)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \qquad \text{(since } n \geq N_1 \text{ and } n \geq N_2\text{)} \\ &= \varepsilon. \end{aligned}$$

Hence, given any $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $|a_n + b_n - (A + B)| < \varepsilon$. Hence $a_n + b_n \to A + B$.

(ii) Let $\varepsilon > 0$ be given. By Proposition 2.13, since b_n is convergent, it is bounded, so there exists $K > 0$ such that $|b_n| \leq K$ for all n. Then $\varepsilon' = \varepsilon/(K + |A|)$ is a positive number, and $a_n \to A$ and $b_n \to B$, so there exist $N_1, N_2 \in \mathbb{Z}^+$ such that $|a_n - A| < \varepsilon'$ for all $n \geq N_1$, and $|b_n - B| < \varepsilon'$ for all $n \geq N_2$. Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$\begin{aligned} |a_n b_n - AB| &= |a_n b_n - A b_n + A b_n - AB| \\ &= |(a_n - A) b_n + A(b_n - B)| \\ &\leq |a_n - A||b_n| + |A||b_n - B| \qquad \text{(Triangle Inequality)} \\ &< \varepsilon' K + |A|\varepsilon' \qquad \text{(since } n \geq N_1 \text{ and } n \geq N_2\text{)} \\ &= \varepsilon. \end{aligned}$$

Hence, $a_n b_n \to AB$.

(iii) Let $\varepsilon > 0$ be given. Since $a_n \to A \neq 0$, and $\frac{|A|}{2}$ is a positive number, there exists $N_1 \in \mathbb{Z}^+$ such that, for all $n \geq N_1$, $|a_n - A| < \frac{|A|}{2}$, and hence, $|a_n| > \frac{|A|}{2}$ by the reverse triangle inequality. Similarly, $\varepsilon' = \frac{\varepsilon}{2}|A|^2$ is also a positive number, so there exists $N_2 \in \mathbb{Z}^+$, such that, for all $n \geq N_2$, $|a_n - A| < \varepsilon'$. Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$\begin{aligned} \left|\frac{1}{a_n} - \frac{1}{A}\right| &= \left|\frac{A - a_n}{a_n A}\right| = \frac{|a_n - A|}{|a_n||A|} \\ &< \frac{\varepsilon'}{|a_n||A|} \qquad \text{(since } n \geq N_2\text{)} \\ &< \frac{\varepsilon'}{(|A|/2)|A|} \qquad \text{(since } n \geq N_1\text{)} \\ &= \varepsilon. \end{aligned}$$

Hence, $\frac{1}{a_n} \to \frac{1}{A}$.

□

Starting from simple known facts, like $1/n \to 0$, we can use Theorem 2.15 to deduce the convergence of a large collection of sequences. For example, $1/n^2$, $1/n^3$, $1/n^4 \ldots$ all converge to 0:

Example 2.16 Claim: for any integer $p \in \mathbb{Z}^+$, the sequence $a_n = \dfrac{1}{n^p}$ converges to 0.

Proof: We will prove this by induction (on p). The claim is true for $p = 1$, by Example 2.5. Assume the claim holds for $p = k$. Then, in the case $p = k + 1$,

$$a_n = \frac{1}{n}\frac{1}{n^k} \to 0 \times 0 = 0$$

by Theorem 2.15 (part (ii)). Hence, by induction, the claim holds for all $p \in \mathbb{Z}^+$. □

Example 2.17 Claim: the sequence $a_n = \dfrac{2n^2+n}{n^2+7}$ converges to 2.

Proof: First note that

$$a_n = \frac{2+\frac{1}{n}}{1+\frac{7}{n^2}} = \left(2+\frac{1}{n}\right) \times \frac{1}{1+\frac{7}{n^2}}.$$

Now the constant sequence 2 converges to 2 (Example 2.9) and $\frac{1}{n} \to 0$ (Example 2.5), so $2+\frac{1}{n} \to 2+0=2$ by Theorem 2.15(i). Similarly, 1 converges to 1, 7 to 7 and $\frac{1}{n^2}$ converges to 0 (Example 2.16), so $7/n^2 \to 7\times 0 = 0$ (Theorem 2.15(ii)), and hence $1+\frac{7}{n^2} \to 1$ (Theorem 2.15(i)). Furthermore, $1+\frac{7}{n^2}$ is never 0, so $1/(1+\frac{7}{n^2}) \to \frac{1}{1} = 1$ (Theorem 2.15(iii)). Hence $a_n \to 2\times 1 = 2$ (Theorem 2.15(ii)). □

Clearly, if we maintain this level of detail in our arguments, we will pass out through exhaustion. So in future, we will use Theorem 2.15 more freely, along the lines of the following example:

Example 2.18 Claim: the sequence $a_n = \dfrac{n^4}{2(n+1)^2(n^2+1)}$ converges to $\frac{1}{2}$.

Proof:

$$a_n = \frac{1}{2(1+(1/n))^2(1+(1/n^2))} \to \frac{1}{2\times(1+0)^2\times(1+0)} = \frac{1}{2}$$

by Theorem 2.15. □

Exercise 2.19 Prove from first principles that the sequence in Example 2.18 converges to $\frac{1}{2}$ (i.e. give a direct ε–N proof). It's good for your (mathematical) soul...

Theorem 2.20 (The Squeeze Rule) *Let (a_n), (b_n) and (c_n) be sequences such that (a_n) and (c_n) converge to the same limit L. If $a_n \le b_n \le c_n$ for all $n \in \mathbb{Z}^+$, then (b_n) also converges to L.*

Proof: Let $\varepsilon > 0$ be given. Since $a_n \to L$, there exists $N_1 \in \mathbb{Z}^+$ such that, for all $n \ge N_1$, $|a_n - L| < \varepsilon$, and hence, $a_n > L - \varepsilon$. Similarly, since $c_n \to L$, there exists $N_2 \in \mathbb{Z}^+$ such that, for all $n \ge N_2$, $|c_n - L| < \varepsilon$, and hence, $c_n < L + \varepsilon$. Let $N = \max\{N_1, N_2\}$. Then for all $n \ge N$,

$$\begin{aligned} & b_n - L \ge a_n - L \\ & \qquad > -\varepsilon \qquad (\text{since } n \ge N_1) \\ \text{and} \quad & b_n - L \le c_n - L \\ & \qquad < \varepsilon \qquad (\text{since } n \ge N_2). \end{aligned}$$

Hence, for all $n \ge N$, $|b_n - L| < \varepsilon$. □

Example 2.21 Claim: $b_n = \dfrac{\cos n}{n^2+1} \to 0$.

Proof: Let $a_n = -\dfrac{1}{n^2}$ and $c_n = \dfrac{1}{n^2}$. Then $a_n \le b_n \le c_n$ for all $n \in \mathbb{Z}^+$ (since $-1 \le \cos n \le 1$) and $a_n \to 0$ and $c_n \to 0$ (by Example 2.16 and Theorem 2.15), so $b_n \to 0$ by the Squeeze Rule. □

Summary

- A **real sequence** is a mapping $a : \mathbb{Z}^+ \to \mathbb{R}$. We denote the image of $n \in \mathbb{Z}^+$ under the mapping a by a_n and call it the nth **term** of the sequence. We usually denote the sequence as a whole by (a_n).
- A sequence (a_n) **converges** to a real number L if, for each $\varepsilon > 0$ there exists a positive integer N such that, for all $n \geq N$, $|a_n - L| < \varepsilon$.
- If (a_n) converges to L, we write $a_n \to L$.
- The number L is called the **limit** of the sequence, sometimes denoted $\lim a_n$ or $\lim\limits_{n\to\infty} a_n$.
- An ε–N proof of convergence is a direct argument showing that, given any positive number ε, there is a positive integer N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.
- The limit of a convergent sequence is unique.
- If a sequence is convergent, then it is bounded. The converse is false.
- The usual **Algebra of Limits** holds: if $a_n \to A$ and $b_n \to B$ then
  - $a_n + b_n \to A + B$,
  - $a_n b_n \to AB$, and
  - $a_n/b_n \to A/B$ (provided $B \neq 0$ and b_n is never 0).
- The **Squeeze Rule**: if $a_n \leq b_n \leq c_n$ and $a_n \to L$ and $c_n \to L$, then $b_n \to L$.

3 Sets of real numbers

You will have studied sets in Core Mathematics over the last few weeks, and we now remind ourselves of some set notation.

- Recall that a **set** is a collection of objects, real or of the imagination.
- Given a set A, we write $x \in A$ to denote that object x is in set A. This is usually translated into words (when reading the expression) as "x is an element of A". The expression $x \notin A$ means x is **not** an element of A.
- Another set B is a **subset** of A if every element of B is an element of A. This is denoted $B \subseteq A$. Note that every set is a subset of itself, $A \subseteq A$. Two sets A, B are equal if they have precisely the same elements, in which case $A \subseteq B$ and $B \subseteq A$.
- We have already seen several sets of numbers such as the real numbers $\mathbb{R}$, the positive integers $\mathbb{Z}^+$, the integers $\mathbb{Z}$ and the rational numbers (or quotients) $\mathbb{Q}$ (see section 1.1).
- A further very important set if the **empty set**, denoted $\emptyset$. This is the set containing no objects. Do not confuse it with $\{\emptyset\}$, which certainly isn't the empty set since it contains an object (namely, the empty set)!

One of the most common ways of specifying a set is by simply listing its elements inside curly brackets, for example

$$A = \{1, 2, 3, 4, 5, 6, 7, 8\} \ .$$

A second if by giving a condition which elements must satisfy,

$$A = \{x \ : \ x \in \mathbb{Z}^+, \ x \leq 8\}.$$

In general $A = \{x \,:\, P(x)\}$ means the set of objects x for which condition $P(x)$ is true. If part of condition $P(x)$ is that x should be in a particular set, we often move that part to the left of the colon. For example, the set above could be written

$$A = \{x \in \mathbb{Z}^+ \,:\, x \leq 8\}$$

and we could write the rational numbers as

$$\mathbb{Q} = \left\{x \in \mathbb{R} : x = \frac{a}{b} \text{ where } a, b \in \mathbb{Z}, b \neq 0\right\} .$$

Some important sets of the real numbers are the following: Given real numbers a, b, with $a < b$, we define several different **intervals**, as follows:

$$\begin{aligned}
(a,b) &= \{x \in \mathbb{R} \,:\, a < x < b\} \\
[a,b] &= \{x \in \mathbb{R} \,:\, a \leq x \leq b\} \\
[a,b) &= \{x \in \mathbb{R} \,:\, a \leq x < b\} \\
(a,b] &= \{x \in \mathbb{R} \,:\, a < x \leq b\} \\
(a,\infty) &= \{x \in \mathbb{R} \,:\, x > a\} \\
[a,\infty) &= \{x \in \mathbb{R} \,:\, x \geq a\} \\
(-\infty,b) &= \{x \in \mathbb{R} \,:\, x < b\} \\
(-\infty,b] &= \{x \in \mathbb{R} \,:\, x \leq b\}.
\end{aligned}$$

So a round bracket denotes that the end point is omitted from the interval, while a square bracket denotes that it is included. Note that ∞ is **not** a number, so expressions like $(a, \infty]$ do not make sense. Intervals which **include** all their endpoints, that is, $[a, b]$, $[a, \infty)$, $(-\infty, b]$ are said to be **closed**. Those which exclude all their endpoints, that is, (a, b), (a, ∞), $(-\infty, b)$ are said to be **open**.

Given two sets A, B, there are several useful ways of constructing new sets from them.

- The **intersection** of A and B is

$$A \cap B = \{x \,:\, x \in A \text{ and } x \in B\}.$$

- The **union** of A and B is

$$A \cup B = \{x \,:\, x \in A \text{ or } x \in B\}.$$

  Note that, in mathematics, we use "or" in the non-exclusive sense, that is, "P or Q" means P is true, or Q is true (or both).

- The **complement** of B in A is

$$A \backslash B = \{x \,:\, x \in A \text{ and } x \notin B\}.$$

- The **cartesian product** of A and B is

$$A \times B = \{(x, y) \,:\, x \in A \text{ and } y \in B\},$$

  that is, the set of **ordered pairs** of objects (x, y), where x comes from A and y comes from B. Do not confuse the symbol $\times$, used in the context of sets, with multiplication.

Example 3.1

$$\begin{aligned}
\{1,2,3,4,5,6,7,8\} &= [1,8] \cap \mathbb{Z} \\
&= (-\infty, 9) \cap \mathbb{Z}^+ \\
&= \{1,3,5,7\} \cup \{2,4,6,8\} \\
\mathbb{R} \backslash \{-1,1\} &= (-\infty,-1) \cup (-1,1) \cup (1,\infty) \\
\{\text{me}, \text{you}\} \times \{1,2,3\} &= \{(\text{me},1), (\text{me},2), (\text{me},3), (\text{you},1), (\text{you},2), (\text{you},3)\} \\
(-2,\infty) \cap (-\infty,0] &= (-2,0]
\end{aligned}$$

3.1 Boundedness and the axiom of completeness

When we first learn about real numbers, it is common to think of them as infinite decimal expansions, that is, expressions of the form

$$20142108.030118151205\ldots$$

with finitely many digits taken from $\{0, 1, \ldots, 9\}$ to the left of a dot (the "decimal point") and infinitely many to the right (possibly all but finitely many of these being 0). This is OK, but it has some disadvantages: the choice of base 10 for the expansion is pretty arbitrary, and there's the rather annoying fact that different expansions can represent the same number, for example,

$$0.999999\ldots = 1.000000\ldots.$$

For our purposes, it is better to take a radically different approach: rather than attempt to **define** the set of real numbers constructively, we will characterize it by stating the properties that we assume it has. These properties come in three stages – we have already seen two of these.

First, we assume that $\mathbb{R}$ is a **field**, that is, a set on which the four basic operations of arithmetic $(+, \times, -, \div)$ are well defined and have the familiar properties (e.g. $x + y = y + x$ and $x \times (y + z) = (x \times y) + (x \times z)$).

Second, we assume that $\mathbb{R}$ is an **ordered** field: it is equipped with an ordering relation $<$. This is a structure which, given a pair of distinct elements $x, y \in \mathbb{R}$, $x \neq y$, determines which of them is the **smaller**. So, for any $x, y \in \mathbb{R}$, exactly one of $x = y$, $x < y$ or $y < x$ is true. The ordering relation $<$ is also assumed to satisfy the usual properties (e.g. if $x < y$ then $x + z < y + z$ for all real numbers z).

These first two structures (the arithmetic operations and ordering relation) have exactly the properties you've been familar with since primary school, so we won't labour them here.[2] In fact, you already know an example of an ordered field, namely the set of **rational numbers** $\mathbb{Q}$.

So what is the difference between $\mathbb{Q}$ and $\mathbb{R}$? For our purposes, the key difference is that $\mathbb{R}$ satisfies an extra property called **completeness**, while $\mathbb{Q}$ does not. This property is not familiar from school level maths, so we will study it in some depth. To define completeness requires some preliminary ideas.

Definition 3.2 Let A be a subset of $\mathbb{R}$. An **upper bound** on A is any $K \in \mathbb{R}$ such that, for all $x \in A$, $x \leq K$. A **lower bound** on A is any $L \in \mathbb{R}$ such that, for all $x \in A$, $x \geq L$. If A has an upper bound, we say that A is **bounded above**. If A has a lower bound, we say that A is **bounded below**. A is **bounded** if it is bounded both above and below.

Example 3.3 The set $A = [-1, 2)$ is bounded above, by 2 for example, and below, by -1 for example. Hence A is bounded. Note that upper and lower bounds are **not** unique. That is, $2, 54, 19345\pi$ are all upper bounds on A. In fact, any $K \geq 2$ is an upper bound on A, and any $L \leq -1$ is a lower bound on A.

Remark 3.4 It is important not to confuse **bounded** with **finite**. A finite set is one containing a finite number of distinct elements. Any finite subset of $\mathbb{R}$ will certainly be bounded, but the converse is clearly false: not every bounded subset of $\mathbb{R}$ is finite, as the above example shows.

Example 3.5 The set of positive integers $\mathbb{Z}^+$ is bounded below, by 1 for example (or -54, or -1000). However, it is **not** bounded above.

Let's think carefully about what it means to say that a set A is **unbounded above**. It means that A has no upper bound. Equivalently, every real number K is **not** an upper bound on A. Hence, given any $K \in \mathbb{R}$, there exists $x \in A$ such that $x > K$.

[2] If you'd like to see a systematic development of the idea of an ordered field, you'll find it in most textbooks on real analysis. *A Sequential Introduction to Real Analysis* by J.M. Speight (World Scientific, Singapore, 2016) would be a great choice (have a look at who is lecturing the second part of this course).

So an alternative formulation of the statement "$\mathbb{Z}^+$ is unbounded above" is "given any real number K there exists a positive integer n such that $n > K$.". We have already seen this property - it is the Archimedean property of $\mathbb{R}$!

Note that $\mathbb{Q}$ also has the Archimedean Property (given any $K \in \mathbb{Q}$ there is some $n \in \mathbb{Z}^+$ such that $n > K$).

Given a set of real numbers, we would like to be able to use properties of the largest and smallest element. Our first attempt at a "largest" and "smallest" value in a set could be the following:

Definition 3.6 The **maximum** of a subset A of $\mathbb{R}$ (if it exists) is the element $a \in A$ such that for any other $b \in A$, $b \leq a$.

The **minimum** of a subset A of $\mathbb{R}$ (if it exists) is the element $a \in A$ such that for any other $b \in A$, $b \geq a$.

This works well for finite sets (i.e. sets with only a finite number of elements).

Example 3.7 Consider the set $A = \{1, 2, 3, 5, 7\}$. Then $\max A = 7$ and $\min A = 1$.

In fact we have already been using this notation - we have written $\max\{N_1, N_2\}$ for the largest of the numbers N_1 and N_2, and we now see that really we were writing them as the maximum value of the 2-element set $\{N_1, N_2\}$.

However, this definition starts to break down pretty quickly - for example, we already get into problems with intervals. We would like to say that $[-1, 2)$ has maximum 2, but in fact, as $2 \notin [-1, 2)$ it has no maximum! Clearly, we want a definition which gives a "maximum" for any set that is bounded above.

Definition 3.8 The **supremum** of a subset A of $\mathbb{R}$ is its least upper bound (if this exists), denoted $\sup A$. So $\sup A$ is a real number with two properties:

(i) $\sup A$ is an upper bound on A.

(ii) No number less than $\sup A$ is an upper bound on A.

Similarly the **infimum** of A is its greatest lower bound (if this exists), denoted $\inf A$.

It follows immediately from the definition that $\sup A$, if it exists, is unique. For if two numbers both satisfy the above two properties and are different, then the smaller of them, K_1 say, is an upper bound on A by (i), and is less than the larger one, K_2 say, contradicting (ii). So it makes sense to speak of **the** supremum and/or infimum of a subset of $\mathbb{R}$, in contrast to upper/lower bounds.

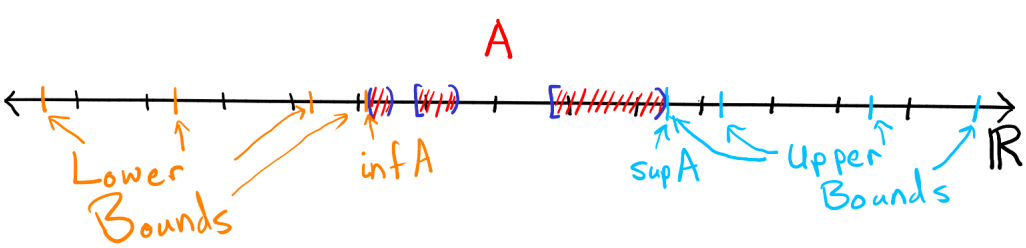


Figure 2: Picture of upper and lower bounds, and the supremum and infimum for a set A (shaded in) on the number line $\mathbb{R}$

Example 3.9 Let $A = [-1, 2)$. Then, immediately from the definition of this kind of interval, $\sup A = 2$ and $\inf A = -1$.

Example 3.10 Let $A = \{\frac{1}{n} : n \in \mathbb{Z}^+\}$. Every element of A is positive, so A is bounded below, by 0. In fact, $\inf A = 0$. Let's prove this. We already know that 0 is a lower bound on A, so it just remains to show that every number bigger than 0 is **not** a lower bound on A. So, let $K > 0$. Then $1/K$ is a real number, so by the Archimedean Property, there exists some $n \in \mathbb{Z}^+$ such that $n > 1/K$. But then $1/n \in A$ and $1/n < K$, so K is not a lower bound on A.

The supremum is a bit less subtle. I claim that $\sup A = 1$. Again, let's prove this. Let $x \in A$. Then $x = 1/n$ for some $n \in \mathbb{Z}^+$. But $n \geq 1$, so $x \leq 1$. Hence 1 is an upper bound on A. Any $K < 1$ is not an upper bound on A, since $1 \in A$.

The existence of suprema (= plural of supremum) is the defining difference between $\mathbb{R}$ and $\mathbb{Q}$.

Axiom 3.11 (The Axiom of Completeness) Every nonempty subset of $\mathbb{R}$ which is bounded above has a supremum in $\mathbb{R}$.

The equivalent statement for $\mathbb{Q}$ (replace $\mathbb{R}$ by $\mathbb{Q}$ in both places) is false. For example $A = \{x \in \mathbb{Q} : x^2 < 2\}$ is a nonempty subset of $\mathbb{Q}$ (e.g. $0 \in A$) which is bounded above (e.g. every $x \in A$ is less than 2), but it has no supremum in $\mathbb{Q}$, that is, there is no **rational** number r with the required properties that (i) r is an upper bound on A and (ii) no rational number $s < r$ is an upper bound on A. The problem is, of course, that $\sqrt{2}$ is not rational, and there is no smallest rational number bigger than $\sqrt{2}$.

The axiom of completeness looks rather lopsided: it says that every nonempty subset of $\mathbb{R}$ which is bounded **above**, has a supremum, but says nothing about sets which are bounded **below**. That's because the equivalent statement for **infima** (= plural of infimum) follows directly from Axiom 3.11. Proving this is a good exercise in keeping clear sight of what upper/lower bound and sup/inf mean:

Proposition 3.12 *Let $A \subset \mathbb{R}$ be nonempty and bounded below. Then A has an infimum.*

Proof: Let $B = \{-x : x \in A\}$, which is certainly nonempty. Let K be a lower bound on A. Then $-K$ is an upper bound on B. (Check: if $y \in B$ then $-y \in A$, and hence $-y \geq K$. Hence $y \leq -K$.) Hence, B is a nonempty subset of $\mathbb{R}$ which is bounded above so, by the Axiom of Completeness, B has a supremum, L say. Now $-L$ is a lower bound on A. (Check: if $x \in A$ then $-x \in B$, so $-x \leq L$. Hence $x \geq -L$.)

Let M be any real number greater than $-L$. Then $-M < L$, and L is the **least** upper bound on B. Hence $-M$ is **not** an upper bound on B, so there exists $y \in B$ such that $y > -M$. But then $-y \in A$ and $-y < M$, so M is not a lower bound on A. Hence, no number greater than $-L$ is a lower bound on A. Hence $-L$ is the infimum of A. $\square$

3.2 Some consequences of the axiom of completeness

In this section, we will deduce some basic properties of the set $\mathbb{R}$ by using Axiom 3.11. The first of these we have already asserted as Fact: the set $\mathbb{Z}^+$ is unbounded above. This looks entirely obvious, but is it really? As far as we're concerned, $\mathbb{R}$ is, **by definition** a complete ordered field (so we are not defining it to be the set of all decimal expansions, for example). Is it really obvious that a complete ordered field cannot have a magic element, x say, which is bigger than every positive integer?

Theorem 3.13 (The Archimedean Property of $\mathbb{R}$) *Given any real number K, there is some positive integer n such that $n > K$.*

Proof: Assume, to the contrary, that $\mathbb{Z}^+$ is bounded above. Then, by Axiom 3.11, $\mathbb{Z}^+$ has a supremum, U, say. Since $U - 1 < U$, it is not an upper bound on $\mathbb{Z}^+$, so there exists $n \in \mathbb{Z}^+$ such

that $n > U - 1$. But then $n + 1 \in \mathbb{Z}^+$, and $n + 1 > U$, contradicting the fact that U is an upper bound on $\mathbb{Z}^+$. $\square$

That's a relief! Note that the only property of $\mathbb{Z}^+$ we used in the above proof was the "successor property": if n is in $\mathbb{Z}^+$, then $n + 1$ is in $\mathbb{Z}^+$ also. By an almost identical argument, we can show that any subset of $\mathbb{R}$ which has the "predecessor property" must be unbounded **below**.

Lemma 3.14 *Let $A \subseteq \mathbb{R}$ be any nonempty subset of $\mathbb{R}$ with the property that whenever $m \in A$ then $m - 1 \in A$. Then A is unbounded below.*

Proof: Assume, to the contrary, that A is bounded below. Then, by Proposition 3.12, A has an infimum, L say. Since $L + 1 > L$, it is not a lower bound on A, so there exists $m \in A$ such that $m < L + 1$. But then, by our assumption about A, $m - 1 \in A$ also, and $m - 1 < L$. This contradicts the fact that L is a lower bound on A. $\square$

As a corollary of these two results, we can prove that the set of rational numbers must be **dense** in the set of reals, in the following sense:

Theorem 3.15 ($\mathbb{Q}$ is dense in $\mathbb{R}$) *Between any pair of distinct real numbers, there is a rational number.*

Proof: Let's call the smaller of the two real numbers x and the larger y. By the Archimedean Property, there exists $n \in \mathbb{Z}^+$ such that $n > \frac{1}{y-x}$, and hence, $ny - 1 > nx$. We will show that there exists a rational number of the form m/n between x and y.

Consider the set

$$E = \{m \in \mathbb{Z} \,:\, m > nx\}.$$

This is nonempty, by the Archimedean Property again (if it weren't, nx would be an upper bound on $\mathbb{Z}$, and hence on $\mathbb{Z}^+ \subset \mathbb{Z}$). We claim there exists $m \in E$ such that $m < ny$. Assume this is false, so every $m \in E$ satisfies $m \geq ny$. Then, given any $m \in E$, $m - 1 \geq ny - 1 > nx$, so $m - 1 \in E$. Hence, E has the "predecessor property", so by Lemma 3.14, E is unbounded **below**. But E is clearly bounded below (by nx) by definition, a contradiction. Hence, the claim is true: there exists $m \in E$ with $m < ny$. But then m is an integer and $nx < m < ny$, so $x < \frac{m}{n} < y$, as required. $\square$

Recall that a real number is **irrational** if it is not rational. Since $\mathbb{Q}$ is countable, we deduce immediately that the set of irrational numbers, $\mathbb{R}\backslash\mathbb{Q}$, is uncountable. In particular, there certainly **are** irrational numbers in $[0, 1]$, although we do not yet have a single explicit example of such a number! In MATH1000 you may have seen that $\sqrt{2}$ can't be rational, but you have **not** yet proved that it exists at all! However, if you have completed the second Problem Set, you will have shown this. For now, we will assume that $\sqrt{2}$ exists, and we will see more on this later in the course. From the existence of a single irrational number, we can easily deduce that the irrational numbers are, like the rational numbers, dense in $\mathbb{R}$:

Theorem 3.16 ($\mathbb{R}\backslash\mathbb{Q}$ is dense in $\mathbb{R}$) *Between any pair of distinct real numbers there is an irrational number.*

Proof: Let's call the smaller of the two real numbers x and the larger y. Let w be an irrational number in $[0, 1]$. Then $w > 0$, so by Theorem 3.15, there exists a rational number $\frac{m}{n}$ such that

$$\frac{x}{w} < \frac{m}{n} < \frac{y}{w}.$$

Hence $\frac{m}{n}w$ lies between x and y and is certainly irrational (since if $\frac{m}{n}w = \frac{p}{q}$ then $w = \frac{pn}{qm}$, which is rational). $\square$

You will also see in Core Maths that while $\mathbb{Q}$ is countable, $\mathbb{R} \setminus \mathbb{Q}$ is not. So, between any pair of

rationals there's an irrational, and between any pair of irrationals there's a rational. But there are more irrationals between 0 and 1 than there are rationals in $\mathbb{R}$! This set $\mathbb{R}$ is very, very strange...

Definition 3.17 A sequence (a_n) is **increasing** if $a_{n+1} \geq a_n$ for all $n \in \mathbb{Z}^+$. It is **decreasing** if $a_{n+1} \leq a_n$ for all $n \in \mathbb{Z}^+$. It is **monotone** if it is increasing or decreasing.

Remarks

- It follows immediately from the definition that, if (a_n) is increasing then $a_n \geq a_m$ whenever $n \geq m$. Similarly, a decreasing sequence has $a_n \leq a_m$ for all $n \geq m$.
- It follows that an increasing sequence is automatically bounded below (by a_1) and a decreasing sequence is automatically bounded above (again, by a_1).
- Note that it's possible for a sequence to be both increasing and decreasing! But only if it's constant.
- We sometimes refer to a sequence as **strictly increasing**, meaning $a_{n+1} > a_n$ for all $n \in \mathbb{Z}^+$.

We have proved that all convergent sequences are bounded, and observed that not all bounded sequences are convergent ($a_n = (-1)^n$ provides a counterexample). We next prove that all **monotone** bounded sequences **are** convergent. This is the first limit theorem in which we make direct use of the Axiom of Completeness.

Theorem 3.18 *Let (a_n) be increasing and bounded above. Then (a_n) converges to*

$$L := \sup\{a_n \,:\, n \in \mathbb{Z}^+\}.$$

Proof: The set $A = \{a_n \,:\, n \in \mathbb{Z}^+\}$ is nonempty and bounded above, and hence, by Axiom 3.11, has a supremum, $L \in \mathbb{R}$. We claim that $a_n \to L$.

Let $\varepsilon > 0$. Then $L - \varepsilon < L$, so by the definition of supremum, $L - \varepsilon$ is **not** an upper bound on A. Hence there exists $N \in \mathbb{Z}^+$ such that $a_N > L - \varepsilon$. But (a_n) is increasing, so for all $n \geq N$, $a_n \geq a_N > L - \varepsilon$. Further, L **is** an upper bound on A, so for all $n \in \mathbb{Z}^+$, $a_n < L < L + \varepsilon$. Putting these two inequalities together, we see that for all $n \geq N$

$$-\varepsilon < a_n - L < \varepsilon$$

and hence $|a_n - L| < \varepsilon$. □

Theorem 3.19 *Let (a_n) be decreasing and bounded below. Then (a_n) converges.*

Proof: Exercise. You can modify the above proof (note you'll need to appeal to Theorem 3.12 rather than Axiom 3.11). Or you can define $b_n = -a_n$ and argue that it is increasing and bounded above. □

Corollary 3.20 (Monotone Convergence Theorem) *Every bounded monotone sequence converges.*

Example 3.21 Let $a_n = \alpha^n$ where $\alpha \in (0, 1)$. Claim: $a_n \to 0$.
Proof: $a_n > 0$ for all n, and $a_{n+1} = \alpha a_n < a_n$ since $0 < \alpha < 1$, so (a_n) is decreasing and bounded below. Hence, by the Monotone Convergence Theorem, $a_n \to L$ for some L. But $b_n = a_{n+1}$ converges to L also (it's the same sequence, but with the first term omitted), and $b_n = \alpha a_n$, so by Theorem 2.15, $b_n \to \alpha L$. But the limit of a convergent sequence is unique (Theorem 2.11), so $L = \alpha L$. If $L \neq 0$, then $\alpha = L/L = 1$, a contradiction. Hence $L = 0$. □

The key trick in the proof above was to think of (a_n) as an inductively defined sequence: $a_1 = \alpha$, $a_{n+1} = \alpha a_n$. We can try the same strategy for other inductively defined sequences.

Example 3.22 (Example 2.2 revisited) Recall we defined a sequence (a_n) inductively by specifying a_1 and using the rule

$$a_{n+1} = \frac{a_n}{1 + a_n^2} \tag{3.2}$$

for all $n \in \mathbb{Z}^+$. Consider this sequence in the case where $a_1 = \alpha > 0$ (the case $\alpha < 0$ can be handled similarly, and the case $\alpha = 0$ is trivial). We claim that $a_n \to 0$.
Proof: First note that $a_n > 0$ for all n, as one can easily show by induction ($a_1 > 0$ and if $a_n > 0$ then $a_{n+1} > 0$ by (3.2)). It follows that

$$\frac{a_{n+1}}{a_n} = \frac{1}{1 + a_n^2} < 1$$

and hence $a_{n+1} < a_n$ for all n. Hence (a_n) is decreasing. We have already shown that (a_n) is bounded below (by 0), so by the Monotone Convergence Theorem, a_n converges to some limit L. Clearly, the sequence $b_n = a_{n+1}$ also converges to L (in fact, this needs to be proven – we will see this shortly in Lemma 4.5). But

$$b_n = \frac{a_n}{1 + a_n^2}$$

so, by Theorem 2.15, b_n converges to $L/(1 + L^2)$. But the limit of a convergent sequence is unique (Theorem 2.11), so

$$L = \frac{L}{1 + L^2}$$

whose only solution is $L = 0$. Hence $a_n \to 0$.

□

Summary

- An **upper bound** on a set $A \subset \mathbb{R}$ is a number $K \in \mathbb{R}$ such that $x \leq K$ for all $x \in A$. A **lower bound** on A is $L \in \mathbb{R}$ such that $x \geq L$ for all $x \in A$. A is **bounded above** if it has an upper bound, **bounded below** if it has a lower bound, and **bounded** if it has both.
- The **supremum** of A is its least upper bound. That is, $K = \sup A$ if (i) K is an upper bound on A and (ii) no $L < K$ is an upper bound on A.
- The **infimum** of A is its greatest lower bound. That is, $K = \inf A$ if (i) K is a lower bound on A and (ii) no $L > K$ is a lower bound on A.
- $\mathbb{R}$ is, by definition, an ordered field which additionally satisfies the **Axiom of Completeness:** every nonempty subset of $\mathbb{R}$ which is bounded above has a supremum in $\mathbb{R}$.
- It follows from this definition that:
  - $\mathbb{R}$ has the Archimedean Property: given any $K \in \mathbb{R}$, there exists $n \in \mathbb{Z}^+$ such that $n > K$.
  - $\mathbb{Q}$ is dense in $\mathbb{R}$: between any pair of distinct real numbers, there is a rational number.
  - $\mathbb{R} \backslash \mathbb{Q}$ is dense in $\mathbb{R}$: between any pair of distinct real numbers, there is an irrational number.
- A sequence is increasing if $a_{n+1} \geq a_n$ for all $n \in \mathbb{Z}^+$. A sequence is decreasing if $a_{n+1} \leq a_n$ for all $n \in \mathbb{Z}^+$. A sequence is monotone if it is either increasing or decreasing.
- **Monotone convergence theorem** states that any bounded monotone sequence converges.

4 Subsequences

4.1 Definition and convergence properties

Definition 4.1 A sequence (b_k) is a **subsequence** of a sequence (a_n) if there exists a strictly increasing sequence of positive integers (n_k) such that $b_k = a_{n_k}$ for all $k \in \mathbb{Z}^+$.

In other words, all the terms of (b_k) must occur in (a_n), and they must occur in the same order, but we are allowed to omit some (possibly infinitely many) terms from (a_n).

Example 4.2 (i) The constant sequence $(b_k) = (1, 1, 1, 1, \ldots)$ is a subsequence of $a_n = (-1)^n$, with $n_k = 2k$. It's the subsequence consisting of only the even-numbered terms of (a_n). From this example we see that a divergent (that is non-convergent) sequence can have a convergent subsequence.

(ii) $b_k = \dfrac{1}{2^k}$ is a subsequence of $a_n = 1/n$, with $n_k = 2^k$. In this case, both (a_n) and (b_k) converge, and their limits are equal.

(iii) $b_k = \dfrac{1}{|3k - 11|}$ is **not** a subsequence of $a_n = 1/n$ even though every term of (b_k) is the reciprocal of a positive integer, so appears somewhere in the sequence (a_n). The problem is that the terms don't occur in the right order:

$$\begin{aligned}(b_k) &= \left(\frac{1}{8}, \frac{1}{5}, \frac{1}{2}, 1, \frac{1}{4}, \frac{1}{7}, \ldots\right) \\ &= (a_8, a_5, a_2, a_1, a_4, a_7, \ldots)\end{aligned}$$

(iv) Every sequence (a_n) is a subsequence of itself! We take n_k to be the increasing sequence $n_k = k$.

(v) Given any sequence (a_n), and any positive integer p, $b_k = a_{k+p}$ is a subsequence of (a_n), with $n_k = k + p$. This is just the subsequence obtained by omitting the first p terms terms of (a_n). $\square$

As we have seen, it is possible for a divergent sequence to have a convergent subsequence. On the other hand, if a sequence is convergent, with limit L, one expects that all its subsequences should also converge to L. This turns out to be true.

Theorem 4.3 *If $a_n \to L$ and (b_k) is a subsequence of (a_n), then $b_k \to L$.*

Proof: Let $\varepsilon > 0$ be given. Since $a_n \to L$, there exists $N \in \mathbb{Z}^+$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. Since (b_k) is a subsequence of (a_n), $b_k = a_{n_k}$ where n_k is a strictly increasing sequence of positive integers. Now $n_1 \geq 1$, and if $n_k \geq k$ then $n_{k+1} \geq n_k + 1 \geq k + 1$. Hence, by induction, $n_k \geq k$ for all k. Hence, for all $k \geq N$, $n_k \geq n_N \geq N$, so $|b_k - L| = |a_{n_k} - L| < \varepsilon$. Hence $b_k \to L$. $\square$

We can use this to give a sneaky proof that $a_n = (-1)^n$ doesn't converge, without having to resort to an ε–N argument:

Example 4.4 (Equation (2.8) revisited) Claim: $a_n = (-1)^n$ doesn't converge to any limit.
Proof: Assume, to the contrary, that $a_n \to L$. Then, by Theorem 4.3, the subsequences a_{2k} and a_{2k+1} also converge to L. But $a_{2k} = 1 \to 1$ and $a_{2k+1} = -1 \to -1$, so $1 = L = -1$, a contradiction. $\square$

Note that we've actually used Theorem 4.3 a couple of times already. In Examples 3.21 and 2.3 (revisited) we asserted that if $a_n \to L$ then $b_n = a_{n+1} \to L$ also. This follows from Theorem 4.3 since (b_n) is a subsequence of (a_n). It's important to realize that Theorem 4.3 does **not** say that if a subsequence of (a_n) converges to L then (a_n) converges to L (we've already seen a counterexample to this: $a_n = (-1)^n$). Just showing that a subsequence of (a_n) converges is not enough (usually) to show that (a_n) converges. There is an important case, however, where convergence of the subsequence does imply convergence of the whole sequence, namely, subsequences obtained by chopping off the first m terms of the sequence.

Lemma 4.5 (Tail Lemma) *Let (a_n) be a real sequence, m be a positive integer and $b_n = a_{n+m}$. If $b_n \to L$ then $a_n \to L$.*

Proof: Assume $b_n \to L$. Let $\varepsilon > 0$ be given. Then there exists $N_1 \in \mathbb{Z}^+$ such that, for all $n \geq N_1$, $|b_n - L| < \varepsilon$. Let $N = N_1 + m$. Then, for all $n \geq N$, $n - m \geq N_1$, so

$$|a_n - L| = |b_{n-m} - L| < \varepsilon.$$

Hence $a_n \to L$. □

This Lemma is very useful in practice, because it allows us to ignore any number of terms at the start of a sequence when attempting to prove convergence.

Example 4.6 Claim $a_n = \dfrac{n^3}{2^n}$ converges.
This isn't immediately obvious, because a_n is a quotient of functions of n both of which grow unbounded as n gets large, so it's not obvious whether (a_n) is even bounded, let alone convergent. To prove convergence, we'd like to use the Monotone Convergence Theorem, but there's a problem: (a_n) is neither increasing nor decreasing! In fact

$$a_1 < a_2 < a_3 < a_4,$$

but $a_n \geq a_{n+1}$ from there on (i.e. for all $n \geq 4$). So although (a_n) isn't decreasing, it is "eventually" decreasing, and by Lemma 4.5, that's good enough.
Proof: First note that a_n is bounded below (by 0, for example). Now

$$\begin{aligned}
\frac{a_{n+1}}{a_n} &= \frac{(n+1)^3}{2n^3} = \frac{n^3 + 3n^2 + 3n + 1}{2n^3} \\
&\leq \frac{n^3 + 3n^2 + 3n^2 + n^2}{2n^3} \qquad \text{(since } n \geq 1\text{)} \\
&= \frac{1}{2} + \frac{7}{2n} < 1 \qquad \text{provided } n > 7.
\end{aligned}$$

Hence, for all $n \geq 8$, $a_{n+1} < a_n$. Hence $b_n = a_{n+7}$ is a decreasing sequence, bounded below. By the Monotone Convergence Theorem, (b_n) converges so, by Lemma 4.5, (a_n) converges also. □

The above argument does **not** tell us what the limit of (a_n) is, only that the limit exists. In fact, $a_n \to 0$, but to show this we must work harder. We'll return to this later, after we've studied convergence of series.

4.2 The Bolzano-Weierstrass Theorem

We have seen that all subsequences of a convergent sequence are convergent, and that non-convergent sequences can sometimes have convergent subsequences. The goal of this section is to prove a famous and very powerful result called the Bolzano-Weierstrass[3] Theorem, which says that every **bounded**

[3] Weierstraß for German purists.

sequence has at least one convergent subsequence. It won't be immediately clear why this fact is such a big deal. Later we will use it to prove the Extreme Value Theorem, which is a crucial result when one comes to define integration carefully, for example.

The strategy of proof is quite straightforward: we first prove that **every** sequence has a monotone subsequence. Having done this, the Bolzano-Weierstrass Theorem follows immediately from the Monotone Convergence Theorem.

Lemma 4.7 *Every sequence has a monotone subsequence.*

Proof: Let the sequence in question be (a_n). We make the following definition: a term a_m is **dominant** if $a_n \leq a_m$ for all $n > m$ (that is, a_m is at least as big as all subsequent terms). Consider the set of all dominant terms of the sequence (a_n). There are exactly two possibilities: either (i) the set of dominant terms is infinite, or (ii) the set of dominant terms is finite (which includes the case where there are no dominant terms).

Case (i): if the set of dominant terms is infinite, then the subsequence consisting of just these terms is, by definition, decreasing. Hence (a_n) has a decreasing subsequence.

Case (ii): if the set of dominant terms is finite (or empty), there is a term in the sequence, a_m say, beyond which there are no dominant terms. Let $n_1 = m + 1$. Since a_{n_1} is **not** dominant, there exists $n > n_1$ such that $a_n > a_{n_1}$. Choose such an n and call it n_2. Then a_{n_2} is also not dominant (since $n_2 > n_1 > m$), so there exists a positive integer greater than n_2, let's call it n_3, such that $a_{n_3} > a_{n_2}$. In this way we construct a strictly increasing sequence of positive integers $n_1, n_2, n_3, \ldots$ such that $a_{n_{k+1}} > a_{n_k}$ for all k. But then (a_{n_k}) is an increasing subsequence of (a_n). Hence (a_n) has an increasing subsequence.

In either case, we see that (a_n) has a monotone subsequence, as was to be proved. □

Theorem 4.8 (The Bolzano-Weierstrass Theorem) *Let (a_n) be a bounded real sequence. Then (a_n) has a convergent subsequence.*

Proof: By Lemma 4.7, (a_n) has a monotone subsequence (a_{n_k}). But the range of (a_{n_k}) is a subset of the range of (a_n), so (a_{n_k}) is also bounded (by the same upper and lower bounds as (a_n)), and hence converges by the Monotone Convergence Theorem (Theorem 3.20). □

Of course, the converse of this Theorem is false: just because a sequence has a convergent subsequence, it doesn't follow that it's bounded. For example

$$a_n = \begin{cases} n & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

has a convergent subseqence, $a_{2k} = 0 \to 0$, but is clearly unbounded above.

Example 4.9 (Equation 2.1 revisited) Recall we defined the sequence $a_n = \sin n$ and noted that its terms "bounce around" in the interval $(-1, 1)$ seemingly at random. By Theorem 4.8 we know that a_n has at least one convergent subsequence, although we have absolutely no idea how to write it down, or what its limit might be. (In fact, given any $L \in [-1, 1]$ it turns out that this sequence has a subsequence converging to L, but proving this goes somewhat beyond the scope of this course).

4.3 The Cauchy property

One slightly inconvenient thing about Definition 2.4 is that, to prove a sequence converges from first principles, we must first know what its limit is. The Monotone Convergence Theorem allows us to prove convergence of some sequences without knowing their limits, but this only works in the special case where the sequence happens to be monotone (increasing or decreasing). In this section we will study a criterion which turns out to be **equivalent** to convergence, for **all** sequences, but which makes no mention of limits.

Definition 4.10 A real sequence (a_n) is **Cauchy** (or **has the Cauchy property**) if, for each $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that, for all $n, m \geq N$, $|a_n - a_m| < \varepsilon$.

Remark This looks very similar to the definition of convergence. It says that, given any $\varepsilon > 0$ (no matter how small), there is a point in the sequence beyond which **all terms lie closer than distance ε from one another**. Note that it makes no mention of any limit.

We will show that real sequences converge if and only if they have the Cauchy property.

Lemma 4.11 *If (a_n) converges then (a_n) is Cauchy.*

Proof: Assume $a_n \to L$. Let $\varepsilon > 0$ be given. Then $\varepsilon/2 > 0$, so there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$,

$$|a_n - L| < \frac{\varepsilon}{2}.$$

But then, for all $n, m \geq N$,

$$|a_n - a_m| = |(a_n - L) - (a_m - L)| \leq |a_n - L| + |a_m - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

by the Triangle Inequality. Hence (a_n) is Cauchy. □

Lemma 4.12 *If (a_n) is Cauchy then (a_n) is bounded.*

Proof: Exercise. Look to the proof of Proposition 2.13 for inspiration. □

Lemma 4.13 *Let (a_n) be Cauchy, and assume some subsequence of (a_n) converges to L. Then (a_n) converges to L.*

Proof: Let (a_{n_k}) be the subsequence which converges to L. Let $\varepsilon > 0$ be given. Since $a_{n_k} \to L$, there exists $N_1 \in \mathbb{Z}^+$ such that, for all $k \geq N_1$, $|a_{n_k} - L| < \varepsilon/2$. Further, since (a_n) is Cauchy, there exists $N_2 \in \mathbb{Z}^+$ such that, for all $n, m \geq N_2$, $|a_n - a_m| < \varepsilon/2$. Let $N = \max\{N_1, N_2\}$. Then, for all $n \geq N$,

$$\begin{aligned}
|a_n - L| &= |(a_n - a_{n_N}) + (a_{n_N} - L)| \\
&\leq |a_n - a_{n_N}| + |a_{n_N} - L| && \text{(Triangle Inequality)} \\
&< \frac{\varepsilon}{2} + |a_{n_N} - L| && \text{(since } n \geq N_2\text{, and } n_N \geq N \geq N_2\text{)} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} && \text{(since } N \geq N_1\text{).}
\end{aligned}$$

Hence, $a_n \to L$. □

Theorem 4.14 *A real sequence converges if and only if it is Cauchy.*

Proof: If (a_n) converges, it is Cauchy, by Lemma 4.11. Conversely, if (a_n) is Cauchy, it is bounded (Lemma 4.12), so has a convergent subsequence (by the Bolzano-Weierstrass Theorem) and so converges (Lemma 4.13). □

Summary

- A **subsequence** of a sequence (a_n) is any sequence of the form (a_{n_k}) where n_k is a strictly increasing sequence of positive integers.
- Non-convergent sequences can have convergent subsequences.
- If (a_n) converges to L, every subsequence of (a_n) also converges to L.
- If (a_{n+m}), where m is a constant postive integer, converges to L, then (a_n) converges to L.
- Every sequence has a monotone subsequence. Together with the Monotone Convergence Theorem, this implies:
- The **Bolzano-Weierstrass Theorem**: every **bounded** sequence has a convergent subsequence.
- A sequence (a_n) is **Cauchy** if, for each $\varepsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that, for all $n, m \geq N$, $|a_n - a_m| < \varepsilon$.
- A sequence converges if and only if it is Cauchy.

5 Series

5.1 Definition and convergence

Informally, a series is an infinite sum

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots ,$$

for example

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots .$$

Before proceeding any further, we should ask ourselves what, exactly, the expression on the right hand side of these equations really means. Why do we even need to bother with this? Am I simply wasting time and paper with this? Well here is a (non) example where our intuition fails us: We could ask what does $\sum_{n=1}^{\infty}(-1)^n$ converge to? Naively we could write (**but you never should**) something like

$$\sum_{n=1}^{\infty}(-1)^n = -1 + 1 - 1 + 1 - \ldots = (-1+1) + (-1+1) + \ldots = 0 + 0 + 0 + \ldots = 0$$

but then again, we could also naively write (**but again, please don't**)

$$\sum_{n=1}^{\infty}(-1)^n = -1 + 1 - 1 + 1 - \ldots = -1 + (1-1) + (1-1) + \ldots = -1 + 0 + 0 + 0 + \ldots = -1 .$$

There is clearly something very wrong here! The problem is that we do not have a clear definition of what a infinite sum is and what it means for this to converge.

Before we get too worried, clearly any *finite* sum is a mathematically well-defined object - we may use all the usual properties of sums on finite sequences as usual. Issues (like the above) appear only when we have infinite sums. To avoid nonsense like the previous example, we now rigorously define what a series is in terms of finite sums. In particular, a series is just the limit of a particular sequence.

This hopefully now seems not too dissimilar from what we have been looking at so far. We make the connection precise, in the following way:

Definition 5.1 An **infinite series** comprises an infinite sequence of **terms** or **summands**, a_n, say: from these terms, we may define the n**-th partial sum**,

$$s_n = \sum_{k=1}^{n} a_k.$$

Then (s_n) is an infinite sequence of real numbers and if it converges, we will say that the **series is convergent**. In this case, define the value of the infinite series or its **sum to infinity** to be):

$$\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} \big(\sum_{k=1}^{n} a_k\big) = \lim_{n\to\infty} s_n.$$

If a series is not convergent we will say that the series is divergent.

That is, **a series is just the limit of a particular sequence, the sequence of partial sums**. We say that a series is divergent or convergent if its partial sums are divergent or convergent respectively. By a very standard abuse of notation, we will often say that the series $\sum_{k=1}^{\infty} a_k$ is convergent or divergent (despite it having no real meaning if the series is divergent!). In our non example above, we were trying to find the value of a divergent series, which clearly doesn't work!

Example 5.2 (Harmonic series) Claim: the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof: We will show that the sequence of partial sums (s_k) has an unbounded, and hence divergent, subsequence. It then follows from Theorem 4.3 that (s_k) itself diverges. Consider s_{2^p}:

$$\begin{aligned}
s_{2^p} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{p-1}+1} + \cdots + \frac{1}{2^p}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\underbrace{\frac{1}{2^p} + \cdots + \frac{1}{2^p}}_{2^{p-1}\text{ times}}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots \frac{1}{2} \\
&= 1 + \frac{p}{2}.
\end{aligned}$$

This is unbounded, so (s_k) is divergent. □

Remark This is the most famous of all divergent series, called the **harmonic series**. Note that the **terms** of the series $a_n = 1/n$ converge to 0. Nonetheless, the series itself does not converge.

Example 5.3 Claim: the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1.

Proof: Note that

$$\begin{aligned}
a_n &= \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \\
\text{so} \qquad s_k &= \sum_{n=1}^{k} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
&= \sum_{n=1}^{k} \frac{1}{n} - \sum_{n=1}^{k} \frac{1}{n+1} \\
&= \sum_{n=1}^{k} \frac{1}{n} - \sum_{m=2}^{k+1} \frac{1}{m} \qquad (\text{where } m = n+1) \\
&= 1 - \frac{1}{k+1}.
\end{aligned}$$

Hence $s_k \to 1$. □

This particular example was cooked up so that an explicit expression for the partial sum s_k could be found. Another important example where we can compute the limit exactly is the **geometric series**:

Example 5.4 (Geometric series) Let $\alpha \in (-1, 1)$ and consider the series

$$\sum_{n=0}^{\infty} \alpha^n.$$

This is the **geometric series with common ratio** α. We claim that it converges to $\dfrac{1}{1-\alpha}$.
Proof:

$$\begin{aligned}
s_k &= 1 + \alpha + \alpha^2 + \cdots + \alpha^k \\
\Rightarrow \qquad \alpha s_k &= \alpha + \alpha^2 + \alpha^3 + \cdots + \alpha^{k+1} \\
\Rightarrow \qquad (1-\alpha) s_k &= 1 - \alpha^{k+1} \\
\Rightarrow \qquad s_k &= \frac{1 - \alpha^{k+1}}{1-\alpha}.
\end{aligned}$$

Now $\alpha^{k+1} \to 0$ (see Example 3.21), so $s_k \to 1/(1-\alpha)$. □

Usually it's not possible to find a simple formula for s_k, so we need to develop ways of determining whether a series converges by dealing directly with its **terms**, a_n, rather than its partial sums, s_k. In the next section we will construct several convergence tests for this purpose. It's important to realize that these tests will only tell us **whether** a series converges, not **what limit** it converges to (if it does converge). For example, we will be able to prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, even though we have absolutely no idea what it converges to.[4]

[4] Actually, it converges to $\pi^2/6$, but to prove this requires some very fancy stuff.

5.2 Convergence tests for series

Recall that a series $\sum_{n=1}^{\infty} a_n$ converges if and only if its sequence of partial sums, $s_k = \sum_{n=1}^{k} a_n$ converges. Our first convergence test gives a very simple **necessary** but **not sufficient** condition for convergence.

Theorem 5.5 (The Divergence Test) *If $\sum_{n=1}^{\infty} a_n$ converges, then (a_n) converges to 0.*

Proof: By assumption, $s_k = \sum_{n=1}^{k} a_n$ converges to some limit L. Hence, $s_{k+1} \to L$ also by Theorem 4.3 (it's a subsequence of (s_k)). Hence $a_{k+1} = s_{k+1} - s_k \to L - L = 0$ by Theorem 2.15, so the sequence $a_n \to 0$ by Proposition 4.5. □

Warning: this theorem says that **if** $\sum a_n$ converges, then $a_n \to 0$. It does **NOT** say that if $a_n \to 0$ then $\sum a_n$ converges. We have already seen a counterexample, the harmonic series: $a_n = 1/n \to 0$ but $\sum_{n=1}^{\infty} \frac{1}{n}$ does **not** converge!

Example 5.6 $\sum_{n=1}^{\infty} \frac{n^2-1}{n^2+1}$ does not converge, by the Divergence Test (since its terms a_n converge to 1, not 0).

Next we state a couple of elementary algebraic properties applying to convergent series:

Proposition 5.7 (Algebra of series) *If $\sum\limits_{n=1}^{\infty} a_n$, $\sum\limits_{n=1}^{\infty} b_n$ are convergent series, then*

(i) $\sum\limits_{n=1}^{\infty} \big(a_n + b_n\big) = \sum\limits_{n=1}^{\infty} a_n + \sum\limits_{n=1}^{\infty} b_n$;

(ii) If C is any real number, then $\sum\limits_{n=1}^{\infty} \big(Ca_n\big) = C \sum\limits_{n=1}^{\infty} a_n$.

Proof: Exercise! Hint: Write the partial sums and then apply the Algebra of Limits □

Theorem 5.8 (The Comparison Test) *Suppose we have series $\sum\limits_{n=1}^{\infty} a_n$, $\sum\limits_{n=1}^{\infty} b_n$ and a real number $C > 0$ where $0 \le a_n \le Cb_n$ for all n. Then*

- *If $\sum\limits_{n=1}^{\infty} b_n$ converges then $\sum\limits_{n=1}^{\infty} a_n$ converges.*
- *If $\sum\limits_{n=1}^{\infty} a_n$ diverges then $\sum\limits_{n=1}^{\infty} b_n$ diverges.*

Proof: As $0 \le a_n \le Cb_n$ then, if we denote (s_n) is the sequence of partial sums of the series $\sum\limits_{n=1}^{\infty} a_n$ and (t_n) is the sequence of partial sums of the series $\sum\limits_{n=1}^{\infty} b_n$, then we have $s_n \le Ct_n$ for all n.

For the first bullet point, as (t_n) converges, it is bounded above, by T, say, so that $s_n \le C \cdot T$. Then (s_n) is a monotonically increasing sequence which is bounded above, and so converges by monotone convergence theorem, Theorem 3.20.

Suppose now that $\sum_{n=1}^{\infty} a_n$ diverges. If $\sum_{n=1}^{\infty} b_n$ were to converge then this would contradict the first bullet point. Therefore we must also have $\sum_{n=1}^{\infty} b_n$ also diverges □

Note that the second bullet point came from some logical trickery rather than a real proof – this is because the second statement is logically equivalent to the first (in fact this is called the contrapositive). We will see more on this in the next section of the notes.

A commonly used alternative version of the above is the following:

Remark 5.9 (Important!) In the above test, it is often useful to replace the condition $0 \leq a_n \leq Cb_n$ with the assumption $a_n > 0$, $b_n > 0$ and $\frac{a_n}{b_n}$ is bounded.

Why is this allowed? Suppose that $a_n > 0$ and $b_n > 0$ and $\frac{a_n}{b_n}$ is bounded. This means that there is a constant $C \in \mathbb{R}$ so that $\frac{a_n}{b_n} < C$. Furthermore, as $a_n > 0$ we must have $C > 0$. Rearranging we have $0 \leq a_n \leq Cb_n$, as required in the original statement.

We use this result straightaway to prove a series converges, from results we have already established.

Example 5.10 The series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Proof: We compare the terms of $\sum\limits_{n=1}^{\infty} \frac{1}{n^2}$ with those of $\sum\limits_{n=1}^{\infty} \frac{1}{n(n+1)}$. Since we are trying to show **convergence**, we set the series we *know* converges as the denominator; that is, take $a_n = \frac{1}{n^2}$ and $b_n = \frac{1}{n(n+1)}$; then

$$\frac{a_n}{b_n} = \frac{n(n+1)}{n^2} = \frac{n^2+n}{n^2} = 1 + \frac{1}{n} \leq 2,$$

so the series $\sum\limits_{n=1}^{\infty} a_n = \sum\limits_{n=1}^{\infty} \frac{1}{n^2}$ converges by the comparison test (and Remark 5.9). □

So, we have proved definitively that the sum $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36} + \dots$ converges to a well-defined real value. What is it? It is not at all easy to find, and is perhaps quite surprisingly equal to $\frac{\pi^2}{6}$. Finding the value of this series was known as the Basel problem, and was solved by Euler in 1734 – if you are curious the Wikipedia page provides details of Euler's method of solution.

We can also straightaway establish covergence for a whole class of infinite series.

Proposition 5.11 *For all $k \in \mathbb{Z}^+$ with $k \geq 2$, the series* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^k}$ *converges.*

Proof: We compare $\sum\limits_{n=1}^{\infty} \frac{1}{n^k}$ with $\sum\limits_{n=1}^{\infty} \frac{1}{n^2}$ where again the *known* convergent series goes as the denominator: let $a_n = \frac{1}{n^k}$ and $b_n = \frac{1}{n^2}$; then

$$a_n = \frac{1}{n^{k-2}} b_n \leq b_n,$$

so $\sum\limits_{n=1}^{\infty} a_n = \sum\limits_{n=1}^{\infty} \frac{1}{n^k}$ converges by the comparison test (with $C = 1$). □

Example 5.12 $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2n^2 + \sin n}$ diverges.

Proof: Let $b_n = n/(2n^2 + \sin n)$ and $a_n = 1/n$. Then $\sum_{n=1}^{\infty} b_n$ diverges, by Example 5.2. Now

$$b_n = \frac{n}{2n^2 + \sin n} \geq \frac{n}{2n^2 + 1} \geq \frac{n}{2n^2 + n^2} = \frac{1}{3} a_n.$$

where we used that $1 \leq n$ for the second inequality. Therefore, by part (ii) of the Comparison Test (with $C = 3$), $\sum_{n=1}^{\infty} a_n$ also diverges as $a_n \leq 3b_n$.

□

Of course applying the comparison test requires both a good idea of whether the series in question actually does converge or not, and an appropriate series whose convergence behaviour is known, to compare it to. We will now work towards a test for convergence referring only to the terms of the series itself. It has, however, the disadvantage that it is not conclusive for *all* series.

5.3 Absolute convergence

Note that the Comparison Test could only be applied to series all of whose terms are non-negative. This may seem like, and indeed is, quite a strong restriction.

Given a series with a mix of positive and negative terms, we can simplify our analysis of this series by considering an accompanying series where we insist all terms are non-negative – that is, we can consider the series consisting of terms $|a_n|$ for each a_n in the original series.

Proposition 5.13 (Absolute convergence) *Given any series $\sum\limits_{n=1}^{\infty} a_n$, if the series $\sum\limits_{n=1}^{\infty} |a_n|$ converges then the series $\sum\limits_{n=1}^{\infty} a_n$ converges. In such a case we say that $\sum\limits_{n=1}^{\infty} a_n$ is* ***absolutely convergent*** *(and that it* ***converges absolutely****).*

Proof: We are given that the series $\sum_{n=1}^{\infty} |a_n|$ converges, to L say. Let

$$b_n = \begin{cases} a_n & \text{if } a_n \geq 0 \\ 0 & \text{if } a_n < 0 \end{cases} \qquad c_n = \begin{cases} 0 & \text{if } a_n \geq 0 \\ a_n & \text{if } a_n < 0 \end{cases}$$

and note that $a_n = b_n + c_n$ for all n. Let s_k, t_k, u_k denote the kth partial sums of the series with terms a_n, b_n and c_n, respectively. Then $s_k = t_k + u_k$, and we seek to prove that the sequence (s_k) converges. Now, (t_k) is increasing, since $b_n \geq 0$ for all n. Furthermore,

$$t_k = \sum_{n=1}^{k} b_n = \sum_{n=1}^{k} |b_n| \leq \sum_{n=1}^{k} |a_n| \leq L,$$

so, by the Monotone Convergence Theorem, (t_k) converges, to L_+ say. Similarly, (u_k) is decreasing, since $c_n \leq 0$ for all n, and

$$u_k = \sum_{n=1}^{k} c_n = -\sum_{n=1}^{k} |c_n| \geq -\sum_{n=1}^{k} |a_n| \geq -L$$

so, by the Monotone Convergence Theorem, (u_k) converges, to L_- say. Hence, by Theorem 2.15, $s_k \to L_+ + L_-$. □

Our next convergence test (everyone's favourite) involves only the terms of the series under consideration, and is very easy to apply. Its only disadvantage is that it very often gives an inconclusive answer.

Theorem 5.14 (The Ratio test) *Let $\sum\limits_{n=1}^{\infty} a_n$ be a series with $a_n \neq 0$ for all n such that the sequence $\left|\dfrac{a_{n+1}}{a_n}\right|$ converges. Then*

- *If $\lim\limits_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right| < 1$, the series $\sum\limits_{n=1}^{\infty} a_n$ converges absolutely; and*
- *If $\lim\limits_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right| > 1$, the series $\sum\limits_{n=1}^{\infty} a_n$ diverges.*

Proof: Suppose $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$. Set $\epsilon = \frac{1-L}{2} > 0$, then by the convergnence of $\left|\frac{a_{n+1}}{a_n}\right|$, there is $N \in \mathbb{Z}^+$ such that for all $n \geq N$,

$$\left|\frac{a_{n+1}}{a_n}\right| < r = L + \epsilon = L + \frac{1-L}{2} = \frac{1+L}{2} < 1 \ .$$

Set $r = \frac{1+L}{2} < 1$. So, for $n \geq N$, $|a_{n+1}| < r|a_n|$. Then it can be seen by induction that $|a_{N+j}| < r^j|a_N|$ for all $j \in \mathbb{Z}^+$. Set

$$b_n = \begin{cases} r^n|a_N| & \text{if } n \geq N \\ |a_n| & \text{otherwise.} \end{cases}$$

Note that as $\sum_{n=N}^{\infty} b_n$ is a geometric sequence with $r < 1$, this converges. Furthermore, as $\sum_{n=1}^{N} b_n$ is some finite number, we see that $\sum_{n=1}^{\infty} b_n$ converges due to the convergence of $\sum_{n=N}^{\infty} b_n$ (exercise: prove this in detail!). Hence as $0 \leq |a_n| \leq b_n$ and $\sum_{n=1}^{\infty} b_n$ converges, by the comparison test $\sum\limits_{n=1}^{\infty} a_n$ converges absolutely.

Now suppose that $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = M > 1$ and take $\epsilon = \frac{M-1}{2} > 0$. Then there is $N \in \mathbb{Z}^+$ so that for all $n \geq N$, $\left|\frac{a_{n+1}}{a_n}\right| > q = M - \epsilon = \frac{M+1}{2} > 1$, so that $|a_{N+j}| > q^j|a_N|$ and in particular $|a_{N+j}| > |a_N| > 0$ for all $j \in \mathbb{Z}^+$, so that (a_n) does not converge to zero and so $\sum\limits_{n=1}^{\infty} a_n$ does not converge. □

The obvious/striking drawback of this convergence test is that it cannot tell us anything about series where the ratio of successive terms converges to 1 – for example, this is that case for any series whose terms are rational functions in n (that is, a fraction whose numerator and denominator are each polynomials in n). For such series the Comparison Test – with an appropriately chosen comparison series – is often the best choice.

Example 5.15 Claim: the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

Proof: Let $a_n = 1/n!$. Then $a_{n+1}/a_n = n!/(n+1)! = 1/(n+1) \to 0 < 1$, so the series converges by the Ratio Test. □

The limit of this series is a very famous number, called **Euler's number** e. Note that the series starts with $n = 0$, and recall that $0! = 1$. Another way to define e is as the limit of the **sequence** $(1 + 1/n)^n$. However, proving that this sequence converges is much trickier than proving that the series $\sum(1/n!)$ converges.

In the next example we give a very sneaky use of the ratio test: we use it to prove that a certain **sequence** converges to 0 by showing that the series whose **terms** are the sequence converges! Recall that in Example 4.6 we used the Montone Convergence Theorem to show that

$$a_n = \frac{n^3}{2^n}$$

converged. This is not entirely obvious because both the numerator and the denominator increase without bound as n gets large, so it's not clear whether the polynomial in the numerator wins (so a_n diverges), the exponential factor in the denominator wins (so $a_n \to 0$) or the two reach a draw (so $a_n \to L > 0$) . While our previous argument showed the sequence converges, so the polynomial doesn't win, it didn't tell us what the limit is. In fact, the limit is 0. This is an example of a more general fact: exponential growth always beats polynomial growth. To deal with Example 4.6 just choose $p = 3$ and $\alpha = 1/2$ in the following example:

Example 5.16 (Exponentials beat polynomials) Let $p \in \mathbb{Z}^+$ and $\alpha \in (0,1)$ be constant. Then $a_n = n^p \alpha^n \to 0$.
Proof: Consider the series $\sum_{n=1}^{\infty} a_n$. This has

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^p \alpha^{n+1}}{n^p \alpha^n} = \left(1 + \frac{1}{n}\right)^p \alpha \to \alpha < 1.$$

Hence $\sum_{n=1}^{\infty} a_n$ converges, by the Ratio Test, so $a_n \to 0$ by the Divergence Test. □

5.4 Alternating series

Whilst we have said, and we will see, changes of sign make the convergence of series tricky to deal with, there is one particularly straightforward case where having both positive and negative terms in a series makes the convergence easy to determine: this is for certain cases of *alternating series*.

Definition 5.17 If $\sum\limits_{n=1}^{\infty} a_n$ is such that $a_n \neq 0$ and $\frac{a_{n+1}}{a_n} < 0$ for all n (so that consecutive terms have opposite signs), then $\sum\limits_{n=1}^{\infty} a_n$ is an **alternating series**.

Lemma 5.18 *A series $\sum\limits_{n=1}^{\infty} a_n$ is an alternating series if and only if either*

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^{n+1} |a_n|, \text{ or}$$
$$\sum_{n=1}^{\infty} a_n = -\sum_{n=1}^{\infty} (-1)^{n+1} |a_n|.$$

Proof: (Exercise) The idea: You would use induction, starting by distinguishing the cases that either the first term is positive, or the first term is negative. □

Theorem 5.19 (Alternating Series Test) *If (a_n) is a monotonically decreasing sequence of positive numbers which converges to zero, then the series*

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges. (Note that the series $-\sum\limits_{n=1}^{\infty} (-1)^{n+1} a_n$ then also converges by Algebra of series)

Proof: We consider two subsequences of the sequence of partial sums: the subsequence $(u_k) = (s_{2k})$ of all even-numbered partial sums, and the subsequence $(v_k) = (s_{2k-1})$ of all odd-numbered partial sums. The sequence (u_k) is monotonically increasing: considering u_k, u_{k+1}, we have

$$u_{k+1} - u_k = s_{2k+2} - s_{2k} = a_{2k+1} - a_{2k+2} \geq 0 \text{ as } a_{2k+1} \geq a_{2k+2},$$

so $u_k \leq u_{k+1}$ as required. Furthermore, for any k,

$$\begin{aligned} u_k = s_{2k} = \sum_{n=1}^{2k} (-1)^{n+1} a_n &= a_1 - a_2 + a_3 - a_4 + \ldots + a_{2k-1} - a_{2k} \\ &= a_1 - (a_2 - a_3) - \ldots - (a_{2k-2} - a_{2k-1}) - a_{2k} \\ &< a_1, \end{aligned}$$

so the sequence (u_k) is bounded above, and so converges, to L, say.

Now

$$v_k = s_{2k-1} = s_{2k} - a_{2k} = u_k - a_{2k},$$

where we have just shown the sequence (u_k) converges to L, and where a_{2k} is a subsequence of the sequence (a_n) which converges to zero by assumption; so by algebra of limits,

$$\lim_{n\to\infty} v_k = \lim_{n\to\infty} u_k - \lim_{n\to\infty} a_{2k} = L - 0 = L.$$

Then for any $\epsilon > 0$, we have N_u and N_v where all even partial sums u_k with $k \geq N_u$ are within ϵ of L, and all odd partial sums v_k with $k \geq N_v$ are within ϵ of L; so for $n \geq \max(2N_u, 2N_v - 1)$ we have $|s_n - L| < \epsilon$ and so (s_n) (and hence $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$) converges. □

Example 5.20 (The alternating harmonic series) The series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

converges.

Proof: Taking $a_n = \frac{1}{n}$ then the Alternating Sign Test applies, as the sequence (a_n) is positive, monotonically decreasing and converges to zero. □

Note once again we are told almost nothing about what this series converges to (we can observe that it follows from facts established in the proof of the Alternating Sign Test that the sum to infinity lies between $s_1 = 1$ and $s_2 = \frac{1}{2}$). Any guesses what it might be (*the answer may surprise you*)?

It is very important to note at this stage

Example 5.21 The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

is not absolutely convergent.

Proof: We have

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

the harmonic series, which we showed diverges in 5.2. □

Definition 5.22 A convergent series which does not converge absolutely is called **conditionally convergent**.

We should be very careful when applying the Alternating series test that the series in question does satisfy the assumptions of the test. Here is an example.

Example 5.23 Consider the sequence

$$a_n = \begin{cases} \frac{1}{n^3} & n \text{ odd} \\ -\frac{1}{n^2} & n \text{ even} \end{cases}$$

I claim that the series $\sum_{n=1}^{\infty} a_n$ converges.

Proof: Let $b_n = 1/n^2$. Then

$$\begin{aligned} \frac{|a_n|}{b_n} &= \begin{cases} \frac{1}{n} & n \text{ odd} \\ 1 & n \text{ even} \end{cases} \\ &\leq 1. \end{aligned}$$

Since $|a_n|/b_n$ is bounded above, and $\sum b_n$ is known to converge (Example 5.10, we conclude that $\sum |a_n|$ converges also, by part (i) of the Comparison Test, that is, $\sum a_n$ converges **absolutely**. Hence $\sum a_n$ converges by Theorem 5.13. □

Remark 5.24 This is an alternating series, but we can't use the Alternating Series Test, because $|a_n|$ is not a decreasing sequence.

Recall that we have defined Euler's constant by the series

$$e := \sum_{k=0}^{\infty} \frac{1}{k!} = \lim_{n\to\infty} s_n, \qquad \text{where} \qquad s_n = \sum_{n=0}^{n} \frac{1}{k!}$$

We now prove an alternative useful characterisation of e, this time as an alternative sequence.

Lemma 5.25

$$x_n := \left(1 + \frac{1}{n}\right)^n \to e$$

as $n \to \infty$.

Proof:
Step 1: We prove that for any $m < n$,

$$\left(1 - \frac{m-1}{n}\right)^m \sum_{k=0}^{m} \frac{1}{k!} \leq x_n \leq \sum_{k=0}^{n} \frac{1}{k!} \ .$$

Proof of Step 1: By the binomial expansion we have

$$x_n := \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} c_{n,k}$$

where

$$\begin{aligned} c_{n,k} &:= \binom{n}{k} \frac{1}{n^k} \\ &= \frac{n!}{k!(n-k)!n^k} \\ &= \frac{n(n-1)(n-2)\dots(n-(k-1))}{k!n^k} \\ &= \frac{1}{k!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\dots\left(1 - \frac{k-1}{n}\right) \ . \end{aligned}$$

So for any $k \leq m < n$, we have that

$$\frac{1}{k!}\left(1 - \frac{m-1}{n}\right)^m \leq c_{n,k} \leq \frac{1}{k!}$$

So,

$$\left(1 - \frac{m-1}{n}\right)^m \sum_{k=0}^{m} \frac{1}{k!} \leq x_n \leq \sum_{k=0}^{n} \frac{1}{k!} = s_n \ .$$

Step 2: We have that $x_n \leq e$.

Proof of Step 2: As $s_n = \sum_{k=0}^{n} \frac{1}{k!}$ is monotone increasing and converges to e, then by Theorem 3.18 $s_n \leq \lim_{n\to\infty} s_n = e$, so the right hand inequality from Step 1 implies that $x_n \leq e$.

Step 3: x_n is monotone and bounded, and so $x_n \to L$ for some L.

Proof of Step 3: Note that $c_{n+1,k} > c_{n,k}$, as $1 - \frac{l}{n} < 1 - \frac{l}{n+1}$ for any positive l. Hence $x_{n+1} - x_n = c_{n+1,n+1} + \sum_{k=0}^{n}(c_{n+1,k} - c_{n,k}) \geq 0$ and so x_n is monotone. Step 2 implies that x_n is bounded. Hence we may apply the Monotone Convergence Theorem, Theorem 3.20, to see that $x_n \to L$.

Step 4: We show that $L = e$.

Proof of Step 4: From Step 2 and Proposition 2.14, we see that $L \leq e$. Furthermore, by Step 1, for any $m < n$

$$\left(1 - \frac{m-1}{n}\right)^m \sum_{k=0}^{m} \frac{1}{k!} - x_n \leq 0$$

and so taking a limit by Proposition 2.14, the Algebra of Limits, Theorem 2.15, and the Tail Lemma, Lemma 4.5, (as we need $m < n$), for any $m \in \mathbb{Z}^+$,

$$\sum_{k=0}^{m} \frac{1}{k!} - L \leq 0 \ , \qquad \text{or equivalently,} \qquad s_m = \sum_{k=0}^{m} \frac{1}{k!} \leq L \ .$$

Sending m to infinity in the sequence s_m and applying Proposition 2.14 again gives us $e \leq L$. So overall we have $e \leq L \leq e$ so $L = e$.

□

5.5 Power series and radius of convergence

A large number of important functions (e.g. sin, cos, exp, ...) are in fact formally defined by their power series. In this section, we study exactly what this means and get to grips with some of the theory of such power series.

Informally, a power series is an expression of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

where $(a_n)_{n\in\mathbb{Z}^+}$ is a sequence of terms and x is a real variable. More accurately:

Definition 5.26 A power series (with variable $x \in \mathbb{R}$), is defined for a given $x \in \mathbb{R}$ as the series $\sum_{n=0}^{\infty} b_n$, where $b_0 = a_0$ and $b_n = a_n x^n$ for $n \in \mathbb{Z}^+$. We write this as $\sum_{n=0}^{\infty} a_n x^n$.

Note that the first term is always just a_0 (there is no x on this term). At each fixed $x \in \mathbb{R}$, the power series defines the infinite series $\displaystyle\sum_{n=0}^{\infty} b_n$ where $b_n = a_n x^n$ for each $n \in \mathbb{Z}^+$. These series may or may not converge, and this will depend on the a_n and the value of x; likewise, where the series does converge, the value it converges to will depend on x (in combination with the a_n).

Examples 5.27 (i) If $a_n = 1$ for all n the power series $\displaystyle\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} x^n$ gives the geometric series with repeated ratio x at each $x \in \mathbb{R}$. Then we know that this power series converges only for those x for which $|x| < 1$, where we will have

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

(ii) If $a_n = (-1)^n$ then the power series $\sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} (-x)^n$ is again a geometric series for each x, only now with repeated ratio $-x$. This converges for $|-x| < 1$ and we will have

$$\sum_{n=0}^{\infty} (-x)^n = \frac{1}{1+x}.$$

(iii) Note that given any power series $\sum_{n=0}^{\infty} a_n x^n$, at $x = 0$ this defines the series $\sum_{n=0}^{\infty} b_n$ where $b_n = a_n x^n = a_n 0^n = 0$, so that $b_0 = a_0$ and $b_n = a_n \cdot 0 = 0$ for all $n \geq 1$, so that $\sum_{n=0}^{\infty} b_n = a_0 + 0 + 0 + 0 + \ldots$: that is, **every** power series converges at $x = 0$ to the value a_0.

(iv) The power series $\sum_{n=0}^{\infty} n^n x^n$ converges if and only if $x = 0$.

Proof: The 'if' claim is established by the previous example; we need to show that if $x \neq 0$ then the series $\sum_{n=0}^{\infty} n^n x^n$ does not converge. At some fixed $x \neq 0$ the power series gives the infinite series $\sum_{n=0}^{\infty} b_n$ where $b_n = n^n x^n = (nx)^n$. Then for $N > \left|\frac{1}{x}\right|$ we have that $|nx| > 1$ for all $n \geq N$, and so furthermore $|b_n| = |nx|^n > 1$ and so the series does not converge at any $x \neq 0$ by the divergence test. □

(v) If there is some $d \in \mathbb{Z}^+$ such that $a_n = 0$ for all $n > d$ then the power series $\sum_{n=0}^{\infty} a_n x^n$ gives the finite sum $a_0 + a_1 x + a_2 x^2 + \ldots + a_d x^d$ at each x and so converges for all $x \in \mathbb{R}$ to the value $a_0 + a_1 x + a_2 x^2 + \ldots + a_d x^d$.

The final example above, as well as examples (i) and (ii), should suggest the idea that power series can be used to define functions of a real variable x, on those regions of the real line where the power series converges. This will actually be an exceptionally useful technique to define those functions mentioned in the previous chapter that we have not been able to provide an 'analytic' justification of their existence or properties. First, however, we want to develop some solid results concerning where in the real line (that is, for which values of $x \in \mathbb{R}$) we can say that a power series converges.

Definition 5.28 We define the **radius of convergence** of a power series $\sum_{n=0}^{\infty} a_n x^n$ to be

$$R := \sup\{|x| : \sum_{n=1}^{\infty} |a_n x^n| \text{ converges } \} .$$

If no supremum exists (the set is unbounded above), then we will say that the series has an infinite radius of convergence.

Lemma 5.29 *Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges at $x = x_1$. Then it converges **absolutely** for all x such that $|x| < |x_1|$.*

Proof: Since $\sum_{n=1}^{\infty} a_n x_1^n$ converges, we have that $a_n x_1^n \to 0$ (Proposition 5.5), and so is bounded: there exists $K > 0$ such that for all n, $|a_n x_1^n| < K$. But then for all n,

$$|a_n x^n| = |a_n x_1^n| \frac{|x|^n}{|x_1|^n} < K \left(\frac{|x|}{|x_1|} \right)^n$$

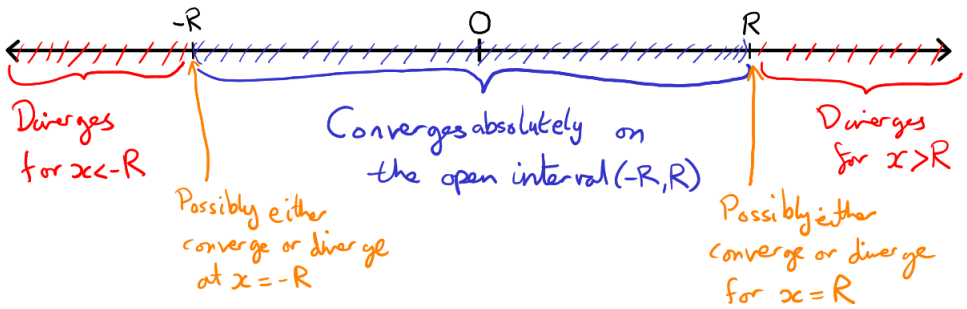


Figure 3: Given a power series with a finite radius of convergence R, the series converges absolutely for $x \in (-R, R)$, diverges for $x \in \mathbb{R} \setminus [-R, R]$ and could either converge or diverge at $x = R$ and $x = -R$.

so if $|x| < |x_1|$, $\sum_{n=0}^{\infty} |a_n x^n|$ converges by comparison with the convergent (geometric) series $\sum_{n=0}^{\infty} (|x|/|x_1|)^n$. $\square$

Theorem 5.30 *Let $\sum_{n=0}^{\infty} a_n x^n$ have radius of convergence R. Then*

$$\sum_{n=0}^{\infty} a_n x^n$$

converges **absolutely** *for $|x| < R$ and* **diverges** *for $|x| > R$.*

Proof: Let $A := \{|x| : \sum_{n=0}^{\infty} |a_n x^n| \text{ converges}\} \subset \mathbb{R}$, so that $R = \sup A$.

If $|x| < R$ then there exists x_1 with $|x| < |x_1| < R$ such that $\sum_{n=0}^{\infty} a_n x_1^n$ is convergent (else R isn't the **least** upper bound on A, $|x|$ being smaller), so $\sum_{n=0}^{\infty} a_n x^n$ is absolutely convergent by Lemma 5.29.

If $|x| > R$ and $\sum_{n=0}^{\infty} a_n x^n$ **converges** then for $x_2 = (|x|+R)/2$, $\sum_{n=0}^{\infty} a_n x_2^n$ converges absolutely by Lemma 5.29, since $|x_2| < |x|$. But $|x_2| > R$, a contradiction (R isn't an upper bound on A at all!), so we conclude that $\sum_{n=0}^{\infty} a_n x^n$ **diverges**. $\square$

So to reiterate, given a finite radius of convergence R, a power series converges absolutely if $x \in (-R, R)$ and diverges for $x \in \mathbb{R} \setminus [R, R]$. If the radius of convergence is infinite, then the power series converges for all $x \in \mathbb{R}$.

Examples 5.31

(i) We saw that the first two examples in Example 5.27 converged for $|x| < 1$; that is, they have radius of convergence equal to 1.

(ii) Define the sequence of terms (a_n) by $a_n = \begin{cases} 0 & n \text{ odd}, \\ \dfrac{k^2}{k+1} & n = 2k \text{ for } k \in \mathbb{Z}^+, \end{cases}$ so that at each $x \in \mathbb{R}$ the power series $\displaystyle\sum_{n=0}^{\infty} a_n x^n$ defines a series $\displaystyle\sum_{k=0}^{\infty} b_k = \sum_{k=0}^{\infty} \frac{k^2}{k+1} x^{2k}$. Applying the ratio test, we find

$$\left|\frac{b_{k+1}}{b_k}\right| = \left|\frac{(k+1)^2(k+1)x^{2(k+1)}}{(k+2)k^2x^{2k}}\right| = \left|\frac{(k+1)^3x^{2k+2}}{(k+2)k^2x^{2k}}\right| = \frac{(k+1)^3}{(k+2)k^2}|x^2|$$

so that

$$\lim_{k\to\infty}\left|\frac{b_{k+1}}{b_k}\right| = \lim_{k\to\infty} \frac{(k+1)^3}{(k+2)k^2}|x^2| = |x^2|,$$

and so $|x| < 1 \implies \lim_{k\to\infty} \left|\frac{b_{k+1}}{b_k}\right| < 1$, so the series converges while for $|x| > 1 \implies \lim_{k\to\infty} \left|\frac{b_{k+1}}{b_k}\right| > 1$ the series diverges. Hence, the radius of convergence of $\sum_{n=0}^{\infty} a_n x^n$ is 1.

This example demonstrates a useful heuristic for finding the radius of convergence of a power series: apply the ratio test and solve for the values of $|x|$ for which the ratio is < 1. Note however that in this example we did not find the radius of convergence by applying the ratio test to the a_n appearing in the definition of the power series but to the (non-zero) terms appearing in a reindexed description of the series. So it would **not** be possible, for example, to *define* the radius of convergence as $R = \frac{1}{C}$ where $C = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$. Unfortunately, quite a few textbooks get this wrong! If you find such a textbook, treat it with mistrust. In the case above this would lead to an undefined radius of convergence.

Example 5.32 The power series $\sum_{n=0}^{\infty} \frac{1}{n!} x^n$ converges for all $x \in \mathbb{R}$.

Proof: Fix any $x \in \mathbb{R}$; then

$$\left|\frac{x^{n+1} \cdot n!}{(n+1)! x^n}\right| = \frac{1}{n+1}|x|,$$

so that

$$\lim_{n\to\infty} \left|\frac{x^{n+1} \cdot n!}{(n+1)! x^n}\right| = 0$$

for all $x \in \mathbb{R}$ and so, by the Ratio Test, the series converges. □

Example 5.33 The power series $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$ and $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$ converge for all $x \in \mathbb{R}$.

Proof: Exercise! □

We may now define some of the most important functions in mathematics, the exponential and trigonometric functions, by their power series. We define $\exp : \mathbb{R} \to \mathbb{R}$, $\sin : \mathbb{R} \to \mathbb{R}$ and $\cos : \mathbb{R} \to \mathbb{R}$ by

$$\exp(x) = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \qquad \sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}, \qquad \cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} .$$

These are of course exactly the functions which with the familiar properties that you have seen in school. However, at this point, we have only proven that these functions exist. More work will be required to show that they satisfy the usual properties.

Example 5.34 What is the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^3}{8^n} x^{3n-1}$?

Solution: Let $b_n = \frac{n^3}{8^n} x^{3n-1}$. Then, for all $x \neq 0$, $|b_n| > 0$, so we can (try to) use the Ratio Test to see whether $\sum |b_n|$ converges. Now

$$\begin{aligned} \frac{|b_{n+1}|}{|b_n|} &= \frac{(n+1)^3}{n^3} \frac{8^n}{8^{n+1}} |x|^3 \\ &= \left(1 + \frac{1}{n}\right)^3 \frac{1}{8} |x|^3 \\ &\to \frac{|x|^3}{8} \end{aligned}$$

So by the Ratio Test, $\sum b_n$ converges absolutely if $|x| < 2$, but not if $|x| > 2$. By the definition of radius of convergence, we deduce that $R = 2$.

Summary

- Given a series $\sum_{n=1}^{\infty} a_n$ we call a_n the nth **term** of the series, and

$$s_k = \sum_{n=1}^{k} a_n$$

  the kth **partial sum** of the series.

- A series **converges** to L if and only if its sequence of partial sums (s_k) converges to L, in the usual sense: for each $\varepsilon > 0$ there exists $K \in \mathbb{Z}^+$ such that for all $k \geq K$, $|s_k - L| < \varepsilon$. The series **diverges** if (s_k) diverges (that is, does not converge).

- The **harmonic** series $\sum_{n=1}^{\infty} 1/n$ **diverges**.

- For all $p \geq 2$, the series $\sum_{n=1}^{\infty} 1/n^p$ **converges**.

- The **geometric series** $\sum_{n=0}^{\infty} \alpha^n$ **converges** to $1/(1-\alpha)$ for all $\alpha \in (-1, 1)$. It diverges if $|\alpha| \geq 1$.

- The **Divergence Test**: if $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \to 0$.

- The **Comparison Test**: if $a_n, b_n > 0$ and

  (i) a_n/b_n is bounded and $\sum b_n$ converges, then $\sum a_n$ converges.

  (ii) b_n/a_n is bounded and $\sum b_n$ diverges, then $\sum a_n$ diverges.

- A series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges.

- Absolute convergence implies convergence. The converse is false.

- The **Ratio Test**: if $a_n > 0$ and $a_{n+1}/a_n \to L$ then

  (i) If $L < 1$, $\sum a_n$ converges.

  (ii) If $L > 1$, $\sum a_n$ diverges.

- The **Alternating Series Test:** if $a_n > 0$, $a_n \to 0$ and (a_n) is decreasing, then $\sum (-1)^{n+1} a_n$ converges.

- A **power series** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n \ ,$$

  where $x \in \mathbb{R}$. Its **radius of convergence** is

$$R = \sup\{|x| : \sum_{n=0}^{\infty} a_n x^n \text{ converges absolutely}\} \ .$$

  If the supremum doesn't exist, we say that the series has an infinite radius of convergence.

- If a power series has radius of convergence R then for all $x < R$, the power series converges absolutely, and for all $|x| > R$, it diverges.

6 Some symbolic logic

6.1 Statements and their symbolic manipulation

In this module we have dealt with many mathematical statements, some of which were quite logically subtle. For example, consider the statement

> "The real sequence (a_n) converges to L."

We learned that this statement really means

> "For each positive real number ε there exists a positive integer N such that, for all $n \geq N$, $|a_n - L| < \varepsilon$."

What does the statement

> "The real sequence (a_n) converges"

mean? It means that (a_n) converges to L for some real number L. That is,

> "There exists a real number L such that, for each positive number ε, there exists a positive integer N such that, for all $n \geq N$, $|a_n - L| < \varepsilon$."

Recall that a sequence **diverges** if it doesn't converge. So what does the statement

> "The real sequence (a_n) diverges"

really mean? We will see that it means precisely

> "For each real number L there exists a positive real number ε such that for all positive integers N there exists $n \geq N$ such that $|a_n - L| \geq \varepsilon$."

This is, to put it mildly, **not** obvious! In this section we will introduce some basic ideas from symbolic logic which will allow us to manipulate and analyze complicated mathematical statements like "(a_n) converges," and obtain from them related statements like "(a_n) diverges."

The key idea is to represent mathematical statements by symbols which can be combined and algebraically manipulated. For our purposes, a **statement** is any declaration which is unambiguously either true or false. For example, the declarations

P:	$(-2)^2 = 4$
Q:	Leeds is the third largest city in the UK by population
R:	$-2 > 2$
S:	Every even integer greater than 2 can be expressed as the sum of two primes

are all statements: P and Q are true, R is false, and no-one knows (at the time of writing) whether S is true or false, but it is clearly one or the other[5]. On the other hand, declarations like

P':	Tough on crime, tough on the causes of crime!
Q':	Lucy in the sky with diamonds

are not statements because neither can be meaningfully said to be true, or false.

Given one or more statements $P, Q, \ldots$, we can produce new statements from them by three basic constructions:

- **Negation**: not P, denoted $\neg P$, is the statement which is true if and only if P is false. For the examples above, we have

[5] This is the infamous Goldbach Conjecture.

$\neg P$:	$(-2)^2 \neq 4$	False
$\neg Q$:	Leeds is not the third largest city in the UK by population	False
$\neg R$	$-2 \leq 2$	True

- **Conjunction**: P and Q, denoted $P \wedge Q$, is the statement which is true if both P and Q are true, and false otherwise. So, for the examples above, $P \wedge Q$ is true, but $P \wedge R$ is false (because R is false).
- **Disjunction**: P or Q, denoted $P \vee Q$, is the statement which is true if P is true, or Q is true, or both, but is false if both P and Q are false. So, for the examples above, $P \vee Q$ is true, as is $P \vee R$ (because P is true). However, $(\neg P) \vee R$ is false (because both $\neg P$ and R are false).

You should bear in mind that in mathematics we use the word "or" in its non-exclusive sense. That is "P or Q" means "P or Q (or both)," which is different from its use in everyday English. For example, a child who is told

"Finish your broccoli or you can't have any ice cream"

could reasonably expect to get some ice cream if he/she/they finishes his/her/their broccoli. If the nagging adult is a mathematician, the child would be wise to seek clarification.

It is useful to summarize these definitions using **truth tables**. These represent the truth/falsity of a constructed statement in terms of the truth/falsity of its constituent pieces. We represent "false" by the value 0 and "true" by the value 1:

P	$\neg P$
0	1
1	0

P	Q	$P \wedge Q$
0	0	0
0	1	0
1	0	0
1	1	1

P	Q	$P \vee Q$
0	0	0
0	1	1
1	0	1
1	1	1

Two statements P, Q are **logically equivalent** if one is true if and only if the other is true. We represent this symbolically by $P \iff Q$. For example

P	$\neg P$	$\neg(\neg P)$
0	1	0
1	0	1

so for any statement,

$$\neg(\neg P) \iff P \tag{6.3}$$

which is sometimes called the Principle of Double Negation. We can use truth tables to verify many other logical equivalences. For example:

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

from which we deduce that for all statements P, Q,

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q). \tag{6.4}$$

As a consequence of (6.3) and (6.4),

$$\begin{aligned} \neg(P \vee Q) &\iff \neg[(\neg\neg P) \vee (\neg\neg Q)] \qquad \text{by (6.3)} \\ &\iff \neg[\neg((\neg P) \wedge (\neg Q))] \qquad \text{by (6.4)} \\ &\iff (\neg P) \wedge (\neg Q) \qquad \text{by (6.3)} \end{aligned}$$

that is, for all statements P, Q,

$$\neg(P \vee Q) \iff (\neg P) \wedge (\neg Q). \tag{6.5}$$

Exercise 6.1 Verify this explicitly using a truth table.

The rules (6.4) and (6.5) are known as **de Morgan's laws**.

Example 6.2 Find the negation of the statement $P \vee (Q \wedge R)$.
Solution:

$$\begin{aligned} \neg[P \vee (Q \wedge R)] \quad &\Longleftrightarrow \quad (\neg P) \wedge \neg(Q \wedge R) \qquad \text{by (6.5)} \\ &\Longleftrightarrow \quad (\neg P) \wedge [(\neg Q) \vee (\neg R)] \qquad \text{by (6.4)} \end{aligned}$$

6.2 Implications

It is convenient to define another connective between two statements P, Q in addition to conjunction $(P \wedge Q)$ and disjunction $(P \vee Q)$, namely **implication**, denoted $P \Rightarrow Q$, pronounced "if P then Q", or "P implies Q". (Actually, the standard logic notation is $P \to Q$, and $P \Rightarrow Q$ is reserved for something subtly different. However, we are already making heavy use of the symbol $\to$ to denote convergence of sequences, so we will stick to $\Rightarrow$ for implication.) The truth table for this statement is

P	Q	$P \Rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

As an example, consider the statement

"If everyone in this class passes the final exam, I will eat my hat."

This is clearly false if everyone passes the exam, and I refuse to eat my hat. On the other hand, it's clearly true if everyone passes and I **do** eat my hat. What if not everyone passes? Then the question of whether I eat my hat becomes moot: the statement is true if I don't, but it's also true if I do (I never said I'd eat my hat **only** if everyone passes - maybe I just like eating hats!). So $P \Rightarrow Q$ is false only when P is true and Q is false. In every other case, it is true.

Unlike $P \wedge Q$ and $P \vee Q$, the order of the statements P, Q in $P \Rightarrow Q$ matters, as can be seen immediately from its truth table. For example, the statement

"If you are vegetarian then you don't eat pork"

is true, whereas

"If you don't eat pork then you are vegetarian"

is false (except in Spain). Given an implication $P \Rightarrow Q$, the implication $Q \Rightarrow P$ is called its **converse**. As we have just seen, an implication and its converse are **not** logically equivalent. We have seen many examples of this fact in this module. For example

"If a real sequence converges then it is bounded"

is true (see Proposition 2.13), but its converse

"If a real sequence is bounded then it converges"

is false (e.g. $a_n = (-1)^n$). Similarly

"If a real sequence is bounded then it has a convergent subsequence"

is true (this is the Bolzano-Weierstrass Theorem), but its converse

"If a real sequence has a convergent subsequence then it is bounded"

is false (e.g. $a_n = (1 + (-1)^n)n$). Another important, and often misused, example is

"If $\sum_{n=1}^{\infty} a_n$ converges then $a_n \to 0$"

which is true (this is the Divergence Test), but its converse

"If $a_n \to 0$ then $\sum_{n=0}^{\infty} a_n$ converges"

is false (e.g. $\sum_{n=1}^{\infty} 1/n$).

It is possible to rewrite $P \Rightarrow Q$ using only $\neg$ and $\vee$, and this is often useful when analyzing the precise meaning of an implication. In fact

$$[P \Rightarrow Q] \quad \Longleftrightarrow \quad [(\neg P) \vee Q]. \tag{6.6}$$

We can check this by computing the truth table of the right hand side:

P	Q	$\neg P$	$(\neg P) \vee Q$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	0	1

From this, we deduce that

$$\begin{aligned}
[P \Rightarrow Q] \quad &\Longleftrightarrow \quad [(\neg P) \vee Q] \\
&\Longleftrightarrow \quad [Q \vee (\neg P)] \\
&\Longleftrightarrow \quad [(\neg(\neg Q)) \vee (\neg P)] \qquad \text{by (6.3)} \\
&\Longleftrightarrow \quad [(\neg Q) \Rightarrow (\neg P)] \qquad \text{by (6.6)}
\end{aligned}$$

So, for any pair of statements P, Q,

$$[P \Rightarrow Q] \quad \Longleftrightarrow \quad [(\neg Q) \Rightarrow (\neg P)].$$

Exercise 6.3 Check this by computing the truth table of $(\neg Q) \Rightarrow (\neg P)$.

Given an implication $P \Rightarrow Q$, the implication $(\neg Q) \Rightarrow (\neg P)$ is called its **contrapositive**. We have just shown that an implication and its contrapositive are logically equivalent. Looking back at the above examples, we see that Proposition 2.13 could equally well have been stated

"If a real sequence is unbounded then it does not converge,"

and the Bolzano-Weierstrass Theorem is equivalent to

"If a real sequence has no convergent subsequence then it is unbounded,"

and the Divergence Test is equivalent to

"If a_n does not converge to 0 then $\sum_{n=1}^{\infty} a_n$ diverges,"

which is, in fact, the form in which one actually uses it.

Example 6.4 Write down the converse and contrapositive of each of the following implications.

(i) *If $x^2 = 2$ then x is irrational*

Converse: If x is irrational then $x^2 = 2$

Contrapositive: If x is rational then $x^2 \neq 2$

(ii) *If $a_n \to A$ and $b_n \to B$ then $a_n + b_n \to A + B$*

Converse: If $a_n + b_n \to A + B$ then $a_n \to A$ and $b_n \to B$

Contrapositive: If $a_n + b_n \nrightarrow A + B$ then $a_n \nrightarrow A$ or $b_n \nrightarrow B$

(iii) *If* $x > 2$ *then* $x^2 > 4$

Converse: If $x^2 > 4$ then $x > 2$

Contrapostive: If $x^2 \leq 4$ then $x \leq 2$

If you are asked to prove an implication, you should always consider whether proving its contrapositive will be easier. Since the contrapositive is logically equivalent to the original implication, this will suffice.

Example 6.5 Claim: Let n be an integer. If n^2 is divisible by 3 then n is divisible by 3.
Proof: We will prove the contrapositive: if n is **not** divisible by 3 then n^2 is **not** divisible by 3. So, assume that n is not divisible by 3. Then its remainder on division by 3 is either 1 or 2, that is, either $n = 3k + 1$ or $n = 3k + 2$, where k is an integer. In the first case

$$n^2 = 9k^2 + 6k + 1 = 3k' + 1$$

where $k' = 3k^2 + 2k \in \mathbb{Z}$, and in the second case

$$n^2 = 9k^2 + 12k + 4 = 3k' + 1$$

where $k' = 3k^2 + 4k + 1 \in \mathbb{Z}$. In either case, we see that n^2 is not divisible by 3 (in fact, its remainder is 1 in both cases), as was to be proved. $\square$

Another important fact follows immediately from (6.6): **the negation of an implication is NOT and implication!**

$$\neg[P \Rightarrow Q] \quad \Longleftrightarrow \quad \neg[Q \vee \neg P] \quad \Longleftrightarrow \quad (\neg Q) \wedge (\neg\neg P) \quad \Longleftrightarrow \quad P \wedge \neg Q.$$

Note that this is entirely different from $(\neg P) \Rightarrow (\neg Q)$. For example, the negation of the statement "if n is prime then n is odd" is the statement "n is prime and n is not odd", **not** the statement "if n is not prime then n is not odd."

6.3 Quantifiers

We have been a little bit sloppy up to now in dealing with statements which contain variables. Consider, for example, the statement

$$x > 2.$$

Is this really a statement? Until one specifies a particular value for x, one cannot say whether it is true. So it's true for $x = 3$, and false for $x = -2$, for example. What about if $x = 1 + 3i$? The "statement" $1+3i > 2$ is pretty meaningless, and it isn't very helpful to declare it either true or false (e.g. if you decide it's false, then you'd expect $1 + 3i \leq 2$ to be true, and this "statement" is equally meaningless). If we're being careful, we should make explicit the condition that x is a real number, and if we denote the statement $x > 2$ with a symbol, we should choose one like $P(x)$, rather than P, to remind ourselves that whether $P(x)$ is true depends on the particular x under consideration. So $P(3)$ is true, in this case, while $P(2)$ is false.

There are two very useful ways to turn a statement with a variable into an unambiguous true-or-false statement, using **quantifiers**. Let $P(x)$ be any statement which makes sense for all values of x in some specified set A (e.g. $A = \mathbb{R}$ and $P(x) : x > 2$, as above). Then

$\forall x\, P(x)$ means "for all $x \in A$, $P(x)$ is true"

and

$\exists x\, P(x)$ means "there exists some $x \in A$ such that $P(x)$ is true".

The symbol $\forall$ is called the **universal quantifier** and the symbol $\exists$ is called the **existential quantifier**. For the example above, $\forall x\, P(x)$ is false, because, for example, $P(-1)$ is false (since -1 is not greater than 2), but $\exists x\, P(x)$ is true because, for example, $P(3)$ is true. Note that, once we've coupled it with a quantifier, the statement no longer contains a variable, and is a simple true or false assertion. Note also that it's entirely irrelevant what we **call** the variable. So $\forall y\, P(y)$ is precisely the same statement as $\forall x\, P(x)$.

In practice, rather than specifying the set A in advance, we often specify it explicitly with the quantifier. In the above example, we would write

$$\forall x \in \mathbb{R},\ x > 2$$

which is false, and

$$\exists x \in \mathbb{R},\ x > 2 \tag{6.7}$$

which is true. You should feel free to add punctuation and/or ordinary English to expressions like this, if you think it clarifies the meaning. For example, I think

$$\exists x \in \mathbb{R} \text{ such that } x > 2$$

is clearer than (6.7). Indeed, you should feel free to avoid using the symbols $\forall$ and $\exists$ entirely if you think they obscure the mathematical meaning of the sentences you are writing. As we will see, they can be extremely useful, but you shouldn't chuck them into your mathematical prose willy-nilly. They won't make your proofs any more rigorous, elegant or impressive.

Example 6.6 Translate the following sentences into symbols:

(i) The square of any even integer is even.

In symbols: $\forall n \in \mathbb{Z}, [(n/2) \in \mathbb{Z} \Rightarrow (n^2/2) \in \mathbb{Z}]$

(ii) Given any real number K there is a positive integer N such that $N > K$.
In symbols: $\forall K \in \mathbb{R},\ \exists N \in \mathbb{Z}^+\ N > K$

(iii) There exists a rational number whose square is 2.
In symbols: $\exists x \in \mathbb{Q},\ x^2 = 2$

(iv) There exists a real number which is bigger than all rational numbers.
In symbols: $\exists x \in \mathbb{R},\ \forall y \in \mathbb{Q}\, x > y$

(v) The sequence (a_n) converges to L.
In symbols: $\forall \varepsilon \in (0, \infty),\ \exists N \in \mathbb{Z}^+,\ \forall n \in \mathbb{Z}^+ \cap [N, \infty),\ |a_n - L| < \varepsilon$

(vi) The sequence (a_n) converges.
In symbols: $\exists L \in \mathbb{R},\ \forall \varepsilon \in (0, \infty),\ \exists N \in \mathbb{Z}^+,\ \forall n \in \mathbb{Z}^+ \cap [N, \infty),\ |a_n - L| < \varepsilon$

Example 6.7 Translate the following "sentences" into English:

(i) $\forall x \in \mathbb{Z},\ x^2 > x$

Translation: The square of every integer x exceeds x.

Better translation: Every integer is less than its square.

(ii) $\exists x \in \mathbb{R},\ \forall y \in \mathbb{R}\ x^2 > y$
Translation: There exists a real number whose square is greater than every real number.

(iii) $\forall y \in \mathbb{R},\ \exists x \in \mathbb{R}\ x^2 > y$

Translation: For each real number y, there exists a real number x whose square exceeds y.

Better translation: The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, is unbounded above.

In the above example, we had a statement depending on two variables, $P(x, y)$, to which we applied a pair of quantifiers, $\exists x$ and $\forall y$. An important point to notice is that **it matters which order we apply the quantifiers in!** That is

$$\exists x\ \forall y\ P(x, y) \qquad \text{is not the same as} \qquad \forall y\ \exists x\ P(x, y).$$

In Example 6.7, we had $P(x, y) : x^2 > y$, from which we constructed

$$S_1 : \qquad \exists x \in \mathbb{R}\, \forall y \in \mathbb{R}\, P(x, y) \qquad\qquad \text{FALSE!}$$

and

$$S_2 : \qquad \forall y \in \mathbb{R}\, \exists x \in \mathbb{R}\, P(x, y) \qquad\qquad \text{TRUE!}$$

Since S_2 is true and S_1 is false, they're clearly not equivalent.

Example 6.8 Here's a non-mathematical example illustrating the same point. Let X be the set of all people (in this room, say), C be the set of all colours and $P(x, c)$ be the statement "x's favourite colour is c". Then

$$\forall x \in X\ \exists c \in C\ P(x, c) \quad \Longleftrightarrow \quad \text{Everyone has a favourite colour,}$$

whereas

$$\exists c \in C\ \forall x \in X\ P(x, c) \quad \Longleftrightarrow \quad \text{There is a colour which is everyone's favourite,}$$

which is clearly a very different statement.

The situation in which quantifiers really come into their own is when you want to understand the **negation** of a statement involving several quantifiers. Let's ask ourselves, what does it mean to say that $\forall x\, P(x)$ is **false**? It means that it is **not** true that for all x in A, $P(x)$ is true. That is, there is at least one element x in A for which $P(x)$ is **not** true. In other words, there exists $x \in A$ such that $\neg P(x)$ is true. That is

$$\neg[\forall x\, P(x)] \qquad \Longleftrightarrow \qquad \exists x\, \neg P(x). \tag{6.8}$$

Example 6.9 Consider the assertion that "all politicians are corrupt." I think this is false. In saying that, I am certainly **not** asserting that "all politicians are not corrupt" (which is equivalent to "no politician is corrupt"). Rather, I am asserting that "there exists at least one politician who is not corrupt." In symbols, let X denote the set of politicians and $P(x)$ be the statement "x is corrupt." Then the negation of $\forall x \in X\, P(x)$ is **not** $\forall x \in X\, \neg P(x)$, but rather $\exists x \in X\, \neg P(x)$.

Note that to prove that $[\forall x \in A\ P(x)]$ is **false**, we need to show that $[\exists x \in A\ \neg P(x)]$ is true, that is, we need to show that there exists (at least) **one** element x in A such that $P(x)$ is **false**. Such an element x is called a **counterexample** to $[\forall x \in A\ P(x)]$.

Example 6.10 Consider the statement $\forall x \in \mathbb{R}\ x^2 < 10$. This is clearly false. To prove that it is false, all we need to do is exhibit a **single** real number whose square is not less than 10. For example, $x = 10$ will do (since $10^2 \geq 10$). To prove the statement is false we do **not** have to find **all** real numbers x such that $x^2 \geq 10$. Giving a single counterexample is quicker, clearer and more elegant. $\square$

The rule for negating $\exists x\, P(x)$ actually follows from (6.8) and rule (6.3), but let's derive it by thinking. What does it mean to say that $\exists x\, P(x)$ is **false**? It means that there does **not** exist x in A such that $P(x)$ is true. Hence, for each and every $x \in A$, $P(x)$ must be **false**. That is

$$\neg[\exists x\, P(x)] \qquad \Longleftrightarrow \qquad \forall x\, \neg P(x).$$

Example 6.11 Consider the assertion "someone in this room speaks fluent Mandarin." Let's assume this is false. What does this mean? It certainly does **not** merely mean "someone in this room does not speak fluent Mandarin." Rather it means "no-one in this room speaks fluent Mandarin," or, equivalently, "everyone in this room does not speak fluent Mandarin." In symbols, let X denote the set of people in this room and $P(x)$ be the statement "x speaks fluent Mandarin." Then the negation of $\exists x \in X\, P(x)$ is **not** $\exists x \in X\, \neg P(x)$, but rather $\forall x \in X\, \neg P(x)$. $\square$

Example 6.12 Negate the following statements. Determine which is true, the statement or its negation.

(i)

$$\begin{aligned} S \quad &: \qquad \forall x \in \mathbb{Z}\ x^2 > x \\ \neg S \quad &: \qquad \exists x \in \mathbb{Z}\ x^2 \leq x \end{aligned}$$

$\neg S$ is true (for example $0^2 \leq 0$).

(ii)

$$\begin{aligned} S \quad &: \qquad \exists x \in \mathbb{R}\ x^2 + 2x + 2 = 0 \\ \neg S \quad &: \qquad \forall x \in \mathbb{R}\ x^2 + 2x + 2 \neq 0 \end{aligned}$$

$\neg S$ is true, since $x^2 + 2x + 2 = (x+1)^2 + 1 \geq 1$.

(iii)

$$\begin{aligned} S \quad &: \qquad \forall x \in (0, \infty)\, \exists y \in \mathbb{Q}\ 0 < y < 1/x \\ \neg S \quad &: \qquad \exists x \in (0, \infty)\, \neg[\exists y \in \mathbb{Q}\ 0 < y < 1/x] \\ &\Longleftrightarrow \ \exists x \in (0, \infty)\, \forall y \in \mathbb{Q}\ \neg[0 < y < 1/x] \\ &\Longleftrightarrow \ \exists x \in (0, \infty)\, \forall y \in \mathbb{Q}\ \neg[0 < y \ \wedge\ y < 1/x] \\ &\Longleftrightarrow \ \exists x \in (0, \infty)\, \forall y \in \mathbb{Q}\ [y \leq 0 \ \vee\ y \geq 1/x] \end{aligned}$$

S is true by Theorem 3.15: given any $x > 0$, $1/x > 0$, so there exists a rational number y between 0 and $1/x$.

You've already seen these negation rules in many different contexts. For example, consider the Archimedean Property of $\mathbb{R}$. Our first statement of this property was simply that "$\mathbb{Z}^+$ is unbounded above" (see Example 3.5). We then reformulated this as

"Given any real number K, there is some positive integer n such that $n > K$."

We can now see symbolically that these two statements are logically equivalent:

$$\begin{aligned} \mathbb{Z}^+ \text{ is unbounded above} \quad &\Longleftrightarrow \quad \neg\,[\mathbb{Z}^+ \text{ is bounded above}] \\ &\Longleftrightarrow \quad \neg[\exists K \in \mathbb{R}\ \forall n \in \mathbb{Z}^+\ n \leq K] \\ &\Longleftrightarrow \quad \forall K \in \mathbb{R} \neg[\forall n \in \mathbb{Z}^+\ n \leq K] \\ &\Longleftrightarrow \quad \forall K \in \mathbb{R}\ \exists n \in \mathbb{Z}^+\, \neg[n \leq K] \\ &\Longleftrightarrow \quad \forall K \in \mathbb{R}\ \exists n \in \mathbb{Z}^+\, n > K. \end{aligned}$$

In this case, using symbolic logic was really overkill: we could figure out the negation just by thinking carefully. A good example where symbolic logic really helps is the statement "(a_n) does not converge to L". To illustrate, consider the following question:

Example 6.13 We consider three statements

> S_1: "the sequence a_n **doesn't** converge to zero",
>
> S_2: "there exists $\varepsilon > 0$ such that, for all $N \in \mathbb{Z}^+$ there is some integer $n \geq N$ such that $|a_n| \geq \varepsilon$",
>
> S_3: "there exists $\epsilon > 0$ and a subsequence (a_{n_j}) of (a_n) with the property that for all $j \in \mathbb{Z}^+$, $|a_{n_j}| \geq \epsilon$ ".

Show that these statements are equivalent.

Proof: We find this by negating the definition of "a_n **does** converge to zero"

$$\begin{aligned}
a_n \nrightarrow 0 \quad &\iff \quad \neg[a_n \to 0] \\
&\iff \quad \neg[\forall \varepsilon \in (0,\infty)\, \exists N \in \mathbb{Z}^+\, \forall n \in \mathbb{Z}^+ \cap [N,\infty)\ |a_n - 0| < \varepsilon] \\
&\iff \quad \exists \varepsilon \in (0,\infty) \neg[\exists N \in \mathbb{Z}^+\, \forall n \in \mathbb{Z}^+ \cap [N,\infty)\ |a_n| < \varepsilon] \\
&\iff \quad \exists \varepsilon \in (0,\infty)\, \forall N \in \mathbb{Z}^+ \neg[\forall n \in \mathbb{Z}^+ \cap [N,\infty)\ |a_n| < \varepsilon] \\
&\iff \quad \exists \varepsilon \in (0,\infty)\, \forall N \in \mathbb{Z}^+\, \exists n \in \mathbb{Z}^+ \cap [N,\infty)\ \neg[|a_n| < \varepsilon] \\
&\iff \quad \exists \varepsilon \in (0,\infty)\, \forall N \in \mathbb{Z}^+\, \exists n \in \mathbb{Z}^+ \cap [N,\infty)\ |a_n| \geq \varepsilon.
\end{aligned}$$

Translating back into English, $a_n \nrightarrow 0$ means

> there exists $\varepsilon > 0$ such that, for all $N \in \mathbb{Z}^+$ there is some integer $n \geq N$ such that $|a_n| \geq \varepsilon$.

So $S_1 \iff S_2$.

Suppose now we know S_2 is true. Then set $N = 1$ and let a_{n_1} be an integer such that $|a_{n_1}| \geq \epsilon$ (this exists by S_2). Then given a_{n_i} set $N = n_i + 1$, and let n_{i+1} be a positive integer $m \geq n_1 + 1$ such that $|a_m| > \epsilon$ (this exists by S_2). Then, by induction we have found a subsequence (a_{n_j}) of (a_n) such that $|a_{n_j}| \geq \epsilon$ for all $j \in \mathbb{Z}^+$ and statement 3 is true.

Now suppose that S_3 is true, that is, there is an $\epsilon > 0$ and a subsequence (a_{n_j}) with $|a_{n_j}| > \epsilon$. Then if we are given $N \in \mathbb{Z}^+$ then as n_j is an increasing sequence of positive integers, $n_{N+1} > N$. Hence $|a_{n_{N+1}}| > \epsilon$. Hence we have shown S_2 holds.

Hence $S_3 \iff S_2 \iff S_1$, as required. □

Example 6.14 Let (a_n) be a real sequence with the property that every subsequence of (a_n) has a subsequence which converges to 0. Show that (a_n) converges to 0.

Proof: We will prove this by contradiction. So, let (a_n) be a sequence with the specified property, but assume that (a_n) does **not** converge to 0. Then we know that there exists $\varepsilon > 0$ and a subsequence $b_j = a_{n_j}$ of (a_n) so that $|b_j| = |a_{n_j}| > \epsilon$. Then let $c_k = b_{j_k}$ be a subsequence of (b_j) which converges to zero, which exists by our assumption. Hence there is an N so that for all $k \geq N$, $|c_k| < \epsilon$. So, $\epsilon < |b_{j_N}| = |c_N| < \epsilon$. But this is false, so $a_n \nrightarrow 0$ must also be false. □

Remark This result is quite subtle. It does **not** say that

> If every subsequence of (a_n) converges to 0 then (a_n) converges to 0

although this is trivially true, since (a_n) is a subsequence of itself! On the other hand, by Theorem 4.3, Example 6.14 has the following corollary:

> If every subsequence of (a_n) has a subsequence which converges to 0, then every subsequence of (a_n) converges to 0.

This is true despite the fact that, in general, having a subsequence which converges to 0 does **not** imply that a sequence converges to 0.

Here are a couple of questions for you to think about.

(i) Consider the statement

> If every subsequence of (a_n) has a subsequence which has a subsequence which converges to 0 then (a_n) converges to 0.

Is this true?

(ii) In Example 6.14, we assumed that all subsequences have a subsequence converging to 0. Presumably there's nothing special about 0. We could equally well have assumed that all subsequences have a subsequence converging to L for any other fixed $L \in \mathbb{R}$. (If this isn't clear, try adapting the above proof to deal with the case of general L.) Now consider the following statement:

> If every subsequence of (a_n) has a convergent subsequence then (a_n) is convergent.

Is this true? $\square$

Summary

- We can represent statements by symbols P, Q etc. We say that P and Q are **logically equivalent** if P is true if and only if Q is true, denoted $P \iff Q$.
- We obtain new statements from existing ones using the following operations
  - $\neg P$ denotes the **negation** of P, "not P", which is true if and only if P is false
  - $P \wedge Q$ denotes the **conjunction** of P and Q, "P and Q", which is true if and only if both P and Q are true
  - $P \vee Q$ denotes the **disjunction** of P and Q, "P or Q", which is true if P is true or Q is true, or both.
- These operations obey algebraic rules:

$$\neg(\neg P) \iff P, \qquad \neg(P \wedge Q) \iff (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \iff (\neg P) \wedge (\neg Q).$$

- $P \Rightarrow Q$ denotes the **implication** "if P then Q". By definition, this is logically equivalent to $(\neg P) \vee Q$.
  - The **converse** of $P \Rightarrow Q$ is $Q \Rightarrow P$.
  - The **contrapositive** of $P \Rightarrow Q$ is $(\neg Q) \Rightarrow (\neg P)$.

  An implication is logically equivalent to its contrapositive, but **not** to its converse.
- The symbols $\forall$ and $\exists$ are the **universal** and **existential quantifiers**.
  - $\forall x \in X \ P(x)$ means "for all $x \in X$, $P(x)$ is true".
  - $\exists x \in X \ P(x)$ means "there exists some $x \in X$ such that $P(x)$ is true".
- These symbols behave under negation as follows

$$\neg[\forall x \in X \ P(x)] \iff \exists x \in X \ \neg P(x), \qquad \neg[\exists x \in X \ P(x)] \iff \forall x \in X \ \neg P(x).$$

7 Limits of functions

7.1 Cluster points and limits of functions

We now have a very precise (and powerful) definition of *limit* for real sequences. In this section we will show how to use this to give a precise definition of the limit of a real function of a real variable. We will also see that basic properties of limits of functions, which are usually asserted without proof in introductory calculus courses, follow very easily from the corresponding limit theorems for sequences, which we proved in section 2.2.

Informally, $\lim_{x\to a} f(x) = L$ means that $f(x)$ is "arbitrarily close" to L for all x "sufficiently close to, but different from" a. So limits concern the behaviour of x arbitrarily close to a, but *not at* a. There is no reason why a has to be in the domain of f, that is, $f(a)$ may, or may not, be defined. If it is, its value should be irrelevant to $\lim_{x\to a} f(x)$. For example,

$$f(x) = \frac{\sin x}{x}$$

is clearly undefined at 0 (its domain is $(-\infty, 0) \cup (0, \infty)$), but $\lim_{x\to 0} f(x)$ is well defined (it's 1).

In order to probe the behaviour of $f(x)$ for values of x close to, but different from, a, we can consider the sequence $f(x_n)$ where (x_n) is any sequence *converging* to a with $x_n \neq a$. From now on, given a set $D \subseteq \mathbb{R}$ we will say that a sequence a_n is in D if for every $n \in \mathbb{Z}^+$, $a_n \in D$.

This leads us to the main definition of this chapter:

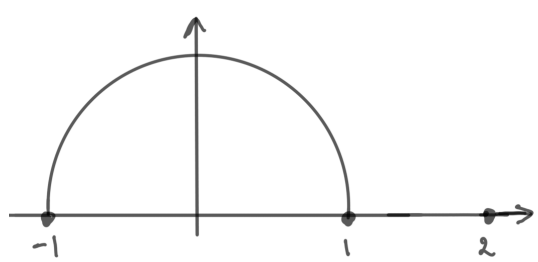
Definition 7.1 Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Then f has **limit** L **at** a if, for all sequences (x_n) in $D\backslash\{a\}$ such that $x_n \to a$, $f(x_n) \to L$. In this case, we usually denote the number L by $\lim_{x\to a} f(x)$.

It's important to note that the definition says that for *all* sequences (x_n) in $D\backslash\{a\}$ converging to a, $f(x_n) \to L$. It's not enough to show that $f(x_n) \to L$ for just one specific sequence. Note that the sequence (x_n) gets arbitrarily close to a (since $x_n \to a$), but never *equals* a (since it takes values in $D\backslash\{a\}$). So the value of f at a, if this exists, is irrelevant.

In order for Definition 7.1 to make sense, there must *exist* at least one sequence in $D\backslash\{a\}$ such that $x_n \to a$. For example, the function

$$f(x) = \sqrt{1 - x^2}$$

has maximal domain $D = [-1, 1]$, so it doesn't make sense to try to define $\lim_{x\to 2} f(x)$, its limit at 2, say, because there don't exist any sequences in $D\backslash\{2\} = [-1, 1]$ which converge to 2.



To be precise, we should include this requirement in our definition of limit. This leads us to the following definition.

Definition 7.2 Let $D \subset \mathbb{R}$ and $a \in \mathbb{R}$. We say that a is a **cluster point** of D if there exists a sequence (x_n) in $D\backslash\{a\}$ such that $x_n \to a$.

Example 7.3 Claim: Every $a \in \mathbb{R}$ is a cluster point of $\mathbb{Q}$.

Proof: Choose any $a \in \mathbb{R}$. By Theorem 3.15, for each $n \in \mathbb{Z}^+$ there exists a rational number x_n between a and $a + 1/n$. Now $x_n \in \mathbb{Q}\backslash\{a\}$ and $x_n \to a$ by the Squeeze Rule. Hence a is a cluster point of $\mathbb{Q}$. $\square$

Example 7.4 Claim: The set of cluster points of $(0, \infty)$ is $[0, \infty)$.

Proof: Let $a \in [0, \infty)$. Then, for each $n \in \mathbb{Z}^+$, $x_n = a + \frac{1}{n}$ is different from a and lies in $(0, \infty)$. Clearly $x_n \to a$. Hence every $a \in [0, \infty)$ is a cluster point of $(0, \infty)$, that is, the set of cluster points of $(0, \infty)$ contains $[0, \infty)$. Conversely, assume $a \notin [0, \infty)$. Then $a < 0$. Let (x_n) be any sequence in $(0, \infty)$. If (x_n) converges at all, its limit must be non-negative, since $x_n > 0$ for all n (Proposition 2.14). Hence $x_n \nrightarrow a$. Since no sequence in $(0, \infty)$ converges to a, a is *not* a cluster point of $(0, \infty)$. $\square$

Many authors call cluster points **limit points**, or **accumulation points**. It makes sense to ask whether $f : D \to \mathbb{R}$ has a limit at a precisely when a is a cluster point of D. With this is mind, we should really tighten up Definition 7.1.

Improved Definition 7.1 Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and a be a cluster point of D. Then f has **limit** L **at** a if, for all sequences (x_n) in $D\backslash\{a\}$ such that $x_n \to a$, $f(x_n) \to L$. In this case, we usually denote the number L by $\lim_{x\to a} f(x)$.

An important fact about limits is that, if they exist, they're unique:

Proposition 7.5 *If* $\lim_{x\to a} f(x)$ *exists, then it is unique.*

Proof: Let $f : D \to \mathbb{R}$ and assume, to the contrary, that K and L are non-equal real numbers satisfying the definition of $\lim_{x\to a} f(x)$. Let (x_n) be a sequence in $D\backslash\{a\}$ such that $x_n \to a$. Then $f(x_n) \to K$ and $f(x_n) \to L$ and $f(x_n) \to L \neq K$. But this contradicts Proposition 2.11. $\square$

So it makes sense to speak of *the* limit of a function at a.

Example 7.6 Let $f : \mathbb{R}\backslash\{1\} \to \mathbb{R}$ such that

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

Let (x_n) be any sequence in $\mathbb{R}\backslash\{1\}$ converging to 1. Then

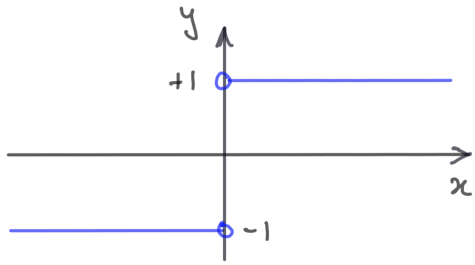
$$f(x_n) \quad = \quad \frac{(x_n - 1)(x_n + 1)}{x_n - 1} = x_n + 1 \to 2.$$

So $\lim_{x\to 1} f(x) = 2$, as you would expect. $\square$

Example 7.7 Let $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ such that

$$f(x) = \frac{x}{|x|}.$$

Then $\lim_{x\to 0} f(x)$ does not exist.



To prove this, note that $x_n = (-1)^n/n$ is a sequence converging to 0, but

$$f(x_n) \quad = \quad \frac{(-1)^n/n}{1/n} = (-1)^n$$

which does not converge at all. $\square$

Example 7.8 Let $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ such that

$$f(x) = \sin\frac{1}{x}.$$

We can prove that $\lim_{x\to 0} f(x)$ does not exist using a similar argument. Assume that $\lim_{x\to 0} f(x)$ exists, and call it L. Let

$$x_n = \frac{1}{\pi n}.$$

Then (x_n) is a sequence in $D\backslash\{0\} = D$ converging to 0, so $f(x_n) \to L$. But

$$f(x_n) = \sin \pi n = 0 \to 0,$$

so $L = 0$. Let

$$y_n = \frac{1}{\pi(2n + \frac{1}{2})}.$$

Again, this is a sequence in $D\backslash\{0\}$ converging to 0, so $f(y_n) \to L = 0$. But

$$f(y_n) = \sin\left(\frac{\pi}{2} + 2n\pi\right) = 1 \to 1,$$

a contradiction. $\square$

Note that $\lim_{x\to a} f(x)$ is a single number, not a function, and the fact that we called the original variable x is of no significance. That is

$$\lim_{x\to a} f(x) = \lim_{y\to a} f(y) = \lim_{\widetilde{\Gamma}'_0 \to a} f(\widetilde{\Gamma}'_0).$$

There is also an alternative (and more commonly used) definition of limits. This is the so-called ε–δ criterion for limits:

Theorem 7.9 *Let a be a cluster point of $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Then $\lim_{x\to a} f(x) = L$ if and only if, for each $\varepsilon > 0$, there exists $\delta > 0$ such that, for all $x \in D$ such that $0 < |x-a| < \delta$, $|f(x)-L| < \varepsilon$.*

Proof: ($\Leftarrow$, the "if" direction): Assume that f, a, L satisfy the ε–δ criterion. We must show that $\lim_{x\to a} f(x) = L$. So, let (x_n) be any sequence in $D\backslash\{a\}$ such that $x_n \to a$. We must show that $f(x_n) \to L$. Let $\varepsilon \in (0,\infty)$ be given. Then, by the ε–δ criterion, there exists $\delta \in (0,\infty)$ such that, for all $x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta)$, $|f(x) - L| < \varepsilon$. But $x_n \to a$ so given any positive real number, for example, δ, there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $x_n \in (a-\delta, a+\delta)$. By definition, $x_n \in D\backslash\{a\}$, so, for all $n \geq N$, $x_n \in (D\backslash\{a\}) \cap (a-\delta, a+\delta)$, whence $|f(x_n) - L| < \varepsilon$. Hence $f(x_n) \to L$.

($\Rightarrow$, the "only if" direction): We will prove the contrapositive, that is, we will show that if f, a, L do *not* satisfy the ε–δ criterion, then $\lim_{x\to a} f(x) \neq L$, that is, there exists a sequence (x_n) in $D\backslash\{a\}$ converging to a whose image under f, $f(x_n)$ does *not* converge to L.

Our first task is to negate the ε–δ criterion:

$$\begin{aligned}
& \neg[\forall\varepsilon \in (0,\infty), \exists\delta \in (0,\infty), \forall x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta), |f(x) - L| < \varepsilon] \\
\Leftrightarrow\ & \exists\varepsilon \in (0,\infty)\neg[\exists\delta \in (0,\infty), \forall x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta), |f(x) - L| < \varepsilon] \\
\Leftrightarrow\ & \exists\varepsilon \in (0,\infty), \forall\delta \in (0,\infty)\neg[\forall x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta), |f(x) - L| < \varepsilon] \\
\Leftrightarrow\ & \exists\varepsilon \in (0,\infty), \forall\delta \in (0,\infty), \exists x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta)\neg[|f(x) - L| < \varepsilon] \\
\Leftrightarrow\ & \exists\varepsilon \in (0,\infty), \forall\delta \in (0,\infty), \exists x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta), |f(x) - L| \geq \varepsilon.
\end{aligned}$$

So, we assume that there exists some positive real number ε such that, for all $\delta > 0$ there is some $x \in (D\backslash\{a\}) \cap (a-\delta, a+\delta)$ such that $|f(x) - L| \geq \varepsilon$. This holds for every $\delta > 0$. In particular, it holds for $\delta = 1/n$ where $n \in \mathbb{Z}^+$: for each $n \in \mathbb{Z}^+$ there exists $x_n \in (D\backslash\{a\}) \cap (a-1/n, a+1/n)$ such that $|f(x_n) - L| \geq \varepsilon$. Consider the sequence (x_n). By definition, it lies in $D\backslash\{a\}$ and

$$a - \frac{1}{n} < x_n < a + \frac{1}{n},$$

so $x_n \to a$ by the Squeeze Rule. But $|f(x_n) - L| \geq \varepsilon$ for all n, so certainly $|f(x_n) - L| \nrightarrow 0$ (this would contradict Proposition 2.14), and so $f(x_n) \nrightarrow L$.

In summary, we have shown that if f, a, L do *not* satisfy the ε–δ criterion, then we can construct a sequence $x_n \in D\backslash\{a\}$ which converges to a such that $f(x_n)$ does *not* converge to L. It follows, in this case, that f does *not* have limit L at a. $\square$

Many (in fact, almost all) textbooks on Real Analysis define limits using the ε–δ criterion, rather than sequences. The Theorem above assures us that this approach is equivalent to ours. One reason to prefer our sequential approach to limits is that, having defined limits purely in terms of convergence of sequences, it is very easy to establish many of their basic properties, since these follow directly from theorems about sequences which we've already proved. We've already seen an example of this: limits of functions are unique (Proposition 7.5) because limits of convergent sequences are unique (Proposition 2.11). Another very useful example is the Algebra of Limits for functions:

Theorem 7.10 (The Algebra of Limits for functions)
Let $f : D \to \mathbb{R}$, $g : D \to \mathbb{R}$, $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = K$. Then

(i) $\lim\limits_{x\to a}(f(x) + g(x)) = L + K$,

(ii) $\lim\limits_{x\to a} f(x)g(x) = LK$,

and if, in addition, $f(x) \neq 0$ for all $x \in D\backslash\{a\}$ and $L \neq 0$, then

(iii) $\lim\limits_{x\to a} \dfrac{1}{f(x)} = \dfrac{1}{L}$.

Proof: Let (x_n) be any sequence in $D\backslash\{a\}$ such that $x_n \to a$. Then $f(x_n) \to L$ and $g(x_n) \to K$ by assumption. Hence

(i) $f(x_n) + g(x_n) \to L + K$ by Theorem 2.15(i).

(ii) $f(x_n)g(x_n) \to LK$ by Theorem 2.15(ii).

(iii) $f(x_n) \neq 0$ and $L \neq 0$ so $1/f(x_n) \to 1/L$ by Theorem 2.15(iii).

□

Note that the theorem follows almost immediately from Theorem 2.15, once we've defined limits of functions in the right way (Definition 7.1).

Hey! You now know what limits are!

It's worthwhile to pause and reflect on what we've achieved in the first semester. We now have a completely precise and rigorous definition of $\lim_{x\to a} f(x)$ for functions $f : D \to \mathbb{R}$, where D is a subset of $\mathbb{R}$. It follows, that we now know, with no hand-waving or weasel words, precisely what the limit

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}$$

means, that is, we have put the definition of the *derivative* of a function on a proper mathematical footing. Note that this is not a limit of $f : D \to \mathbb{R}$, but of the associated function

$$h : D\backslash\{a\} \to \mathbb{R}, \qquad h(x) = \frac{f(x) - f(a)}{x - a}.$$

The interesting, and somewhat curious, thing about this is that we managed to do this by studying *sequences* and, in particular, making precise what it means for sequences to converge. So from the rather modest and unassuming definition of convergence of sequences (Definition 2.4) something very large and powerful has grown.

Summary

- Let $f : D \to \mathbb{R}$ where $D \subseteq \mathbb{R}$. We say that f **has limit** L **at** a, written $\lim_{x\to a} f(x) = L$, if for all sequences (x_n) in $D\backslash\{a\}$ which converge to a, $f(x_n)$ converges to L.
- This definition only makes sense if a is a **cluster point** of D, that is, a point for which there exists at least one sequence in $D\backslash\{a\}$ converging to a.
- A subset D of $\mathbb{R}$ may contain points which are not cluster points of D. It may also have cluster points which are not in D.
- The **Algebra of Limits** for functions follows immediately from the Algebra of Limits for sequences:

$$\begin{aligned}
\lim_{x\to a}(f(x) + g(x)) &= \lim_{x\to a} f(x) + \lim_{x\to a} g(x), \\
\lim_{x\to a} f(x)g(x) &= [\lim_{x\to a} f(x)][\lim_{x\to a} g(x)], \\
\lim_{x\to a} \frac{f(x)}{g(x)} &= \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)},
\end{aligned}$$

  under suitable assumptions.

8 Continuous functions

In this second half of the module, we will take all the theory about convergence of sequences (and series) we've developed in Semester 1, and use it to develop a precise and rigorous theory of calculus (both differential and integral) of real-valued functions of a single real variable. Since we now have a precise definition of limit (see Definition 7.1) we're already halfway there: we know precisely what we mean by the derivative $f'(a)$ of a differentiable function f. (Of course, just defining the derivative isn't the end of the story: we also want to prove that derivatives have all the properties you know and love, that they obey the Product and Chain rules and so on.) Before embarking on that endeavour we will retrench a little, however. Instead of diving straight into differentiability, it will turn out to be important to thoroughly understand a more rudimentary property of (some) functions: continuity.

8.1 Sequential continuity

Given a function $f : \mathbb{R} \to \mathbb{R}$, what does it mean to say that the function is **continuous**? It's common, when first introduced to the notion of continuity, to be told that a function is continuous "if you can draw its graph without taking your pencil off the paper". Clearly this is **not** an acceptable mathematical definition: what if you have no pencil, or paper, or hands? Does the continuity or otherwise of f depend on your skill as a draughtsman? More importantly, can you imagine ever being able to *prove* statements about continuous functions starting from such a "definition"?

So our first job will be to define continuity precisely. In fact it's quite easy to give a precise (and rather elegant) definition of continuity using nothing more than the notion of convergence of sequences. The definition works equally well when the domain of f is a subset of $\mathbb{R}$, not necessarily the whole of $\mathbb{R}$, so that is how we will formulate it.

Given a set $D \subseteq \mathbb{R}$ we will say that a sequence a_n is in D if for every $n \in \mathbb{Z}^+$, $a_n \in D$.

Definition 8.1 Let $D \subseteq \mathbb{R}$. A function $f : D \to \mathbb{R}$ is **continuous at a point** $a \in D$ if, **for all sequences** (x_n) in D such that $x_n \to a$, $f(x_n) \to f(a)$. If f is not continuous at $a \in D$, we say that f is **discontinuous** at a. The function is **continuous** if it is continuous at a for all $a \in D$.

So what is this definition saying? Roughly speaking, this is saying that for points that get arbitrarily close to $a \in D$, the images of these points under f get arbitrarily close to $f(a)$.

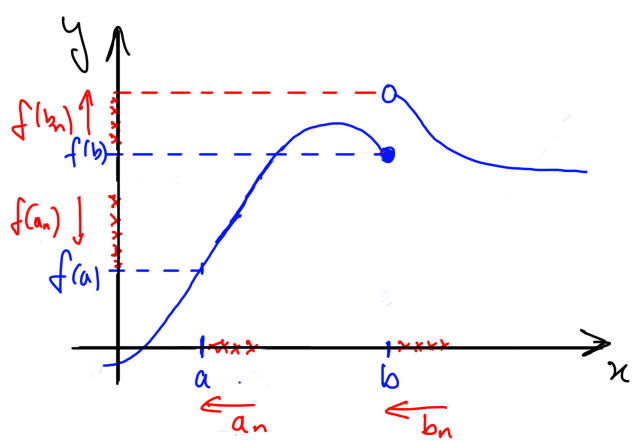


Figure 4: In this picture f is continuous at a since, for any sequence a_n converging to a, $f(a_n)$ converges to $f(a)$. But f is *discontinuous* at b since there exist sequences b_n converging to b such that $f(b_n)$ does *not* converge to $f(b)$. One such sequence is depicted.

Example 8.2

(i) $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = c$ (for c any constant), is continuous.

Proof: Let $a \in \mathbb{R}$ and $x_n \to a$. Then $f(x_n) = c \to c = f(a)$. $\square$

(ii) $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x$ is continuous.

Proof: Let $a \in \mathbb{R}$ and $x_n \to a$. Then $f(x_n) = x_n \to a = f(a)$. $\square$

(iii) $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ such that $f(x) = 1/x$ is continuous.

If you've been taught in the past that a continuous function is "one whose graph you can draw without taking your pen off the paper" then this example will be surprising at first sight. Nonetheless, this function really is continuous according to our (precise) definition:

Proof: Let $a \in \mathbb{R}\backslash\{0\}$ (i.e., let a be a nonzero real number) and x_n be a sequence in $\mathbb{R}\backslash\{0\}$ such that $x_n \to a$. Then $x_n \neq 0$, so

$$f(x_n) = \frac{1}{x_n} \to \frac{1}{a} = f(a)$$

by Theorem 2.15(iii). $\square$

(iv) $f : \mathbb{Z} \to \mathbb{R}$ such that $f(x) = (-1)^x$ is continuous.

This one is also counterintuitive. In fact, it gets worse:

(v) **Every** function $f : \mathbb{Z} \to \mathbb{R}$ is continuous!

Proof: Let $a \in \mathbb{Z}$ and (x_n) be a sequence in $\mathbb{Z}$ converging to a. Then, by the definition of convergence, there exists $N \in \mathbb{Z}^+$ such that for all $n \geq N$, $|x_n - a| < 1/2$. But x_n and a are both integers, so if they lie distance less than $1/2$ away from one another, they must be equal. Hence, for all $n \geq N$, $x_n = a$. Now let $\varepsilon > 0$ be given. Then for all $n \geq N$, $|f(x_n) - f(a)| = |f(a) - f(a)| = 0 < \varepsilon$. Hence, $f(x_n) \to f(a)$. $\square$

Don't worry, however: there **are** discontinuous functions:

(vi) Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 0$ if $x < 0$ and $f(x) = 1$ for $x \geq 0$. Then f is discontinuous at 0.

Proof: The sequence $x_n = -1/n$ converges to 0 but

$$f(x_n) = f(-1/n) = 0 \to 0 \neq f(0).$$

$\square$

This is an example of a function which is **discontinuous** at a single isolated point. It's also possible to construct functions which are **continuous** at a single isolated point (and discontinuous everywhere else):

(vii) Let $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ x & x \notin \mathbb{Q}. \end{cases}$$

Then f is continuous at 0 and discontinuous everywhere else.

Proof: Let $a \in \mathbb{R}$ and assume that f is continuous at a. By Theorems 3.15 and 3.16, for each $n \in \mathbb{Z}^+$, there exist $r_n \in \mathbb{Q}$ and $i_n \in \mathbb{R}\backslash\mathbb{Q}$ such that $a < r_n < a + 1/n$ and $a < i_n < a + 1/n$.

Clearly $r_n \to a$ and $i_n \to a$ by the Squeeze Rule. Hence, by the definition of continuity, $f(r_n) \to f(a)$ and $f(i_n) \to f(a)$. But r_n is rational, so $f(r_n) = 0 \to 0$, and i_n is irrational, so $f(i_n) = i_n \to a$. Hence $a = 0$.

Aside: so what have we proved? Have we proved that f is continuous at 0? No we flipping well have **NOT!** *We have proved that <u>if</u> f is continuous at a <u>then</u> $a = 0$. We have* **not** *proved that <u>if</u> $a = 0$ <u>then</u> f is continuous at a! Another way to say what we have proved is that <u>if</u> $a \neq 0$ <u>then</u> f is <u>not</u> continuous at a. That is, up to this point, we have proved that f is* **discontinuous** *at every $a \neq 0$.*

Let (x_n) be any sequence in $\mathbb{R}$ such that $x_n \to 0$. Let $\varepsilon > 0$ be given. Then, since $x_n \to 0$, there exists $N \in \mathbb{Z}^+$ such that for all $n \geq N$, $|x_n - 0| < \varepsilon$. But then, for all $n \geq N$,

$$|f(x_n) - 0| = |f(x_n)| \leq |x_n| < \varepsilon$$

since $f(x_n) = x_n$ if x_n is irrational, and is 0 otherwise. Hence $f(x_n) \to 0 = f(0)$. Hence, f is continuous at 0. □

8.2 Properties of continuous functions

If we are given a complicated function such as

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \frac{x^3 - 7x}{x^2 + 1}$$

we can show directly from Definition 8.1 that f is continuous, by considering an arbitrary sequence (x_n) converging to a. To do this we will have to appeal at some point to Theorem 2.15 (the Algebra of Limits), to argue that

$$f(x_n) = \frac{x_n^3 - 7x_n}{x_n^2 + 1} \qquad \text{converges to} \qquad \frac{a^3 - 7a}{a^2 + 1} = f(a).$$

The alternative to this is to prove, once and for all, that continuity is "preserved" under sums, products and (under suitable restrictions) quotients.

Theorem 8.3 (Algebra Property of Continuous Functions) *Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ be continuous at $a \in D$. Then*

(i) $f + g$ is continuous at a.

(ii) fg is continuous at a.

If, in addition, $f(x) \neq 0$ for all $x \in D$, then

(iii) $1/f$ is continuous at a.

Proof: Given any sequence (x_n) in D such that $x_n \to a$, we know that $f(x_n) \to f(a)$ and $g(x_n) \to g(a)$ (since f, g are continuous at a). Hence

$$\begin{aligned} (f+g)(x_n) &= f(x_n) + g(x_n) \to f(a) + g(a) \qquad \text{(by Theorem 2.15(i))} \\ (fg)(x_n) &= f(x_n)g(x_n) \to f(a)g(a) \qquad \text{(by Theorem 2.15(ii))} \\ \text{and} \qquad (1/f)(x_n) &= \frac{1}{f(x_n)} \to \frac{1}{f(a)} \qquad \text{(by Theorem 2.15(iii)} \end{aligned}$$

where, in the final line, we have assumed $f(x_n) \neq 0$ and $f(a) \neq 0$. Hence $f + g$, fg and $1/f$ are continuous at a. □

Recall that a **polynomial** function is a function of the form

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m$$

where $a_0, \ldots, a_m$ are (real) constants. If $a_m \neq 0$, we say that p has **degree** m. Polynomials are a very nice class of functions. In particular they are always continuous:

Proposition 8.4 *Every polynomial function $p : \mathbb{R} \to \mathbb{R}$ is continuous.*

Proof: We prove this by induction on m, the degree of the polynomial p. So, if $m = 0$, then $p(x)$ is a degree 0 polynomial, that is, $p(x) = a_0$, constant. Hence $p(x)$ is continuous (see Example 8.2). Now assume that every polynomial of degree k is continuous. Let $p(x)$ be a degree $k+1$ polynomial. Then $p(x) = xq(x) + a_0$ where $q(x)$ is a polynomial of degree k and so is, by assumption, continuous. Now $f(x) = x$ is continuous (see Example 8.2), so $p(x) = xq(x) + a_0$ is continuous by Theorem 8.3(i),(ii). Hence, by induction every polynomial is continuous. □

A nice thing about the set of polynomials is that it is **closed** under function composition. That is, if p and q are polynomials, so is $q \circ p$ (recall that $(q \circ p)(x) = q(p(x))$). For example:

$$\begin{aligned} p(x) &= x^3 + 1, \qquad q(x) = 3x^2 - 2 \\ \Rightarrow (q \circ p)(x) &= q(x^3 + 1) \\ &= q(x^3 + 1) \\ &= 3(x^6 + 2x^3 + 1) - 2 \\ &= 3x^6 + 6x^3 + 1. \end{aligned}$$

So given a pair of polynomials (certainly continuous), their composition is a polynomial (and hence is continuous). Is this a special feature of polynomials? No: it follows almost immediately from our definition of continuity that the composition of two continuous functions is continuous.

Theorem 8.5 *Let $D, E \subseteq \mathbb{R}$, $f : D \to E$ and $g : E \to \mathbb{R}$. If f is continuous at $a \in D$ and g is continuous at $f(a) \in E$, then $g \circ f$ is continuous at a.*

Proof: Let (x_n) be any sequence in D which converges to a. Then $f(x_n) \to f(a)$ (since f is continuous at a), so $(g \circ f)(x_n) = g(f(x_n)) \to g(f(a)) = (g \circ f)(a)$ (since g is continuous at $f(a)$). Hence $g \circ f$ is continuous at a. □

8.3 The Intermediate Value Theorem

To say that this theorem is "important" would be something of an understatement. It is the basis of a useful method for approximately solving equations (the bisection method) and, as we will see, implies the existence of square roots and other similar functions.

Theorem 8.6 (Intermediate Value Theorem) *Let $f : [a, b] \to \mathbb{R}$ be continuous, and y be any real number between $f(a)$ and $f(b)$. Then there exists $c \in [a, b]$ such that $f(c) = y$.*

Proof: There are three possibilities: (i) $f(a) = f(b)$, (ii) $f(a) < f(b)$ and (iii) $f(a) > f(b)$.
(i) If $f(a) = f(b)$ then the only real number between them is $y = f(a)$, so $c = a$ will do.
(ii) Assume $f(a) < f(b)$. If $f(b) = y$ then $c = b$ will do. So assume $y < f(b)$. Define the set

$$A = \{x \in [a, b] \ : \ f(x) \leq y\},$$

which is nonempty (since $a \in A$) and bounded above (by b). Hence by Axiom 3.11, there exists c, the supremum of A. We will prove that $c \in [a, b]$ and that $f(c) = y$, which completes case (ii). c is an upper bound on A and $a \in A$, so $a \leq c$. Further, b is also an upper bound on A, and c is the **least** upper bound on A, so $c \leq b$. Hence $c \in [a, b]$. Consider $c - 1/n$, where $n \in \mathbb{Z}^+$. Since this is less than c, it is **not** an upper bound on A, so there exists $x_n \in A$ with $x_n > c - 1/n$. But then $c - 1/n \leq x_n \leq c$, so $x_n \to c$ by the Squeeze Rule. f is continuous at c, so $f(x_n) \to f(c)$. But each

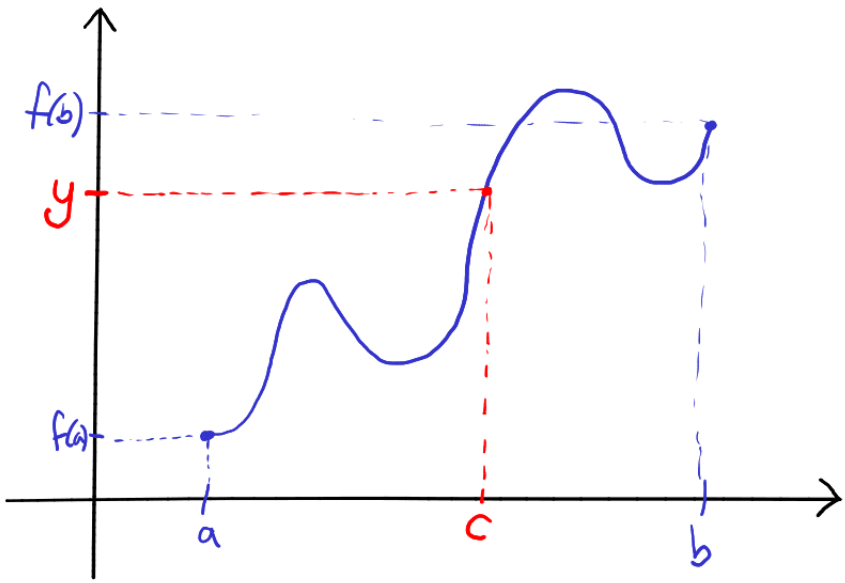


Figure 5: A picture of the Intermediate Value Theorem in the case $f(a) < f(b)$. Here we see that for $y \in [f(a), f(b)]$ we have a $c \in [a, b]$ with $f(c) = y$.

$x_n \in A$, so $f(x_n) \leq y$ for all n, and hence its limit cannot exceed y, by Proposition 2.14. Hence $f(c) \leq y$. Hence $c \in A$ and $c \neq b$, so $c < b$. Consider now the sequence $z_n = c + (b - c)/n$. This is a sequence in $[a, b]$ each term of which exceeds c, so for all n, $z_n \notin A$ (recall c is an upper bound on A). Hence $f(z_n) \geq y$ for all n. But $z_n \to c$ and f is continuous, so $f(z_n) \to f(c)$. Hence, by Proposition 2.14, $f(c) \geq y$. Since we've already shown that $f(c) \leq y$, it follows that $f(c) = y$.

(iii) If $f(a) > f(b)$, define $\widetilde{f} : [a, b] \to \mathbb{R}$ such that $\widetilde{f}(x) = -f(x)$ and $\widetilde{y} = -y$. Then $\widetilde{f}$ is continuous, $\widetilde{f}(a) < \widetilde{f}(b)$, and $\widetilde{y}$ is between $\widetilde{f}(a)$ and $\widetilde{f}(b)$, so, as proved in part (ii), there exists $c \in [a, b]$ such that $\widetilde{f}(c) = \widetilde{y}$. But then $f(c) = y$.

□

Note that the assumption that the domain of f is an interval is crucial:

Example 8.7 $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$, $f(x) = 1/x$ is continuous (by Example 8.2). Let $a = -1$ and $b = 1$. Then $f(a) = -1$ and $f(b) = 1$ and so 0 is a number between $f(a)$ and $f(b)$. But there is no c such that $f(c) = 0$ ($1/c = 0$ has no solution). This is not a counterexample to Theorem 8.6, since f is **not** continuous on $[-1, 1]$.

Example 8.8 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2 - 3$. Then f is continuous. In particular, it is continuous on $[0, 2]$. Now $f(0) = -3$ and $f(2) = 1$, so 0 is a number between $f(0)$ and $f(2)$. Hence, by the Intermediate Value Theorem, there exists $c \in [0, 2]$ such that $f(c) = 0$. That is, there exists $c \in [0, 2]$ such that $c^2 = 3$.

Note that the Intermediate Value Theorem only tells us that c exists, not what it is, or whether it is unique (maybe there are hundreds of different c's!). However, by repeatedly applying the theorem, we can find an approximate solution to $f(c) = 0$ to arbitrarily high accuracy. We know that $f(0) < 0$ and $f(2) > 0$, so 0 is a number between $f(0)$ and $f(2)$, and hence, there is some $c \in [0, 2]$ such that $f(c) = 0$. Now test the midpoint of $[0, 2]$:

$$f(1) \quad = \quad 1 - 3 < 0$$

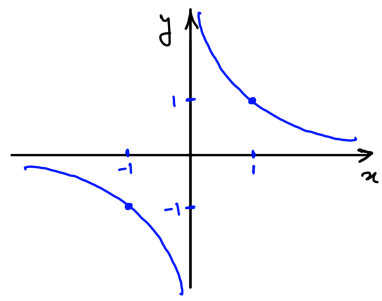


Figure 6: $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$, $f(x) = 1/x$, is continuous, and 0 is a number between $f(-1)$ and $f(1)$. But there is no number $c \in [-1, 1]$ such that $f(c) = 0$.

So 0 is a number between $f(1)$ and $f(2)$, and f is continuous on $[1, 2]$, so by the IVT, there exists $c \in [1, 2]$ such that $f(c) = 0$. (Had $f(1)$ been positive, we'd have deduced that $c \in [0, 1]$. Had it been 0, we'd have deduced that $c = 1$ exactly!) Now test the midpoint of $[1, 2]$, and so on:

$$\begin{aligned} f(1.5) &= -0.75 < 0 \quad \Rightarrow \quad c \in [1.5, 2] \\ f(1.75) &= 0.0625 > 0 \quad \Rightarrow \quad c \in [1.5, 1.75] \\ f(1.625) &= -0.359375 < 0 \quad \Rightarrow \quad c \in [1.625, 1.75] \end{aligned}$$

This is called the "bisection method". Each time we test the midpoint and apply the IVT, we halve the width of the interval in which we've "trapped" our solution c. We initially know c is in $[0, 2]$ an interval of width 2, so after n iterations of the method, we know that c is in some interval of width $2 \times (1/2)^n = 1/2^{n-1}$ (check: after $n = 4$ iterations, we should have c trapped in an interval of width $1/8$ — correct!). Let's say we need to know c to within an error of 5×10^{-5}. Then we need to trap it in an interval of width less than 0.00005. How many iterations of the method do we need? The answer is any n for which

$$\frac{1}{2^{n-1}} < 0.00005.$$

will do. We know that such an n exists because the sequence $1/2^{n-1}$ converges to 0. If we allow ourselves to use logarithms, we can solve the above inequality. Otherwise, trial and error (substituting values of n) shows that $n = 16$ is the smallest positive integer satisfying the inequality. So, not only will the bisection method give us the (approximate) solution to the accuracy we require, we can figure out before we start how many iterations we'll need to make to get this accuracy.

By the way, we've just proved that $\sqrt{3}$ exists! We can easily generalize the argument above to show that every $x \geq 0$ has a square root $\sqrt{x} \geq 0$, and that this number is unique.

Proposition/Definition 8.9 *For each $x \in [0, \infty)$ there exists a unique $y \in [0, \infty)$ such that $y^2 = x$. We denote this number $\sqrt{x}$ and call it* **the square root** *of x.*

Proof: Consider the function $f : [0, \infty) \to \mathbb{R}$,

$$f(y) = y^2 - x,$$

noting that it is polynomial and hence continuous. Then $f(0) = -x \leq 0$ and

$$f(x + \frac{1}{2}) = \left(x + \frac{1}{2}\right)^2 - x = x^2 + \frac{1}{4} \geq \frac{1}{4},$$

so 0 is a number between $f(0)$ and $f(x+\frac{1}{2})$. Hence, by the Intermediate Value Theorem, there exists $y \in [0, x+\frac{1}{2}]$ such that $f(y) = 0$. Then $y^2 = x$, by the definition of f, so $\sqrt{x}$ exists. It remains to prove that it is unique.

Let $z \in [0, \infty)$ satisfy $f(z) = 0$ also. Then

$$0 = f(y) - f(z) = y^2 - z^2 = (y-z)(y+z),$$

so $z = y$ or $z = -y$. But since $z \geq 0$ and $y \geq 0$, $z = -y$ implies $z = 0 = y$. So $f(z) = f(y)$ implies that $z = y$. Hence the square root of x is unique. $\square$

We just proved that there exists a function

$$f : [0, \infty) \to [0, \infty), \qquad f(x) = \sqrt{x}.$$

What properties does it have? Is it injective? Surjective? Continuous? Can you **prove** your answers?

8.4 Maxima, minima and the Extreme Value Theorem

We have seen that continuous functions on closed bounded intervals $[a, b]$ have the very useful and fundamental "intermediate value" property: if they attain two values y_1 and y_2, they attain all values between y_1 and y_2. In this section we will prove they have another fundamental property: they are bounded (above and below), and attain both a maximum and a minimum value. Before proceeding further, we need to make all these terms precise. Recall that the **range** of a function $f : D \to E$ is the subset

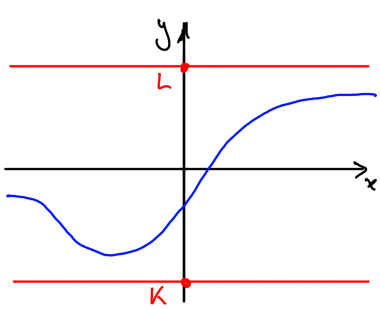
$$f(D) = \{f(x) \in E : x \in D\}$$

of E.

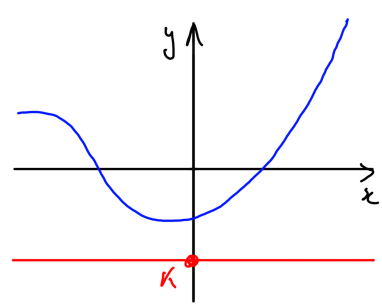
Definition 8.10 A function $f : D \to \mathbb{R}$ is **bounded above** if its range $f(D) \subseteq \mathbb{R}$ is bounded above, that is, if there exists $L \in \mathbb{R}$ such that, for all $x \in D$, $f(x) \leq L$. We call any such $L \in \mathbb{R}$ an **upper bound** on f.

Similarly, f is **bounded below** if $f(D) \subseteq \mathbb{R}$ is bounded below, that is, if there exists $K \in \mathbb{R}$ such that, for all $x \in D$, $f(x) \geq K$. We call any such $K \in \mathbb{R}$ a **lower bound** on f.

The function f is **bounded** if it is bounded above and below.



A bounded function.
K is a lower bound and L is an upper bound.



An unbounded function.
This has a lower bound (for example K)
but no upper bound.

Definition 8.11 A function $f : D \to \mathbb{R}$ **attains a maximum** at $c \in D$ if, for all $x \in D$, $f(x) \leq f(c)$. It **attains a minimum** at $c \in D$ if, for all $x \in D$, $f(x) \geq f(c)$.

Remark Note that these definitions have absolutely nothing to do with calculus! In fact, they make sense even if D is not a subset of $\mathbb{R}$. For example, D could be the set of books on my bookshelf, and f could be the map which assigns to each book x the number of pages it contains. Clearly this function will attain both a maximum and a minimum (my bookshelf will contain both a longest and shortest book), but it's absurd to try to define its derivative.

Note also that, if a function attains a maximum, at c say, it is certainly bounded above, by $f(c)$. Similarly, if it attains a minimum at b, it is bounded below by $f(b)$. But it's quite possible for a function to be bounded above but not attain a maximum, or bounded below and not attain a minimum. For example

$$f : (0, \infty) \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

is bounded below, by 0 for example, but does not attain a minimum: whatever $c \in (0, \infty)$ we try, there is always some other point $x \in (0, \infty)$ with $f(x) < f(c)$: we could take $x = c + 1$, for example.

If f is bounded above, then its range $f(D)$ is a nonempty subset of $\mathbb{R}$ that is bounded above which, by the Axiom of Completeness, has a supremum. We often refer to this number as the **supremum** of the function f. Similarly, if f is bounded below, we define its infimum to tbe the infimum of its range.

Finally, note that if f attains a maximum at c, then its supremum is precisely $f(c)$: for all $x \in D$, $f(x) \leq f(c)$, so $f(c)$ is an upper bound on f, and any number $K < f(c)$ is **not** an upper bound on f, since $f(c) > K$! Similarly, if f attains a minimum at b then its infimum is $f(b)$.

We may now prove our next big theorem:

Theorem 8.12 (Extreme Value Theorem) *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f is bounded and attains both a maximum and a minimum.*

Proof: We first prove that f is bounded above. Assume, towards a contradiction that f is *not* bounded above. Then every $n \in \mathbb{Z}^+$ is not an upper bound on f, so there exists $x_n \in [a, b]$ such that $f(x_n) > n$. But (x_n) is a bounded sequence (above by b, below by a), so has a convergent subsequence (x_{n_k}) by the Bolzano-Weiertsrass Theorem. Since $a \leq x_{n_k} \leq b$, its limit c, say, is also in $[a, b]$. Since $x_{n_k} \to c \in [a, b]$ and f is continuous (at c, in particular), $f(x_{n_k}) \to f(c)$. Since $f(x_{n_k})$ converges, it is bounded above. But $f(x_{n_k}) > n_k \geq k$, a contradiction. Hence, f is bounded above.

We now prove that f attains its supremum.

Since the set $A = \{f(x) : x \in [a, b]\}$ is bounded above, and is nonempty, it has a supremum L, say, by the Axiom of Completeness. Now, for each $n \in \mathbb{Z}^+$, $L - 1/n < L$, so is not an upper bound on A. Hence, for each $n \in \mathbb{Z}^+$, there exists $y_n \in [a, b]$ such that $f(y_n) > L - 1/n$. Clearly $f(y_n) \leq L$ so, by the Squeeze Rule, $f(y_n) \to L$. Since (y_n) is bounded, it too has a convergent subsequence y_{m_k} say, converging to some limit $d \in [a, b]$, by the Bolzano-Weiertsrass Theorem. Since f is continuous at d, $f(y_{m_k}) \to f(d)$. But $f(y_{m_k}) \to L$, since it is a subsequence of $f(y_n)$. Limits are unique, so $f(d) = L = \sup A$. Hence, for all $x \in [a, b]$, $f(x) \leq f(d)$, that is, f attains a maximum (at d).

It follows immediately that f attains a minimum also: let $g : [a, b] \to \mathbb{R}$, $g(x) = -f(x)$. This is continuous so, by the work above, attains a maximum at some point $d \in [a, b]$. But then f attains a minimum at d. $\square$

Note that all the hypotheses are important for this theorem. If $f : [a, b] \to \mathbb{R}$ is not continuous, it may fail to be bounded:

$$f : [0, 1] \to \mathbb{R}, \qquad f(x) = \begin{cases} \frac{1}{x}, & x \in (0, 1] \\ 0, & x = 0, \end{cases}$$

or be bounded but fail to attain a maximum or minimum:

$$f : [0, 1] \to \mathbb{R}, \qquad f(x) = \begin{cases} x, & x \in (0, 1) \\ 1/2, & x \in \{0, 1\}. \end{cases}$$

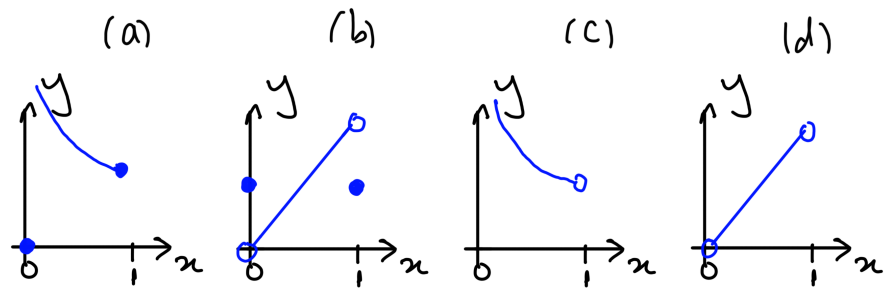
Similarly, if the domain is not a closed bounded interval, the conclusion may fail, even if f is continuous:

$$f : (0,1) \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

is unbounded, and

$$f : (0,1) \to \mathbb{R}, \qquad f(x) = x$$

is bounded but attains neither a maximum nor a minimum. These examples are depicted below:



(a) A function $f : [0,1] \to \mathbb{R}$ that is unbounded above. Note that f is discontinuous.

(b) A bounded function $f : [0,1] \to \mathbb{R}$ that attains neither a maximum nor a minimum value. Again, note that f is discontinuous.

(c) A continuous function $f : (0,1) \to \mathbb{R}$ that is unbounded above. Note that $(0,1)$ is not closed.

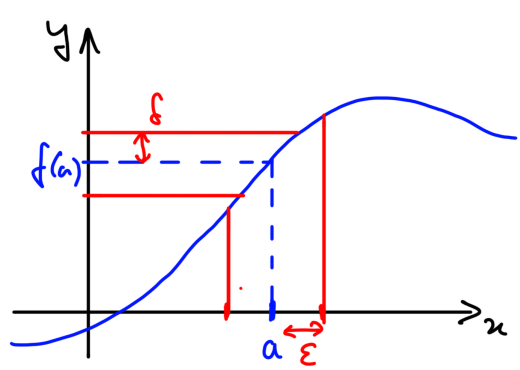
(d) A continuous function $f : (0,1) \to \mathbb{R}$ that is bounded, but attains neither a maximum nor a minimum value. Again, note that $(0,1)$ is not closed.

8.5 An equivalent definition of continuity

We have defined continuity using convergence of sequences: $f : D \to \mathbb{R}$ is continuous at $a \in D$ if, for all sequences (x_n) in D that converge to a, $f(x_n) \to f(a)$. Many mathematicians call this property "sequential continuity", and define continuity at $a \in D$ in a rather different way, by means of what is known as an "ε–δ criterion."

Definition 8.13 (The ϵ–δ definition of continuity) A function $f : D \to \mathbb{R}$ is **ϵ–δ continuous at a point** $a \in D$ if for each $\epsilon > 0$ there exists $\delta > 0$ so that, for all $x \in D$, if $|x - a| < \delta$ then

$$|f(x) - f(a)| < \epsilon \ .$$



It turns out that these two ways to define continuity are exactly equivalent.

Theorem 8.14 *A function $f : D \to \mathbb{R}$ is continuous at $a \in D$ if and only it is ϵ–δ continuous at a.*

Proof: Assume f is ϵ–δ continuous at $a \in D$, and let (x_n) be any sequence in D such that $x_n \to a$. We must show that $f(x_n) \to f(a)$, so f is continuous at a. We will do this by formulating a direct ε–N proof of convergence.

So, let $\varepsilon \in (0, \infty)$ be given. Since f is ϵ–δ continuous at a, there exists $\delta > 0$ such that, for all $x \in D$ with $|x - a| < \delta$, $|f(x) - f(a)| < \varepsilon$. Choose such a number $\delta > 0$. Since $x_n \to a$, by definition there exists $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $|x_n - a| < \delta$. But then, for all $n \geq N$, $x_n \in D$ and $|x_n - a| < \delta$, so $f(x_n) - f(a)| < \varepsilon$ (recall we chose δ specifically to ensure this was true!). Hence $f(x_n) \to f(a)$, as was to be proved.

Conversely, assume that f is *not* ε–δ continuous at $a \in D$. Then there exists some $\varepsilon_* > 0$ such that, for all $\delta > 0$, there is some $x \in D$ with $|x - a| < \delta$ such that $|f(x) - f(a)| > \varepsilon_*$. Such an $x \in D$ exists for each $\delta > 0$ so, in particular, for each $\delta = 1/n$, where $n \in \mathbb{Z}^+$, there exists $x_n \in D$ such that $|x_n - a| < 1/n$ but $|f(x_n) - f(a)| > \varepsilon_*$. Clearly $x_n \to a$ (by the Squeeze Rule) but, equally clearly $f(x_n) \nrightarrow f(a)$ (it fails the ε–N criterion for convergence when $\varepsilon = \varepsilon_*$). Hence f is not continuous at a. $\square$

Exercise 8.15 Prove that the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is ϵ–δ continuous everywhere on $\mathbb{R}$. Is this easier, or harder, than proving that it's continuous using our original (sequential) definition of continuity?

Summary

- A function $f : D \to \mathbb{R}$ is **continuous at** $a \in D$ if, given any sequence (x_n) in D converging to a, $f(x_n)$ converges to $f(a)$.
- A function $f : D \to \mathbb{R}$ is **continuous** if it is continuous at every point $a \in D$.
- Sums, products, quotients and compositions of continuous functions are continuous. It follows that all polynomials and rational functions (quotients of polynomials) are continuous.
- A second equivalent definition of continuity is the following: $f : D \to \mathbb{R}$ is continuous at a point $a \in D$ if for any $\epsilon > 0$ there is a $\delta > 0$ so that for any $x \in D$, if $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$.
- The **Intermediate Value Theorem**: If $f : [a, b] \to \mathbb{R}$ is continuous and $y \in \mathbb{R}$ is a number between $f(a)$ and $f(b)$ then there exists $c \in [a, b]$ such that $f(c) = y$.
- We can use the Intermediate Value Theorem to show that square (and other) roots exist. In fact, for any $p \in \mathbb{Z}^+$, the function $f(x) = x^{1/p}$ is continuous on $[0, \infty)$.
- We can also use Intermediate value theorem to see that every cubic polynomial has at least one root.
- The **Extreme Value Theorem**: If $f : [a, b] \to \mathbb{R}$ is continuous, then f is bounded and attains both a maximum and a minimum value.
- Many mathematicians define continuity differently, using an ε–δ criterion. This definition is exactly equivalent to ours.

9 Differentiable functions

9.1 The main definition

OK, let's build calculus. Now that we have a rigorous definition of the limit of a function, it is straightforward to define derivatives.

Definition 9.1 Let $f : D \to \mathbb{R}$, where D is some subset of $\mathbb{R}$, and $a \in D$ be a cluster point of D. Then f is **differentiable at** a if the limit

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

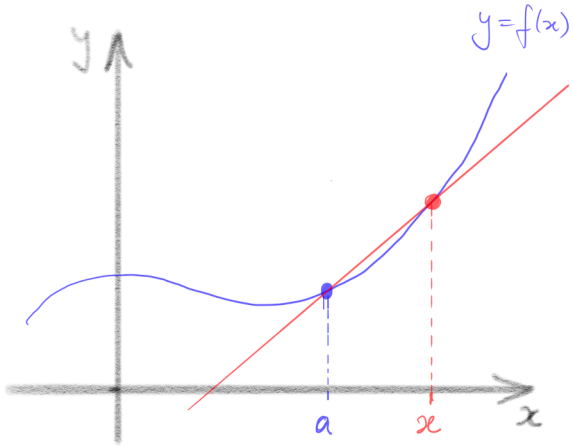
exists. In this case, we denote the limit $f'(a)$ and call it the **derivative of** f **at** a. We say that f is **differentiable** if it is differentiable at a for all $a \in D$.

Remarks

- In general, in order to define $\lim_{x \to a} g(x)$, we only need a to be a *cluster point* of the domain of g: it isn't necessary in general for a to be in the domain of g, so $g(a)$ may or may not exist. For example
$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$
exists, although the function is undefined at $x = 1$. Note, however, that the limit defining $f'(a)$ contains the number $f(a)$, so to be differentiable at a, the point a must be both an *element* and a *cluster point* of the domain of f. For example, a function $f : (0, \infty) \to \mathbb{R}$ cannot be differentiable at 0 (since $0 \notin (0, \infty)$), and a function $f : \mathbb{Z} \to \mathbb{R}$ cannot be differentiable anywhere (since $\mathbb{Z}$ has no cluster points).

- The limit in Definition 9.1 is defined as in Definition 7.1. That is, $f : D \to \mathbb{R}$ is differentiable at a if for all sequences (x_n) in $D \backslash \{a\}$ which converge to a,
$$\frac{f(x_n) - f(a)}{x_n - a} \to f'(a).$$

- Geometrically, the quantity
$$s_n = \frac{f(x_n) - f(a)}{x_n - a}$$
is the slope of the straight line passing through the points $(a, f(a))$ and $(x_n, f(x_n))$. Such a straight line is called a **chord** on the graph $y = f(x)$. As x_n approaches a, the points defining the chord get "arbitrarily close" to one another, and the chords approach a straight line through $(a, f(a))$ with slope $f'(a)$, the so-called **tangent line**. This is why the derivative $f'(a)$ is often interpreted as the slope of the graph $y = f(x)$ at the point $(a, f(a))$. Note, however, that this is just an *interpretation*. The *definition* of $f'(a)$ is in terms of limits which are, in turn, defined precisely in Definition 7.1.



- If $f : D \to \mathbb{R}$ is differentiable, then its derivative is also a function $f' : D \to \mathbb{R}$, mapping D to $\mathbb{R}$. There is a very popular alternative notation for the derivative in this case, namely

$$f'(x) = \frac{df}{dx}.$$

  This notation is convenient in some circumstances, but it tends to blur the distinction between a function (in this case f') and the *value* of the function at a particular point (in this case $f'(x)$), so we will tend to avoid it.

Let's verify that some simple, familiar functions have the derivatives we expect.

Example 9.2 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = c$, a constant. Then f is differentiable (everywhere) and $f'(a) = 0$ for all $a \in \mathbb{R}$.

Proof: Choose any $a \in \mathbb{R}$ and let x_n be any sequence in $\mathbb{R}\backslash\{a\}$ which converges to a. Then

$$\frac{f(x_n) - f(a)}{x_n - a} = \frac{c - c}{x_n - a} = 0 \to 0.$$

Hence

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = 0$$

as was to be proved. □

Example 9.3 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x$. Then f is differentiable (everywhere) and $f'(a) = 1$ for all $a \in \mathbb{R}$.

Proof: Choose any $a \in \mathbb{R}$ and let x_n be any sequence in $\mathbb{R}\backslash\{a\}$ which converges to a. Then

$$\frac{f(x_n) - f(a)}{x_n - a} = \frac{x_n - a}{x_n - a} = 1 \to 1.$$

Hence

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = 1$$

as was to be proved. □

Example 9.4 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2$. Then f is differentiable and $f'(a) = 2a$ for all $a \in \mathbb{R}$.

Proof: Choose any $a \in \mathbb{R}$ and let x_n be any sequence in $\mathbb{R}\backslash\{a\}$ which converges to a. Then

$$\frac{f(x_n) - f(a)}{x_n - a} = \frac{x_n^2 - a^2}{x_n - a} = \frac{(x_n - a)(x_n + a)}{x_n - a} = x_n + a \to a + a.$$

Hence

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = 2a$$

as was to be proved. □

Example 9.5 Let $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ such that $f(x) = 1/x$. Then f is differentiable and $f'(a) = -1/a^2$ for all $a \in \mathbb{R}\backslash\{0\}$.

Proof: Choose any $a \neq 0$ and let x_n be any sequence in $\mathbb{R}\backslash\{0, a\}$ which converges to a. Then

$$\frac{f(x_n) - f(a)}{x_n - a} = \frac{1/x_n - 1/a}{x_n - a} = \frac{a - x_n}{x_n a(x_n - a)} = -\frac{1}{x_n a} \to -\frac{1}{a^2}$$

by the Algebra of Limits (Theorem 2.15). Hence

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} = -\frac{1}{a^2}$$

as was to be proved.

□

OK, all is as we expected. What about functions which are *not* differentiable? How do we *prove* they aren't? A function $f : D \to \mathbb{R}$ is *not* differentiable at a precisely if the limit

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

does not exist. To establish this. it suffices to find **one** example of a sequence (x_n) in $D\backslash\{a\}$ which converges to a for which

$$\frac{f(x_n) - f(a)}{x_n - a}$$

does **not** converge.

Example 9.6 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = |x|$. Then f is not differentiable at 0.

Proof: Assume, towards a contradiction, that f is differentiable at 0 with derivative $f'(0)$. Then

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = f'(0).$$

Let $x_n = (-1)^n/n$. Note that this is a sequence in $\mathbb{R}\backslash\{0\}$ converging to 0. Hence,

$$\frac{f(x_n) - f(0)}{x_n - 0} \to f'(0).$$

But

$$\frac{f(x_n) - f(0)}{x_n - 0} = \frac{|(-1)^n/n|}{(-1)^n/n} = \frac{1}{(-1)^n} = (-1)^n$$

which does not converge, a contradiction. Hence f is *not* differentiable at 0. $\square$

Remark Alternatively, we could have defined $y_n = 1/n$ and $z_n = -1/n$ and noted that both are sequences in $\mathbb{R}\backslash\{0\}$ converging to 0, so if f is differentiable at 0, then both

$$\frac{f(y_n) - f(0)}{y_n - 0} \quad \text{and} \quad \frac{f(z_n) - f(0)}{z_n - 0}$$

must converge to $f'(0)$. But the first converges to 1, while the second converges to -1, which contradicts the uniqueness of limits (Theorem 7.5).

Exercise 9.7 Prove that $f : [0, \infty) \to \mathbb{R}$, $f(x) = \sqrt{x}$ is **not** differentiable at 0. If you get stuck, it may help to watch this [VIDEO]

There is a link between differentiablity and continuity, as we now show.

Proposition 9.8 *Let $f : D \to \mathbb{R}$ be differentiable at $a \in D$. Then f is continuous at a.*

Proof: Let (x_n) be any sequence in D which converges to a. Choose any sequence (y_n) in $D\backslash\{a\}$ which converges to a (such a sequence must exist since a is a cluster point of D) and let

$$z_n := \begin{cases} x_n & \text{if } x_n \neq a \\ y_n & \text{if } x_n = a. \end{cases}$$

I leave it as an exercise to prove that $z_n \to a$. Now

$$|f(x_n) - f(a)| = \left\{ \begin{array}{ll} \left|\frac{f(z_n)-f(a)}{z_n - a}\right| |z_n - a| & \text{if } x_n \neq a \\ 0 & \text{if } x_n = a \end{array} \right\} \leq \left| \frac{f(z_n) - f(a)}{z_n - a} \right| |z_n - a|.$$

Since f is differentiable at a and $z_n \to a$,

$$\frac{f(z_n) - f(a)}{z_n - a} \to f'(a).$$

Hence, by the Algebra of Limits,

$$\left| \frac{f(z_n) - f(a)}{z_n - a} \right| |z_n - a| \to |f'(a)| \times 0 = 0,$$

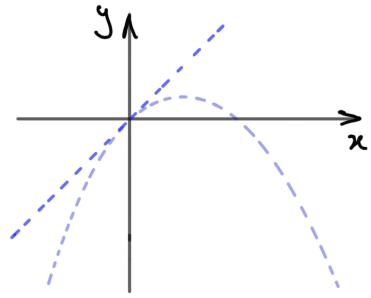
so, by the Squeeze Rule $|f(x_n) - f(a)| \to 0$, that is, $f(x_n) \to f(a)$. Since this holds for all such sequences (x_n), f is continuous at a. $\square$

Remark. The converse of Proposition 9.8 is false: if f is continuous at a, it does not follow that f is differentiable there. We've already seen a counterexample: $f(x) = |x|$ is continuous at 0 but is not differentiable at 0 (see Example 9.6). In this case, the function fails to be differentiable only at a single isolated point. It is straightforward to construct functions which are *differentiable* only at a single isolated point.

Example 9.9 Let $f : \mathbb{R} \to \mathbb{R}$ be defined such that

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ x - x^2 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

I claim that f is differentiable at 0, with $f'(0) = 1$, but is not differentiable anywhere else.



Proof: Let $a \in \mathbb{R}$ and assume that f is differentiable at a. Then, by Proposition 9.8, f is continuous at a. For each $n \in \mathbb{Z}^+$ there exist $r_n \in \mathbb{Q}$ and $i_n \in \mathbb{R}\backslash\mathbb{Q}$ such that $a < r_n < a + 1/n$ and $a < i_n < a + 1/n$. Clearly $r_n \to a$ and $i_n \to a$ by the Squeeze Rule. Hence, by the definition of continuity, $f(r_n) \to f(a)$ and $f(i_n) \to f(a)$. But r_n is rational, so $f(r_n) = r_n \to a$, and i_n is irrational, so $f(i_n) = i_n - i_n^2 \to a - a^2$. Both these limits equal $f(a)$, so $a = a - a^2$, whence $a = 0$. So *if* f is differentiable at a, then $a = 0$. Equivalently, f is *not* differentiable at any $a \neq 0$.

It remains to show that f is differentiable at 0 with $f'(0) = 1$, that is,

$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = 1.$$

So, let (x_n) be any sequence in $\mathbb{R}\backslash\{0\}$ converging to 0, and consider

$$s_n = \frac{f(x_n) - f(0)}{x_n - 0} = \frac{f(x_n)}{x_n}.$$

We must show that $s_n \to 1$. If x_n is rational, then $|s_n - 1| = |x_n/x_n - 1| = 0$, whereas if x_n is irrational then

$$|s_n - 1| = \left| \frac{x_n - x_n^2}{x_n} - 1 \right| = |x_n|.$$

Hence, for all n, $0 \leq |s_n - 1| \leq |x_n|$, so $|s_n - 1| \to 0$ by the Squeeze Rule. Hence $s_n \to 1$. It follows that $f'(0) = 1$, as claimed. □

So this (admittedly rather bizarre) function is differentiable at 0, and has a positive derivative there. Naive intuition would suggest, therefore, that the function should be increasing, at least for values of x sufficiently close to 0. In fact, this is **false!**

Example 9.9 continued Claim: the function f is not increasing on any neighbourhood of 0. That is, there does not exist $\varepsilon > 0$ such that f is increasing on $(-\varepsilon, \varepsilon)$.
Proof: Let $\varepsilon > 0$ be given. Then there exists an irrational number x such that $0 < x < \varepsilon$. By definition, $f(x) = x - x^2 < x$. Similarly, there exists a *rational* number y such that $x - x^2 < y < x$. But then $y < x$ and $f(y) = y > x - x^2 = f(x)$. Hence, f is not increasing on the interval $(-\varepsilon, \varepsilon)$. This is true no matter which positive number ε we choose, which establishes the claim. □

9.2 The rules of differentiation

In principle, if we are given an explicit function, like

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = (x^3 + 1)^2,$$

we can show it is differentiable and compute its derivative by directly applying Definition 9.1. This quickly becomes complicated and tedious, however. So we next develop some "derivative theorems" which will allow us to reduce differentiation of many functions to an algorithmic process. You are (I would hope) already familiar with these and quite fluent in applying them. The point of this section is to show that, now we have a mathematically precise definition of the derivative, we can rigorously *prove* them, thus putting differential calculus on a solid foundation.

Proposition 9.10 (Linearity) *Let $f : D \to \mathbb{R}$, $g : D \to \mathbb{R}$ be differentiable at $a \in D$ with derivatives $f'(a)$ and $g'(a)$, respectively, and c be a real constant. Then*

(i) $cf : D \to \mathbb{R}$ is differentiable at a with derivative $cf'(a)$.

(ii) $f + g : D \to \mathbb{R}$ is differentiable at a with derivative $f'(a) + g'(a)$.

Proof: These both follow immediately from the Algebra of Limits for functions (Theorem 7.10):

$$\begin{aligned}
\lim_{x\to a} \frac{cf(x) - cf(a)}{x - a} &= \lim_{x\to a} c\left(\frac{f(x) - f(a)}{x - a}\right) = cf'(a) \\
\lim_{x\to a} \frac{f(x) + g(x) - (f(a) + g(a))}{x - a} &= \lim_{x\to a} \left(\frac{f(x) - f(a)}{x - a} + \frac{g(x) - g(a)}{x - a}\right) = f'(a) + g'(a)
\end{aligned}$$

□

Combining parts (i) and (ii) of Proposition 9.10, we see that *linear combinations* of differentiable functions are differentiable: for f, g differentiable at a and any constants α, β,

$$h(x) = \alpha f(x) + \beta g(x)$$

is differentiable at a, with derivative

$$h'(a) = \alpha f'(a) + \beta g'(a).$$

What about *products* of differentiable functions? If f and g are differentiable at a, is $h(x) = f(x)g(x)$ differentiable at a? And if so, is $h'(a) = f'(a)g'(a)$? Yes (to the first) and no (to the second):

Proposition 9.11 (Product Rule) *Let $f : D \to \mathbb{R}$, $g : D \to \mathbb{R}$ be differentiable at $a \in D$ with derivatives $f'(a)$ and $g'(a)$, respectively. Then $fg : D \to \mathbb{R}$ is differentiable at a with derivative $f'(a)g(a) + f(a)g'(a)$.*

Proof: Let $h : D \to \mathbb{R}$, $h(x) = f(x)g(x)$. We must show that

$$\lim_{x\to a} \frac{h(x) - h(a)}{x - a} = f'(a)g(a) + f(a)g'(a).$$

So, let (x_n) be any sequence in $D\backslash\{a\}$ converging to a. Then

$$\begin{aligned}
\frac{h(x_n) - h(a)}{x_n - a} &= \frac{f(x_n)g(x_n) - f(a)g(a)}{x_n - a} \\
&= \frac{f(x_n)g(x_n) - f(a)g(x_n) + f(a)g(x_n) - f(a)g(a)}{x_n - a} \\
&= \frac{f(x_n) - f(a)}{x_n - a} g(x_n) + f(a)\frac{g(x_n) - f(a)g(a)}{x_n - a}.
\end{aligned}$$

Now f is differentiable at a, so

$$\frac{f(x_n) - f(a)}{x_n - a} \to f'(a).$$

Similarly, g is differentiable at a, so

$$\frac{g(x_n) - g(a)}{x_n - a} \to g'(a).$$

Furthermore, since g is differentiable at a, it is continuous at a, so

$$g(x_n) \to g(a).$$

Hence, by the Algebra of Limits (for sequences, Theorem 2.15),

$$\frac{h(x_n) - h(a)}{x_n - a} \to f'(a) \times g(a) + f(a) \times g'(a).$$

Since this holds for all such sequences (x_n), the claim is proved. □

Having established the Linearity Property and the Product Rule, it's not hard to prove that all polynomial functions are differentiable, with the derivative we expect.

Proposition 9.12 *Every polynomial function $p : \mathbb{R} \to \mathbb{R}$, $p(x) = a_0 + a_1 x + \cdots + a_m x^m$ is differentiable, and its derivative is another polynomial function $p' : \mathbb{R} \to \mathbb{R}$, namely*

$$p'(x) = a_1 + 2a_2 x + \cdots + m a_m x^{m-1}.$$

Proof: Exercise. Try proof by induction (on the degree of p). If you get stuck, watch this [VIDEO] □

Propositions 9.10 and 9.11 followed easily from the Algebra of Limits. Our next differentiation rule, and its proof, are considerably more subtle. Recall that the *composition* of two functions $f : A \to B$ and $g : B \to C$, denoted $g \circ f : A \to C$ is the function which first "does" f, then feeds the result into g, that is

$$(g \circ f)(x) = g(f(x)).$$

This only makes sense if the range of f is a subset of the domain of g. We've already proved that the composition of two continuous functions is continuous (Theorem 8.5). There's a very famous rule which tells us that the composition of two *differentiable* functions is *differentiable*, and gives a formula for its derivative.

Theorem 9.13 (Chain Rule) *Let $f : D \to E$ be differentiable at $a \in D$ and $g : E \to \mathbb{R}$ be differentiable at $f(a) \in E$. Then $g \circ f : D \to \mathbb{R}$ is differentiable at a, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Proof: We consider two cases separately: (i) $f'(a) \neq 0$ and (ii) $f'(a) = 0$.

(i) Let (x_n) be any sequence in $D \backslash \{a\}$ which converges to a and consider the sequence

$$s_n = \frac{(g \circ f)(x_n) - (g \circ f)(a)}{x_n - a}.$$

We must show that $s_n \to g'(f(a))f'(a)$.

I claim there must exist $N \in \mathbb{Z}^+$ such that, for all $n \geq N$, $f(x_n) \neq f(a)$. (More informally, there is a point in the sequence $f(x_n)$ after which it never takes the value $f(a)$.) For if not, there is a subsequence (x_{n_k}) with $f(x_{n_k}) = f(a)$ for all k, which would imply that

$$\frac{f(x_{n_k}) - f(a)}{x_{n_k} - a} = 0 \to 0.$$

But $x_{n_k} \to a$ so the above sequence must converge to $f'(a)$ which is **not** 0, a contradiction.

Now, consider the *tail* of the sequence (s_n) starting at its $(N+1)^{\text{th}}$ term, that is, the subsequence (s_{N+n}). Recall that, by the Tail Lemma (Lemma 4.5), if (s_{N+n}) converges, then (s_n) converges to the same limit. So it suffices to show that $s_{N+n} \to g'(f(a))f'(a)$. But

$$s_{N+n} = \frac{g(f(x_{n+N})) - g(f(a))}{f(x_{n+N}) - f(a)} \frac{f(x_{n+N}) - f(a)}{x_{n+N} - a},$$

where we have used the fact that for all $n \in \mathbb{Z}^+$, $f(x_{N+n}) \neq f(a)$. Since f is differentiable at a,

$$\frac{f(x_{n+N}) - f(a)}{x_{n+N} - a} \to f'(a),$$

and f is continuous at a (Proposition 9.8), so $f(x_{n+N}) \to f(a)$. But g is differentiable at $f(a)$, so for all sequences (y_n) in $E \backslash \{f(a)\}$ which converge to $f(a)$,

$$\frac{g(y_n) - g(f(a))}{y_n - f(a)} \to g'(f(a)).$$

Using this fact in the case $y_n = f(x_{N+n})$, we see that

$$\frac{g(f(x_{n+N})) - g(f(a))}{f(x_{n+N}) - f(a)} \frac{f(x_{n+N}) - f(a)}{x_{n+N} - a} \to g'(f(a)).$$

Hence

$$s_{N+n} \to g'(f(a))f'(a)$$

by the Algebra of Limits.

(ii) Assume $f'(0) = 0$. Further, assume towards a contradiction that

$$\lim_{x \to a} \frac{(g \circ f)(x) - (g \circ f)(a)}{x - a} \neq 0.$$

Then there exists a sequence (x_n) in $D \backslash \{a\}$ which converges to a such that

$$s_n = \frac{(g \circ f)(x_n) - (g \circ f)(a)}{x_n - a} \nrightarrow 0.$$

But recall (Example 6.13) this implies there exists a constant $\varepsilon > 0$ and a subsequence (x_{n_k}) such that $|s_{n_k}| \geq \varepsilon$ for all k. Clearly, for all k, $f(x_{n_k}) \neq f(a)$ (since otherwise $s_{n_k} = 0$). Hence

$$s_{n_k} = \frac{g(f(x_{n_k})) - g(f(a))}{f(x_{n_k}) - f(a)} \frac{f(x_{n_k}) - f(a)}{x_{n_k} - a} \to g'(f(a))f'(a) = 0$$

by the Algebra of Limits. But $|s_{n_k}| \geq \varepsilon$ for all k, a contradiction. □

Let's do an example where we compute a derivative using the Chain Rule. The point is not so much to calculate the derivative, but rather to show how the Chain Rule (now that we've established it as a theorem) gives us a *proof* that the claimed formula is correct.

Example 9.14 Let $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = (x^2 + 1)^2$. Then f is differentiable and $f'(x) = 4(x^2 + 1)x$.

Proof: $f = h \circ g$ where $h, g : \mathbb{R} \to \mathbb{R}$

$$\begin{aligned} g(x) &= x^2 + 1 \\ h(x) &= x^2. \end{aligned}$$

Now, by Examples 9.4 and 9.2 and the Linearity Property (Proposition 9.10) g, h are differentiable at every $x \in \mathbb{R}$, with derivatives

$$\begin{aligned} g'(x) &= 2x + 0 \\ h'(x) &= 2x. \end{aligned}$$

Hence, by the Chain Rule, f is differentiable at every $x \in \mathbb{R}$, and

$$f'(x) = h'(g(x))g'(x) = 2g(x)2x = 4(x^2 + 1)x.$$

□

Exercise 9.15 Let $g : \mathbb{R} \to \mathbb{R}$ be $g(x) = f(f(\sqrt{2}x))$, where $f : \mathbb{R} \to \mathbb{R}$ is the bizarre function defined in Example 9.9. Prove that g is differentiable at 0 and compute $g'(0)$. Check you answer by watching this [VIDEO]

Example 9.16 Let m be a positive integer and $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ be the function $f(x) = 1/x^m$. Then f is differentiable, and $f'(x) = -m/x^{m+1}$.

Proof: $f = g \circ h$ where $g : \mathbb{R}\backslash\{0\} \to \mathbb{R}\backslash\{0\}$, $g(x) = 1/x$ and $h : \mathbb{R}\backslash\{0\} \to \mathbb{R}\backslash\{0\}$, $h(x) = x^m$. By Examples 9.5 and 9.12, both g and h are differentiable with derivatives

$$\begin{aligned} g'(x) &= -\frac{1}{x^2} \\ h'(x) &= mx^{m-1}. \end{aligned}$$

Hence, by the Chain Rule, f is diffentiable, and

$$\begin{aligned} f'(x) &= g'(h(x))h'(x) = -\frac{1}{h(x)^2} mx^{m-1} \\ &= -\frac{mx^{m-1}}{x^{2m}} \\ &= -\frac{m}{x^{m+1}} \end{aligned}$$

by Example 9.5 and Proposition 9.12. □

Generalizing the trick used in this proof, we can obtain another useful rule of differentiation:

Proposition 9.17 (Quotient Rule) *Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}\backslash\{0\}$, be differentiable at $a \in D$. Then f/g is differentiable at a and*

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Proof: Exercise. If you get stuck, watch this [VIDEO] □

So we now know that the function $f(x) = x^k$ is differentiable for all integers k (everywhere it is well-defined, i.e. everywhere except 0 if $k < 0$), and that its derivative is $f'(x) = kx^{k-1}$.

Remark You are probably more familiar with the Chain Rule expressed something like the following:

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}. \qquad (*)$$

Certainly, this has a look of plausibility about it (don't the "du's" just cancel?) and is arguably more memorable than

$$(g \circ f)'(a) = g'(f(a))f'(a). \qquad (\dagger)$$

Nonetheless, it is vastly inferior. Ask yourself what, exactly, does $(*)$ mean? The derivative of a function y (where? At x?) equals the derivative (where? At u?) of the same (?) function – but with respect to a different variable, whatever that means — times the derivative of some other function u (but isn't u a variable? Can we use the same symbol to denote both a function and a variable?) evaluated somewhere (x? Maybe?). Note that "the function y thought of as a function of u instead of x," where "u is a function of x," is actually a *different* function! And to interpret $(*)$ we're forced to talk about how we *think* about the function y, instead of what it actually *is*. This is the hallmark of imprecise, badly formulated mathematics.

By contrast (†) is a completely unambiguous, precise statement standing on its own. We don't need to explain separately "how we're thinking of" the terms it contains in order to make sense of it. Be warned: if ever I ask for a statement of the Chain Rule, it is Theorem 9.13 I want, not wooly, ambiguous stuff in the style of (∗). □

Summary

- A function $f : D \to \mathbb{R}$ is **differentiable at** $a \in D$ if

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

  exists. In this case, we denote the limit $f'(a)$ and call it the **derivative** of f at a. We say that f is **differentiable** if it is differentiable at a for all $a \in D$.

- The limit in $f'(a)$ is defined as in Definition 7.1.

- If a function is differentiable at a, it is continuous at a. The converse is false.

- Using this definition of $f'(a)$, we can prove that derivatives obey the usual rules of differential calculus, to wit:

$$\begin{array}{lrcl}
\text{Linearity:} & (\alpha f + \beta g)'(a) & = & \alpha f'(a) + \beta g'(a), \\
\text{Product Rule:} & (fg)'(a) & = & f'(a)g(a) + f(a)g'(a), \\
\text{Quotient Rule:} & (f/g)'(a) & = & \dfrac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}, \\
\text{Chain Rule:} & (g \circ f)'(a) & = & g'(f(a))f'(a).
\end{array}$$

10 The Mean Value Theorem

10.1 Maxima and minima

Recall (Definition 8.11) that a function $f : D \to \mathbb{R}$ attains a **maximum** at $c \in D$ if, for all $x \in D$, $f(x) \leq f(c)$, and attains a **minimum** at $d \in D$ if, for all $x \in D$, $f(x) \geq f(d)$. These definitions don't make any reference to the derivative of f. You may find this surprising: after all, in your previous studies of calculus, you've been trained, when asked to find the maximum or minimum values of a function, to find and examine its critical points (also called "turning" or "stationary" points), that is points x were $f'(x) = 0$. The next theorem[6] explains why.

Theorem 10.1 (Interior Extremum Theorem) *Let $f : (a, b) \to \mathbb{R}$ be differentiable, and attain a maximum or a minimum at $c \in (a, b)$. Then $f'(c) = 0$.*

Proof: Assume first that f attains a maximum at $c \in (a, b)$. Consider the sequences

$$x_n = c - \frac{(c-a)}{n+1} \qquad \text{and} \qquad y_n = c + \frac{b-c}{n+1}.$$

These are sequences in $(a, b)\backslash\{c\}$ which converge to c, so

$$s_n = \frac{f(x_n) - f(c)}{x_n - c} \to f'(c) \qquad \text{and} \qquad t_n = \frac{f(y_n) - f(c)}{y_n - c} \to f'(c)$$

by the definition of $f'(c)$. But f attains a maximum at c, so $f(x_n) \leq f(c)$ and $f(y_n) \leq f(c)$, and $x_n < c$ while $y_n > c$. Hence $s_n \geq 0$ (it's a non-positive number divided by a negative number) while $t_n \leq 0$ (it's a non-positive number divided by a positive number). But then

$$\lim_{n\to\infty} s_n \geq 0 \qquad \text{and} \qquad \lim_{n\to\infty} t_n \leq 0$$

by Theorem 2.14. Hence $f'(c) \geq 0$ and $f'(c) \leq 0$, so $f'(c) = 0$.

If f attains a minimum at c, we apply the argument above to $g(x) = -f(x)$, which attains a maximum at c. Then $g'(c) = 0$, so $f'(c) = -g'(c) = -0$. □

Example 10.2 Find the maximum and minimum values attained by

$$f : [0, 2] \to \mathbb{R}, \qquad f(x) = x^3 - x^2.$$

Note that f does attain both a maximum and a minimum, by the Extreme Value Theorem (the domain is a closed bounded interval, and the function is continuous).

Solution: Either the extrema occur at the endpoints $\{0, 2\}$ or one or both occur in $(0, 2)$. If an extremum occurs at $c \in (0, 2)$ then $f(c)$ also an extremum of the function restricted to $(0, 2)$, so $f'(c) = 0$ by the Interior Extremum Theorem. Hence, the extrema occur either at an endpoint 0, 2, or at some interior critical point, where $f'(c) = 0$. Now

$$f'(c) = 3c^2 - 2c = c(3c - 2) = 0$$

if $c = 0$ or $c = 2/3$ (in this case, one of the critical points is actually also an endpoint). Now

$$f(0) = 0, \qquad f(2/3) = -\frac{4}{27}, \qquad f(2) = 4.$$

Hence, the maximum value of f is 4, attained at the endpoint 2, and the minimum value is $-\frac{4}{27}$ attained at the interior critical point $\frac{2}{3}$. Note that in this case, the critical point 0 (also an endpoint) is neither a maximum nor a minimum of f. □

[6] Extremum means "maximum or minimum". Its plural is "extrema."

Remark 10.3 Having shown that, for all $x \in [0,2]$, $-4/27 = f(2/3) \leq f(x) \leq f(2) = 4$, it follows from the Intermediate Value Theorem that the range of this function is

$$f([0,2]) = [-4/27, 4].$$

Do you see why?

We can combine the Interior Extremum Theorem with the Extreme Value Theorem to give a surprisingly powerful result traditionally called Rolle's Theorem:

Theorem 10.4 (Rolle's Theorem) *Let f be a real function which is continuous on $[a,b]$ and differentiable on (a,b), and assume that $f(a) = f(b)$. Then there exists $c \in (a,b)$ such that $f'(c) = 0$.*

Proof: By the Extreme Value Theorem, f attains both a maximum and a minimum on $[a,b]$. If both the maximum and the minimum occur at endpoints of $[a,b]$, then since $f(a) = f(b)$, the maximum value equals the minimum value, so f is constant on $[a,b]$ and hence $f'(c) = 0$ for *all* $c \in (a,b)$, and we're done. Otherwise, either the maximum or the minimum occurs at an interior point $c \in (a,b)$. But then $f'(c) = 0$ by the Interior Extremum Theorem. □

10.2 Increasing and decreasing functions

It is common to interpret the derivative of a function f as the **rate of change** of $f(x)$ as its input variable x is increased. So, if $f'(a) > 0$, we expect that f is an increasing function, at least for x sufficiently close to a. We've already seen that this naive expectation can actually be wrong in general (see Example 9.9). To convert information about the sign of f' into information about the increasing or decreasing behaviour of f, we need to be a bit more careful. We start (of course!) by carefully defining our terms.

Definition 10.5 A function $f : D \to \mathbb{R}$ (where $D \subseteq \mathbb{R}$) is **increasing** if, for all $x, y \in D$, if $y > x$ then $f(y) \geq f(x)$. It is **strictly increasing** if, for all $x, y \in D$, if $y > x$ then $f(y) > f(x)$. It is **decreasing** if, for all $x, y \in D$, if $y > x$ then $f(y) \leq f(x)$, and **strictly decreasing** if, for all $x, y \in D$, if $y > x$ then $f(y) < f(x)$.

Remark Note once again that these definitions have absolutely nothing to do with calculus! They use only the ordering relation on $\mathbb{R}$, and make sense even if the function is not differentiable (or, indeed, continuous). Note also that a function can be both increasing and decreasing – but only if it is constant! (Can you see why?)

Example 10.6 The step function

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \begin{cases} 0, & x < 0. \\ 1, & x \geq 0. \end{cases}$$

is an increasing function.

The key result allowing us to deduce information about the decreasing/increasing behaviour of differentiable functions is called the Mean Value Theorem.

Theorem 10.7 (Extended Mean Value Theorem) *Let f and g be real functions which are continuous on $[a,b]$ and differentiable on (a,b), and assume that, for all $x \in (a,b)$, $g'(x) \neq 0$. Then there exists $c \in (a,b)$ such that*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof: First note that the conditions on g imply that $g(a) \neq g(b)$ (since if $g(b) = g(a)$ then $g'(x) = 0$ for some $x \in (a, b)$ by Rolle's Theorem). Hence,

$$\alpha = \frac{f(b) - f(a)}{g(b) - g(a)}$$

is a well-defined real constant. Let $h : [a, b] \to \mathbb{R}$ such that $h(x) = f(x) - \alpha g(x)$. Then h is continuous on $[a, b]$ and differentiable on (a, b), and, as one may easily verify,

$$h(a) = \frac{f(a)g(b) - g(a)f(b)}{g(b) - g(a)} = h(b).$$

Hence h satisfies the hypotheses of Rolle's Theorem, and we conclude that there exists $c \in (a, b)$ such that $h'(c) = 0$. But then $f'(c) = \alpha g'(c)$, whence the result immediately follows. □

The name "Mean Value Theorem" is usually reserved for a special case of Theorem 10.7:

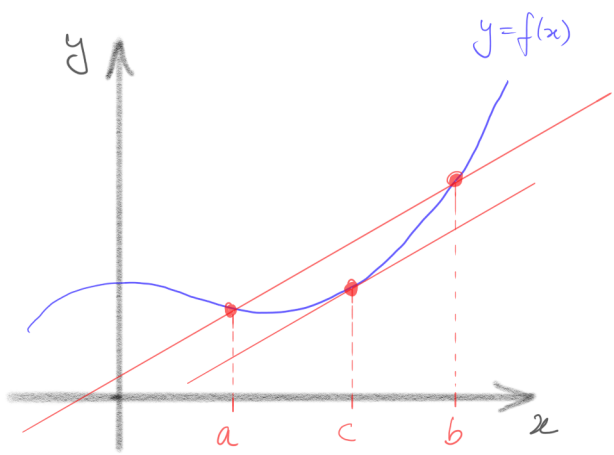
Corollary 10.8 (Mean Value Theorem) *Let f be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

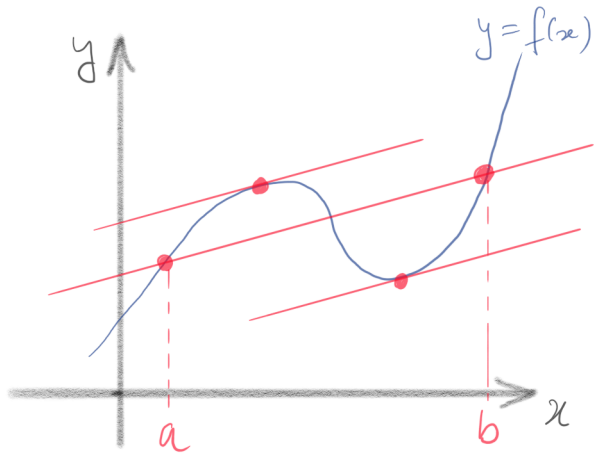
Proof: Just apply Theorem 10.7 in the case where $g : [a, b] \to \mathbb{R}$, $g(x) = x$. □

Remarks

- The Mean Value Theorem has a nice geometric interpretation. Consider the graph $y = f(x)$ of a differentiable function $f : \mathbb{R} \to \mathbb{R}$. For each pair of distinct numbers $a < b$, we can construct the chord (straight line) passing through the points $(a, f(a))$ and $(b, f(b))$ on the graph. Its slope is $(f(b) - f(a))/(b - a)$. The Mean Value Theorem asserts that, at some point $(c, f(c))$ on the graph between these two points, the tangent line to the graph is parallel to the chord.



- The Mean Value Theorem guarantees the existence of a point c with the required $f'(c)$. It says nothing about uniqueness: there could be more than one such point.



The Mean Value Theorem allows us (among many other things) to relate the sign of f' to the increasing/decreasing behaviour of f.

Proposition 10.9 *Let I be an interval and $f : I \to \mathbb{R}$ be differentiable.*

(a) If $f'(x) \geq 0$ for all $x \in I$ then f is increasing.

(b) If $f'(x) > 0$ for all $x \in I$ then f is strictly increasing.

(c) If $f'(x) \leq 0$ for all $x \in I$ then f is decreasing.

(d) If $f'(x) < 0$ for all $x \in I$ then f is strictly decreasing.

(e) f is constant if and only if $f'(x) = 0$ for all $x \in I$.

Proof: The proofs of parts (a)–(d) are similar, so we will do part (a), and leave (b), (c), (d) as an exercise.

(a) Assume, towards a contradiction, that $f'(x) \geq 0$ but f is *not* increasing. Then there exist $a, b \in I$ with $a < b$ such that $f(a) > f(b)$. But then, by the Mean Value Theorem applied to f on $[a, b]$, there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} < 0,$$

a contradiction.

(e) If $f(x) = c$, some constant, then $f'(x) = 0$ for all $x \in I$ by Example 9.2. Conversely, assume $f'(x) = 0$ for all $x \in I$. Then f is increasing by part (a) and decreasing by part (b), so must be constant. $\square$

Remarks

(i) The converses of parts (a) and (c) are also true. That is, if $f : I \to \mathbb{R}$ is increasing and differentiable, then $f'(x) \geq 0$ for all $x \in I$. Similarly if $f : I \to \mathbb{R}$ is decreasing and differentiable, then $f'(x) \leq 0$ for all $x \in I$. One can prove this by adapting the idea of the proof of the Interior Extremum Theorem.

(ii) The converses of parts (b) and (d), by contrast, are **false**! For example

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^3$$

is strictly increasing, but its derivative is not strictly positive.

(iii) It's important to note that Proposition 10.9 concerns functions on *intervals*. For example

$$f : \mathbb{R}\backslash\{0\} \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

has $f'(x) = -1/x^2$ which is negative everywhere, but $f(-1) < f(1)$, so f is not a decreasing function. This doesn't contradict Proposition 10.9 since $\mathbb{R}\backslash\{0\}$ is not an interval.

(iv) Strictly increasing (or decreasing) functions are always injective, so Proposition 10.9 gives us a sneaky trick to prove that (some) functions are injective.

Example 10.10 $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + \sin x$ has $f'(x) = 2 + \cos x \geq 2 - 1 > 0$ for all x, so is strictly increasing by Proposition 10.9. Hence, if $f(x) = f(y)$ then $x \not< y$ and $y \not< x$, so $x = y$. That is, f is injective.

10.3 The Inverse Function Theorem

Recall that in section 8.3 we used the Intermediate Value Theorem to prove that

$$f : [0, \infty) \to [0, \infty), \qquad f(x) = x^2$$

has an inverse function

$$g : [0, \infty) \to [0, \infty), \qquad g(x) = \sqrt{x}.$$

What can we prove about g? Is it continuous? Differentiable? Our next theorem[7] helps us answer these questions.

Theorem 10.11 (The Inverse Function Theorem) *Let f be continuous on $[a, b]$ and differentiable on (a, b), and assume that, for all $x \in (a, b)$, $f'(x) > 0$. Then f is a bijective mapping*

$$f : [a, b] \to [f(a), f(b)],$$

and its inverse function

$$g : [f(a), f(b)] \to [a, b]$$

is continuous on $[f(a), f(b)]$, and differentiable on $(f(a), f(b))$, with

$$g'(y) = \frac{1}{f'(g(y))}.$$

Proof: I first claim that f is strictly increasing. Assume this is false. Then there exist $c, d \in [a, b]$ such that $c < d$ but $f(c) \geq f(d)$. But then, by the Mean Value Theorem applied to f on $[c, d]$, there exists $x \in (c, d)$ such that

$$f'(x) = \frac{f(d) - f(c)}{d - c} < 0,$$

contradicting the fact that f' is positive on (a, b).

Since f is strictly increasing, it is injective and bounded below by $f(a)$ and above by $f(b)$. Further, given any $y \in [f(a), f(b)]$, there exists $x \in [a, b]$ such that $f(x) = y$, by the Intermediate Value Theorem, so f is surjective. Hence f is bijective, and it has an inverse function

$$g : [f(a), f(b)] \to [a, b].$$

Recall that this function, by definition, satisfies

$$g(f(x)) = x, \qquad f(g(y)) = y$$

[7] This is actually a very special case of a much more general theorem of the same name, which works for functions $f : \mathbb{R}^n \to \mathbb{R}^n$ even (when suitably interpreted) in the case $n = \infty$!

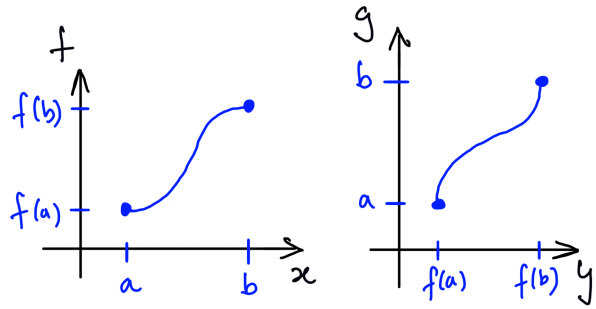


Figure 7: A picture to illustrate the statement of the Inverse Function Theorem. The function f is continuous on $[a, b]$ and has $f' > 0$ on (a, b). We conclude that f is actually a bijective map $[a, b] \to [f(a), f(b)]$ and that its inverse function is continuous on $[f(a), f(b)]$ and differentiable on $(f(a), f(b))$. The most difficult part of the proof is showing that g is continuous.

for all $x \in [a, b]$ and $y \in [f(a), f(b)]$.

I claim that g is continuous. Let $y \in [f(a), f(b)]$ and y_n be any sequence in $[f(a), f(b)]$ that converges to y. We must show that $g(y_n) \to g(y)$. Since f is bijective, there is a unique $c \in [a, b]$ such that $y = f(c)$ and $c = g(y)$. So we must show that $g(y_n) \to c$ (see figure 8). Assume that this is *false*, $g(y_n) \nrightarrow c$, or equivalently, $g(y_n) - c \nrightarrow 0$. Then, as we showed in Example 6.13, there exists $\varepsilon > 0$ and a subsequence $g(y_{n_k})$ such that $|g(y_{n_k}) - c| \geq \varepsilon$ for all k. But $g(y_{n_k})$ is bounded (it's a sequence in $[a, b]$) so, by the Bolzano-Weierstrass Theorem, it has a convergent subsequence $g(y_{n_{k_l}}) \to c_*$ for some $c_* \in \mathbb{R}$. In fact, since $a \leq g(y_{n_{k_l}}) \leq b$, we know that $c_* \in [a, b]$ by Proposition 2.14, and since $|g(y_{n_{k_l}}) - c| \geq \varepsilon$ we know that $|c_* - c| \geq \varepsilon > 0$ and hence $c_* \neq c$. Now f is continuous at c_*, so $f(g(y_{n_{k_l}})) \to f(c_*)$. But $f(g(y_{n_{k_l}})) = y_{n_{k_l}} \to f(c)$ by Theorem 4.3 (since this is a subsequence of y_n, which converges to $f(c)$), and limits are unique, so $f(c_*) = f(c)$. Since f is injective, we conclude that $c_* = c$, a contradiction. Hence $g(y_n) \to c$, as was to be proved.

Finally, we must prove that g is differentiable at every $y \in (f(a), f(b))$ with the claimed derivative. So, choose $y \in (f(a), f(b))$ and let (y_n) be any sequence in $(f(a), f(b)) \backslash \{y\}$ converging to y. We must prove that the sequence of difference quotients

$$s_n := \frac{g(y_n) - g(y)}{y_n - y}.$$

converges to $1/f'(g(y))$. Since g is bijective, there is a unique $c \in (a, b)$ such that $y = f(c)$. Let $x_n := g(y_n) \in (a, b)$ and note that $y_n = f(x_n)$, so $x_n \neq c$ (since $y_n \neq f(c)$). Now g is continuous, as we just proved, and $y_n \to f(c)$, so $x_n = g(y_n) \to g(f(c)) = c$. So (x_n) is a sequence in $(a, b) \backslash \{c\}$ converging to c, and f is differentiable at c, so

$$\frac{f(x_n) - f(c)}{x_n - c} \to f'(c),$$

whence

$$s_n = \frac{x_n - c}{f(x_n) - f(c)} \to \frac{1}{f'(c)} = \frac{1}{f'(g(y))}$$

by the Algebra of Limits. □

Remark 10.12 We can easily deduce an equivalent theorem in the case where $f'(x) < 0$ for all $x \in (a, b)$. We just apply Theorem 10.11 to $g(x) = -f(x)$. Try to formulate this theorem carefully, then prove it.

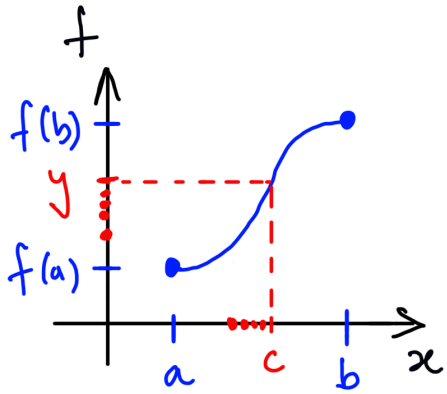


Figure 8: To prove that g is continuous at y we must show that, given any sequence $y_n \in [f(a), f(b)]$ converging to $y = f(c)$, the sequence $x_n = g(y_n)$ converges to c.

The Inverse Function Theorem allows us to deduce useful facts about lots of functions that are naturally defined as the inverse of some function we already understand, for example, square and other roots.

Corollary 10.13 *Let p be a fixed positive integer. Given any $x \in [0, \infty)$, there exists a unique number $x^{1/p} \in [0, \infty)$, its* $\mathbf{p^{th}}$ **root**, *whose p^{th} power is x. The function*

$$g : [0, \infty) \to [0, \infty), \qquad g(x) = x^{1/p}$$

is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$, and its derivative is

$$g'(x) = \frac{1}{p(x^{1/p})^{p-1}}.$$

Proof: Choose any $R > 0$ and consider the function $f(x) = x^p$ on $[0, R]$. This satisfies the hypotheses of the Inverse Function Theorem (on $[a, b] = [0, R]$), so has an inverse function

$$g : [0, R^p] \to [0, R]$$

which is continuous on $[0, R^p]$ and differentiable on $(0, R^p)$, with derivative

$$g'(x) = \frac{1}{f'(g(x))}.$$

Now $g(x) = x^{1/p}$ by definition, and $f'(x) = px^{p-1}$, so

$$g'(x) = \frac{1}{p(g(x))^{p-1}} = \frac{1}{p(x^{1/p})^{p-1}}.$$

Hence the claims hold on $[0, R^p]$ for all $R > 0$, so hold on $[0, \infty)$. $\square$

10.4 Higher derivatives and Taylor's Theorem

If a function $f : D \to \mathbb{R}$ is differentiable (everywhere on D) then its derivative defines another function $f' : D \to \mathbb{R}$, and we can ask whether this function, in turn, is differentiable. If it is, then its derivative is denoted $f'' : D \to \mathbb{R}$ and called the *second derivative* of f. Similarly, if $f'' : D \to \mathbb{R}$ is differentiable, its derivative $f''' : D \to \mathbb{R}$ is the *third derivative* of f. Proceeding inductively, we

can define the nth derivative of f, denoted $f^{(n)}$ to be the derivative (if it exists), of $f^{(n-1)}$, where $f^{(0)}$ is, by definition, just the function f itself (so $f^{(1)} = f'$, $f^{(2)} = f''$ etc.). If $f^{(n)} : D \to \mathbb{R}$ exists, we say that f is n times differentiable, and if f is n times differentiable for all $n \in \mathbb{Z}^+$, we say that f is *smooth*.

Example 10.14 Every polynomial function $p : \mathbb{R} \to \mathbb{R}$, $p(x) = a_0 + a_1 x + \cdots + a_m x^m$, is smooth. This follows immediately from Proposition 9.12.

Example 10.15 The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x|^3$ is twice differentiable but not three times differentiable. To see this, note that f coincides with the smooth function $f_+(x) = x^3$ on $(0, \infty)$ and with the smooth function $f_-(x) = -x^3$ on $(-\infty, 0)$, so it suffices to consider the differentiability of f at 0. Let x_n be a sequence in $\mathbb{R}\backslash\{0\}$ converging to 0. Then

$$\left| \frac{f(x_n) - f(0)}{x_n - 0} \right| = |x_n|^2 \to 0$$

so $f'(0) = 0$, and it follows that f is once differentiable. In fact

$$f'(x) = \begin{cases} f'_+(x) = 3x^2 & x > 0 \\ 0 & x = 0 \\ f'_-(x) = -3x^2 & x < 0 \end{cases},$$

so

$$\left| \frac{f'(x_n) - f'(0)}{x_n - 0} \right| = |3x_n| \to 0,$$

whence $f''(0) = 0$. Hence, f is twice differentiable, and

$$f''(x) = \left\{ \begin{array}{ll} f''_+(x) = 6x & x > 0 \\ 0 & x = 0 \\ f'_-(x) = -6x & x < 0, \end{array} \right\} = 6|x|.$$

But f'' is not differentiable at 0, by Example 9.6, so f is not three times differentiable.

An interesting interpretation of the Mean Value Theorem can be given as follows. Let $I \subseteq \mathbb{R}$ be an open interval, $f : I \to \mathbb{R}$ be differentiable, and choose and fix some $a \in I$. Then for any other $x \in I$, we may apply the Mean Value Theorem to $f : [a, x] \to \mathbb{R}$ (if $x > a$) or $f : [x, a] \to \mathbb{R}$ (if $x < a$) to deduce that there exists c between a and x such that

$$f(x) = f(a) + f'(c)(x - a).$$

Note that the same equation holds trivially in the case where $x = a$ (with $c = x = a$). So, for x close to a (meaning that $|x - a|$ is small) we can approximate f by the constant $f(a)$, and the error in this approximation is of size $|f'(c)||x - a|$, where c is somewhere between a and x (and, of course, depends on x).

The moral is that differentiability (once) of f on an open interval allows one to approximate f by a degree 0 polynomial function (i.e. a constant), with an error controlled by f'. It turns out that if f is n times differentiable one can do better: one can approximate f by a degree $(n-1)$ polynomial function, with an error controlled by $f^{(n)}$.

Theorem 10.16 (Taylor's Theorem) *Let $I \subseteq \mathbb{R}$ be an open interval, $f : I \to \mathbb{R}$ be $(n+1)$ times differentiable, and $a, x \in I$. Then there exists c between a and x such that*

$$\begin{aligned} f(x) &= f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n \\ &\quad + \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}. \end{aligned}$$

Proof: First note that the claimed equation holds trivially (with $c = a$) in the case where $x = a$. Choose and fix $a, x \in I$, $x \neq a$, and consider the function $F : I \to \mathbb{R}$,

$$F(t) = f(t) + f'(t)(x-t) + \frac{f''(t)}{2!}(x-t)^2 + \cdots + \frac{f^{(n)}(t)}{n!}(x-t)^n.$$

Then the claim to be proved is that there exists c between a and x such that

$$f(x) = F(a) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}. \tag{10.9}$$

Now, by the Product Rule, F is differentiable, and, as is easily verified (exercise!),

$$F'(t) = \frac{f^{(n+1)}(t)}{n!}(x-t)^n.$$

Let $G(t) = (x-t)^{n+1}$. Then F, G satisfy the hypotheses of the Extended Mean value Theorem (Theorem 10.7) on the interval with endpoints a, x (note that $G'(t) = 0$ only if $t = x$), so there exists c between a and x such that

$$\begin{aligned} \frac{F'(c)}{G'(c)} &= \frac{F(x)-F(a)}{G(x)-G(a)} = \frac{f(x)-F(a)}{G(a)} \\ \text{i.e.} \qquad \frac{f^{(n+1)}(c)(x-c)^n}{n![-(n+1)(x-c)^n]} &= \frac{f(x)-F(a)}{(x-a)^{n+1}}, \end{aligned}$$

whence (10.9) immediately follows. □

The polynomial

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is often called the nth **Taylor approximant** of the function f about a, and

$$\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

is called the **remainder**. Note that in the proof above, F is p_n but thought of as a function of the expansions point a rather than the variable x.

Example 10.17 Let $f : [0, \infty) \to \mathbb{R}$ be the function $f(x) = \sqrt{x}$. Construct the third Taylor approximant for f about 4, and hence find an approximation to $f(5) = \sqrt{5}$. Use Taylor's Theorem to find upper and lower bounds on $\sqrt{5}$.

Solution: The first four derivatives of f are

$$\begin{aligned} f'(x) &= \frac{x^{-1/2}}{2}, \\ f''(x) &= -\frac{x^{-3/2}}{4}, \\ f'''(x) &= \frac{3x^{-5/2}}{8}, \\ f^{(4)}(x) &= -\frac{15x^{-7/2}}{16}, \end{aligned}$$

so $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -2^{-5}$, and $f'''(4) = 3/2^8$. It follows that the third Taylor approximant for f about 4 is

$$\begin{aligned} p_3(x) &= f(4) + f'(4)(x-4) + \frac{f''(4)}{2!}(x-4)^2 + \frac{f'''(4)}{3!}(x-4)^3 \\ &= 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3. \end{aligned}$$

Using the approximation $f(x) \approx p_3(x)$ gives

$$\sqrt{5} = f(5) \approx p_3(5) = 2 + \frac{1}{4} - \frac{1}{64} + \frac{1}{512} = 2.236328125.$$

By Taylor's Theorem, the true value of $f(5)$ is

$$f(5) = p_2(5) + \frac{f^{(4)}(c)}{4!}(5-4)^4 = p_2(5) - \frac{5}{2^7 c^{7/2}}$$

for some $c \in (4, 5)$. It follows that the true value of $\sqrt{5}$ is strictly less than $p_2(5)$, but strictly greater than $p_2(5) - 5/(2^7 4^{7/2})$. That is, we have established that

$$2.236022949 < \sqrt{5} < 2.236328125.$$

For comparison, my pocket calculator tells me that $\sqrt{5} \approx 2.236067977$. □

Summary

- The **Extreme Value Theorem**: A continuous function on a closed, bounded interval $f : [a, b] \to \mathbb{R}$ is bounded above and below, and attains both a maximum and a minimum value.
- The **Interior Extremum Theorem**: If a differentiable function $f(a, b) \to \mathbb{R}$ attains a maximum or a minimum at $c \in (a, b)$ then $f'(c) = 0$.
- The **Mean Value Theorem**: If a function f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

- We can use the Mean Value Theorem to show that functions $f : I \to \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an interval, are increasing if $f'(x) \geq 0$ for all $x \in I$ (and decreasing if $f'(x) \leq 0$ for all $x \in I$).
- The **definitions** of increasing, decreasing, maximum and minimum have nothing to do with calculus. But we can use calculus to find/understand them.
- A function $f : D \to \mathbb{R}$ is n **times differentiable** if all its derivatives up to the nth exists everywhere in D. It is **smooth** if it is n times differentiable for all $n \in \mathbb{Z}^+$.
- A function which is $(n+1)$ times differentiable can be approximated by a polynomial of degree n, with an error controlled by $f^{(n+1)}$, its $(n+1)^{\text{th}}$ derivative. This follows from **Taylor's Theorem**.

11 Differentiablity of power series

11.1 Power series define smooth functions

Recall that a **power series** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

where x is a real variable and $a_0, a_1, a_2, \ldots$ is a real sequence (often called the **coefficients** of the power series). So this is actually a collection of infinitely many series, one for each choice of value for x. As usual, the series **converges** at a particular fixed value of x if its sequence of partial sums

$$\begin{aligned} s_0 &= a_0 \\ s_1 &= a_0 + a_1 x \\ s_2 &= a_0 + a_1 x + a_2 x^2 \\ &\vdots \end{aligned}$$

converges in the sense of Definition 2.4, and **converges absolutely** if the associated series $\sum_{n=0}^{\infty} |a_n x^n|$ converges. Certainly this sequence converges if $x = 0$, since it is the constant sequence $(a_0, a_0, a_0, \ldots)$, but it may diverge for other values of x. The set of values $x \in \mathbb{R}$ for which the series converges is almost completely determined by its **radius of convergence** (Definition 5.28):

$$R = \sup\{|x| \in [0, \infty) : \textstyle\sum_{n=0}^{\infty} |a_n x^n| \text{ converges}\}.$$

This follows from Theorem 5.30: the series converges absolutely for all $|x| < R$ and diverges for all $|x| > R$.

So, a power series with radius of convergence R defines a function

$$f : (-R, R) \to \mathbb{R}, \quad f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Switching viewpoint, its partial sums can be considered to be a sequence of polynomial functions:

$$\begin{aligned} s_0(x) &= a_0 \\ s_1(x) &= a_0 + a_1 x \\ s_2(x) &= a_0 + a_1 x + a_2 x^2 \\ &\vdots \end{aligned}$$

Each of these is certainly a differentiable function (on the whole of $\mathbb{R}$), so one might expect that their limit, $f(x) = \lim_{k\to\infty} s_k(x)$, is also a differentiable function, and that

$$f'(x) = \lim_{k\to\infty} s_k'(x) = \lim_{k\to\infty} a_1 + x_2 x + \cdots k a_k x^{k-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

This turns out to be true, at least on $(-R, R)$ where R is the radius of convergence of f, but proving it is no simple matter. The problem is that $f'(x)$ is really a *double* limit

$$f'(x) = \lim_{y\to x} \frac{f(y) - f(x)}{y - x} = \lim_{y\to x} \lim_{k\to\infty} \sum_{n=0}^{k} a_n \frac{y^n - x^n}{y - x},$$

and our expected formula for $f'(x)$ results from computing the double limit in the *opposite order*

$$\lim_{k\to\infty} \lim_{y\to x} \sum_{n=0}^{k} a_n \frac{y^n - x^n}{y - x}.$$

But swapping the order in which we evaluate a double limit can, in general, change its value!

Example 11.1 Consider the sequence of functions

$$s_k : \mathbb{R} \to \mathbb{R}, \qquad s_k(x) = \frac{kx^2 - 1}{kx^2 + 1}.$$

Let's compare the double limits

$$\lim_{x\to 0} \lim_{k\to\infty} s_k(x) \quad \text{and} \quad \lim_{k\to\infty} \lim_{x\to 0} s_k(x).$$

We first note that, for all $x \neq 0$, the real sequence

$$s_k(x) = \frac{x^2 - 1/k}{x^2 + 1/k} \to \frac{x^2}{x^2} = 1$$

by the Algebra of Limits. On the other hand, if $x = 0$,

$$s_k(0) = \frac{0-1}{0+1} = -1 \to -1.$$

So

$$\lim_{k\to\infty} s_k(x) = S(x) = \begin{cases} 1, & x \neq 0, \\ -1, & x = 0. \end{cases}$$

Hence

$$\lim_{x\to 0} \lim_{k\to\infty} s_k(x) = \lim_{x\to 0} S(x) = 1.$$

Turning to the second double limit, for each $k \in \mathbb{Z}^+$,

$$\lim_{x\to 0} s_k(x) = s_k(0) = -1,$$

since the function s_k is continuous. Hence

$$\lim_{k\to\infty} \lim_{x\to 0} s_k(x) = \lim_{k\to\infty} -1 = -1.$$

The two limits are different! □

So computing

$$\lim_{k\to\infty} \lim_{y\to x} \sum_{n=0}^{k} a_n \frac{y^n - x^n}{y - x}$$

in principle tells us precisely nothing about

$$\lim_{y\to x} \lim_{k\to\infty} \sum_{n=0}^{k} a_n \frac{y^n - x^n}{y - x}.$$

We will need to work considerably harder!

We begin by showing that a power series and the obvious candidate for its derivative have the same radius of convergence.

Lemma 11.2 *The power series* $f(x) = \sum_{n=0}^{\infty} a_n x^n$ *and* $g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ *have the same radius of convergence.*

Proof: We define the following subsets of $\mathbb{R}$,

$$A = \{|x| \,:\, \textstyle\sum |a_n x^n| \text{ converges}\}, \qquad B = \{|x| \,:\, \textstyle\sum |n a_n x^{n-1}| \text{ converges}\},$$

and recall that $\sup A$ and $\sup B$ are the radii of convergence of f and g, respectively. We will prove that

(i) for all x with $|x| < \sup B$, $f(x)$ converges absolutely (so $|x| \in A$, and hence $\sup A \geq \sup B$), and

(ii) for all x with $|x| < \sup A$, $g(x)$ converges absolutely (so $|x| \in B$, and hence $\sup B \geq \sup A$).

It follows immediately that $\sup A = \sup B$, which is what we seek to prove.
(i) Let $x \in \mathbb{R}$ with $|x| < \sup B$. Then $s_k = \sum_{n=1}^{k} |na_n x^{n-1}|$ converges (Theorem 5.30), and hence is bounded above, and

$$t_k = \sum_{n=0}^{k} |a_n x^n| = |a_0| + |x| \sum_{n=1}^{k} |a_n x^{n-1}| \leq |a_0| + |x| s_k.$$

So (t_k) is increasing and bounded above, and hence converges, by the Monotone Convergence Theorem. Hence, $|x| \in A$.
(ii) Let $x \in \mathbb{R}$ with $|x| < \sup A$. If $x = 0$ then $g(x)$ certainly converges, so we may assume $|x| > 0$. Choose $\rho \in (|x|, \sup A)$. Then $t_k = \sum_{n=0}^{k} |a_n| \rho^n$ converges (Theorem 5.30), and hence is bounded above. Now

$$s_k = \sum_{n=1}^{k} |na_n x^{n-1}| = |x|^{-1} \sum_{n=1}^{k} n \left(\frac{|x|}{\rho} \right)^n |a_n| \rho^n.$$

We have seen that the sequence $n(|x|/\rho)^n \to 0$, since $(|x|/\rho) < 1$ (Example 5.16), so it must be bounded, that is, there exists $K > 0$ such that, for all n, $n(|x|/\rho)^n \leq K$. But then

$$s_k \leq |x|^{-1} \sum_{n=0}^{k} K |a_n| \rho^n = \frac{K}{|x|} t_k.$$

Hence, s_k is increasing and bounded above, so converges, by the Monotone Convergence Theorem (Corollary 3.20). Hence, $|x| \in B$. □

We now show that $g(x)$ really is the derivative of $f(x)$.

Theorem 11.3 *Suppose that* $f(x) = \sum_{n=0}^{\infty} a_n x^n$ *has radius of convergence* $R > 0$. *Then* f *is differentiable on* $(-R, R)$, *and*

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Proof: Choose and fix $y \in (-R, R)$. Let

$$g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad \text{and} \quad h(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

By Lemma 11.2, $g(x)$ has radius of convergence R, so by Lemma 11.2 again (applied to $g(x)$), $h(x)$ also has radius of convergence R. Now choose and fix some $\rho \in (|y|, R)$, for example $\rho = (|y| + R)/2$ will do. Then, by Theorem 5.30, $h(\rho)$ converges absolutely, meaning that the sequence of partial sums

$$\sum_{n=2}^{k} n(n-1) |a_n| \rho^{n-2}$$

converges. But convergent sequence are bounded, so there exists $C > 0$ such that, for all $k \in \mathbb{Z}^+$,

$$\sum_{n=2}^{k} n(n-1) |a_n| \rho^{n-2} \leq C. \tag{11.10}$$

We wish to show that

$$\lim_{x \to y} \frac{f(x) - f(y)}{x - y} = g(y),$$

that is, for all sequences $x_j \in (-R, R) \backslash \{y\}$ such that $x_j \to y$,

$$s_j := \frac{f(x_j) - f(y)}{x_j - y} - g(y) \to 0.$$

So, let (x_j) be any such sequence and denote by s_j the corresponding sequence defined above. By the Algebra of Limits, for each fixed $j \in \mathbb{Z}^+$,

$$s_j = \lim_{k \to \infty} P_{jk}$$

where

$$P_{jk} := \sum_{n=0}^{k} a_n \left\{ \frac{x_j^n - y^n}{x_j - y} - ny^{n-1} \right\}.$$

Note that the $n = 0$ and $n = 1$ terms in this sum are identically 0, so actually

$$P_{jk} = \sum_{n=2}^{k} a_n \left\{ \frac{x_j^n - y^n}{x_j - y} - ny^{n-1} \right\}.$$

Let us consider P_{jk} for some fixed choice of $j, k \in \mathbb{Z}^+$. By the Mean Value Theorem applied to the function $p(x) = x^n$ on the interval from x_j to y, there exists z_j, strictly between x_j and y such that

$$\begin{aligned} \frac{p(x_j) - p(y)}{x_j - y} &= p'(z_j) \\ \frac{x_j^n - y^n}{x_j - y} &= n z_j^{n-1}. \end{aligned}$$

Hence

$$P_{jk} = \sum_{n=2}^{k} n a_n (z_j^{n-1} - y^{n-1}).$$

Note that $0 < |z_j - y| < |x_j - y|$ (since z_j is strictly between x_j and y).

By the Mean Value Theorem applied to the function $q(x) = x^{n-1}$ on the interval from z_j to y, there exists w_j, strictly between z_j and y such that

$$\begin{aligned} \frac{q(z_j) - q(y)}{z_j - y} &= q'(w_j) \\ \frac{z_j^{n-1} - y^{n-1}}{x_j - y} &= (n-1) w_j^{n-2} \\ z_j^{n-1} - y^{n-1} &= (n-1) w_j^{n-2} (z_j - y). \end{aligned}$$

Hence

$$P_{jk} = \sum_{n=2}^{k} n(n-1) a_n w_j^{n-2} (z_j - y) = (z_j - y) \sum_{n=2}^{k} n(n-1) a_n w_j^{n-2}.$$

Note that $0 < |w_j - y| < |z_j - y| < |x_j - y|$ (since w_j is strictly between z_j and y). Hence, by the Triangle Inequality,

$$|P_{jk}| \leq |z_j - y| \sum_{n=2}^{k} n(n-1) |a_n| |w_j|^{n-2} \leq |x_j - y| \sum_{n=2}^{k} n(n-1) |a_n| |w_j|^{n-2}.$$

To summarize: for each j there exists w_j with $0 < |w_j - y| < |x_j - y|$ such that, for all $k \in \mathbb{Z}^+$,

$$|P_{jk}| \leq |x_j - y| \sum_{n=2}^{k} n(n-1)|a_n||w_j|^{n-2}.$$

Since $x_j \to y$, $w_j \to y$ also, by the Squeeze Rule. Hence, there exists $J \in \mathbb{Z}^+$ such that, for all $j \geq J$, $|w_j| < \rho$. But then, for all $j \geq J$ and all k,

$$|P_{jk}| \leq |x_j - y| \sum_{n=2}^{k} n(n-1)|a_n|\rho^{n-2} \leq C|x_j - y|$$

by (11.10). Hence, by Proposition 2.14, for all $j \geq J$

$$|s_j| = \lim_{k\to\infty} |P_{jk}| \leq C|x_j - y|.$$

But $|x_j - y| \to 0$, so the sequence $|s_j| \to 0$ by the Squeeze Rule (and the Tail Lemma). This completes the proof. □

So the derivative of a power series exists on its open interval of convergence and is just the power series obtained by termwise differentiation. Since $f'(x)$ is also a power series with radius of convergence R, we can apply Theorem 11.3 to $f'(x)$ and deduce that $f : (-R, R) \to \mathbb{R}$ is actually *twice* differentiable. Further, $f''(x)$ also has radius of convergence R, so is differentiable on $(-R, R)$, that is, f is *three times* differentiable. In fact, we can keep applying Theorem 11.3 as often as we like, and we conclude that f is *smooth*.

Corollary 11.4 *Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius convergence $R > 0$. Then $f : (-R, R) \to \mathbb{R}$ is a smooth function, and*

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

Proof: For each integer $k \geq 0$, define

$$g_k(x) = \sum_{n=0}^{\infty} \frac{(n+k)!}{n!} a_{n+k} x^n.$$

I claim that each power series $g_k(x)$ has radius of convergence R and that, for all $x \in (-R, R)$, $f^{(k)}(x) = g_k(x)$. We prove this by induction on k.

Certainly the claim holds for $k = 0$, since $g_0 = f$. So, assume that the claim holds for some value $k \geq 0$, and consider g_{k+1}. Defining coefficients $b_n = \frac{(n+k)!}{n!} a_{n+k}$ so that $g_k(x) = \sum_{n=0}^{\infty} b_n x^n$, then

$$\begin{aligned} g_{k+1}(x) &= \sum_{n=0}^{\infty} \frac{(n+k+1)!}{n!} a_{n+k+1} x^n = \sum_{m=1}^{\infty} \frac{(m+k)!}{(m-1)!} a_{m+k} x^{m-1} \\ &= \sum_{m=1}^{\infty} m b_m x^{m-1}. \end{aligned}$$

Hence, by Lemma 11.2, g_{k+1} has the same radius of convergence as g_k, and by Theorem 11.3, $g_{k+1}(x)$ coincides with the derivative of g_k. But, by our induction hypothesis, g_k has radius of convergence R and coincides with $f^{(k)}$, so g_{k+1} has radius of convergence R and coincides with $f^{(k+1)}$. Hence, if the claim holds for some $k \geq 0$, it also holds for $k + 1$. Hence, by induction, the claim holds for all integers $k \geq 0$.

It follows that $f^{(k)}(0) = g_k(0) = k! a_k$, which completes the proof. □

Corollary 11.4 applies to any power series with non-zero radius of convergence so, in particular, it applies to the series defining exp, sin, and cos.

Proposition 11.5 *The functions* $\exp : \mathbb{R} \to \mathbb{R}$, $\sin : \mathbb{R} \to \mathbb{R}$, $\cos : \mathbb{R} \to \mathbb{R}$ *are smooth, and their derivatives are*

$$\exp' = \exp, \qquad \sin' = \cos, \qquad \cos' = -\sin.$$

Proof: That these functions are smooth follows immediately from Corollary 11.4. Furthermore, by Theorem 11.3

$$\exp'(x) = \sum_{n=1}^{\infty} n \left(\frac{1}{n!}\right) x^{n-1} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{m=0}^{\infty} \frac{x^m}{m!} = \exp(x)$$

for all $x \in \mathbb{R}$, so $\exp' = \exp$. The formulae for $\sin'$ and $\cos'$ follow from similar arguments. □

Smooth functions are differentiable, and hence continuous, so this restrospectively justifies our assumption that $\sin : \mathbb{R} \to \mathbb{R}$ (for example) is continuous.

11.2 Exponentials and logarithms

Now that we know that exp is differentiable, and its derivative is exp, we can prove that it has all the nice properties you're familiar with.

Proposition 11.6 *For all* $x, y \in \mathbb{R}$, $\exp(x + y) = \exp(x)\exp(y)$.

Proof: Choose any constant $a \in \mathbb{R}$ and consider the function

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \exp(x)\exp(a - x).$$

By the Product and Chain Rules, this is differentiable with derivative

$$f'(x) = \exp'(x)\exp(a - x) + \exp(x)(-\exp'(a - x)) = \exp(x)\exp(a - x) - \exp(x)\exp(a - x) = 0.$$

Hence by Proposition 10.9, f is constant, that is, for all $x \in \mathbb{R}$, $f(x) = f(a)$, that is

$$\exp(x)\exp(a - x) = \exp(a)\exp(0) = \exp(a).$$

This holds for all $a \in \mathbb{R}$ and, in the case $a = x + y$, yields the claimed identity. □

Proposition 11.7 exp *is a bijective map* $\mathbb{R} \to (0, \infty)$.

Proof: We first note that there is no $x \in \mathbb{R}$ such that $\exp(x) = 0$ since, if there were, then Proposition 11.6 with $y = -x$ would imply

$$1 = \exp(0) = \exp(x - x) = \exp(x)\exp(-x) = 0 \times \exp(-x) = 0.$$

Since exp is continuous and $\exp(0) = 1 > 0$, it follows that $\exp(x) > 0$ for all $x \in \mathbb{R}$, by the Intermediate Value Theorem (if $\exp(x) < 0$ there would exist c between 0 and x with $\exp(c) = 0$, contradicting the fact just established). Hence, $\exp'(x) = \exp(x) > 0$ for all $x \in \mathbb{R}$, so exp is strictly increasing by Proposition 10.9, and hence is injective.

It remains to prove that exp is surjective, that is, its range, $\exp(\mathbb{R})$, is $(0, \infty)$. Now $\exp(0) = 1$ so $1 \in \exp(\mathbb{R})$. By the Mean Value Theorem, for each $x > 0$ there exists $c \in (0, x)$ such that

$$\begin{aligned} \frac{\exp(x) - \exp(0)}{x - 0} &= \exp'(c) = \exp(c) \\ \Rightarrow \qquad \exp(x) &= 1 + x\exp(c) \geq 1 + x\exp(0) = 1 + x \end{aligned}$$

since, as we have observed, exp is increasing. Hence exp is unbounded above. So for each $y > 1$, there exists $a \in \mathbb{R}$ such that $\exp(a) > y$. But then, by the Intermediate Value Theorem, there exists

$x \in [0, a]$ such that $\exp(x) = y$. Hence every $y > 1$ is in $\exp(\mathbb{R})$. Finally, given any $y \in (0, 1)$, $1/y > 1$ so, as we just showed, there exists $x \in \mathbb{R}$ such that $\exp(x) = 1/y$. But then $\exp(-x) = 1/(1/y) = y$ by Proposition 11.6, so $y \in \exp(\mathbb{R})$ also. $\square$

Since exp is a bijective function $\mathbb{R} \to (0, \infty)$ it has an inverse function $(0, \infty) \to \mathbb{R}$ which is conventionally called the **natural logarithm** ln.

Proposition 11.8 *The natural logarithm function* $\ln : (0, \infty) \to \mathbb{R}$ *is differentiable and*

$$\ln'(x) = \frac{1}{x}.$$

Proof: Given any $\rho > 0$, exp is continuous on $[-\rho, \rho]$, differentiable on $(-\rho, \rho)$ and has strictly positive derivative. Hence, by the Inverse Function Theorem (Theorem 10.11), its inverse function

$$\ln : [\exp(-\rho), \exp(\rho)] \to [-\rho, \rho]$$

exists and is differentiable, with derivative

$$\ln'(x) = \frac{1}{\exp'(\ln(x))} = \frac{1}{\exp(\ln(x))} = \frac{1}{x}.$$

Since this holds for all $\rho > 0$, the claim is established. $\square$

So, for each positive number y, $\ln y$ is the real number whose exponential is y. For example, $\exp(0) = 1$ (just evaluate the power series defining $exp(x)$ at $x = 0$) and so $\ln 1 = 0$. Every property of exp has a counterpart for ln. In particular, the identity $\exp(x + y) \equiv \exp(x) \exp(y)$ is equivalent to

Proposition 11.9 *For all* $x, y \in (0, \infty)$, $\ln(xy) = \ln x + \ln y$.

Proof: Choose and fix $y \in (0, \infty)$ and consider the function

$$f : (0, \infty) \to \mathbb{R}, \qquad f(x) = \ln(xy) - \ln x - \ln y.$$

By the Chain Rule and Proposition 11.8,

$$f'(x) = \frac{y}{xy} - \frac{1}{x} = 0$$

so f is constant (Proposition 10.9). Hence, for all x, $f(x) = f(1) = \ln y - 0 - \ln y = 0$. $\square$

Exercise 11.10 Prove Proposition 11.9 by using Proposition 11.6 directly.

A second useful identity quickly follows from this.

Proposition 11.11 *For all* $x \in (0, \infty)$ *and* $n \in \mathbb{Z}$, $\ln x^n = n \ln x$.

Proof: Exercise. (*Hint*: proof by induction.) $\square$

Recall we defined *Euler's number* to be

$$e = \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

It follows that $\ln e = 1$. There is an alternative way to define e, as the limit of the sequence $(1 + \frac{1}{n})^n$. We are now in a position to prove this.

Proposition 11.12 *The sequence* $x_n = \left(1 + \frac{1}{n}\right)^n$ *converges to* e.

Proof: It is clear that $x_n > 0$ for all n. Now, for all n, by Proposition 11.11,

$$\begin{aligned} \ln x_n &= n \ln\left(1 + \frac{1}{n}\right) \\ &= \frac{\ln\left(1 + \frac{1}{n}\right) - \ln 1}{\left(1 + \frac{1}{n}\right) - 1} \\ &= \frac{\ln y_n - \ln 1}{y_n - 1} \end{aligned}$$

where $y_n = 1 + \frac{1}{n}$. Note that $y_n \to 1$. Since ln is differentiable (everywhere, and, in particular, at 1), the sequence of difference quotients

$$\frac{\ln y_n - \ln 1}{y_n - 1}$$

converges to $\ln'(1) = 1/1 = 1$, that is,

$$\ln x_n \to 1.$$

Now exp is continuous (everywhere,a nd, in particular, at 1), so

$$\exp(\ln x_n) \to \exp(1) = e.$$

But $\exp(\ln x_n) = x_n$ by definition of ln, so $x_n \to e$. □

11.3 Exactly summing (some) series

We finish this section with a nice trick. Sometimes we can exactly sum a series by thinking of it as a power series evaluated at a specific value x. Here's a simple example:

Example 11.13 Exactly sum $\sum_{n=1}^{\infty} \frac{n}{2^n}$.

Solution: It's easy to show (using the Ratio Test, for example) that the series converges, but this doesn't tell us what its limit is. We need to be sneakier.

Consider the power series

$$f(x) = \sum_{n=0}^{\infty} x^n.$$

This is the Geometric Series with common ratio x. We showed previously that it has radius of convergence $R = 1$, and for all $x \in (-1, 1)$, its limit is

$$f(x) = \frac{1}{1 - x}. \tag{11.11}$$

By Theorem 11.3, this function is differentiable on $(-1, 1)$ and

$$f'(x) = \sum_{n=1}^{\infty} n x^{n-1}.$$

But, by the Chain Rule, we already know from (11.11) that f is differentiable with derivative

$$f'(x) = \frac{1}{(1 - x)^2}.$$

Hence, for all $x \in (-1, 1)$,

$$\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

In the special case $x = 1/2$, we deduce that

$$\sum_{n=1}^{\infty} \frac{n}{2^{n-1}} = \frac{1}{(1/2)^2} = 4,$$

and hence

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1}{2} \times 4 = 2.$$

□

Summary

- A power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ with radius of convergence $R > 0$ is differentiable on $(-R, R)$, and its derivative is

  $$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

  This power series also has radius of convergence R.

- It follows that $f : (-R, R) \to \mathbb{R}$ is actually **smooth** (infinitely differentiable), and that

  $$a_n = \frac{f^{(n)}(0)}{n!}.$$

- These facts apply to all convergent power series, including those defining exp, sin and cos.

12 Integration

12.1 Dissections and Riemann sums

How do we define the area of a bounded subset of the plane $\mathbb{R}^2$? If the subset is a rectangle, the answer is easy: its area is its length times its width. Similarly, if the subset is a union of non-overlapping rectangles it's easy: we just add up the areas of all the constituent rectangles. But what if the subset is more complicated: the region bounded by the x-axis, the vertical lines $x = a$ and $x = b > a$ and the graph $y = f(x)$ of some non-constant function, for example? One approach is to define the area of such a region to be the unique real number (if it exists) which is no bigger than the total area of any collection of rectangles which covers the region, and no smaller than the total area of any collection of rectangles which is covered by the region. This is the underlying idea that leads to the *Riemann integral*. We begin by identifying the collections of rectangles we will use. These are determined by *dissecting* the interval $[a, b]$ into a finite collection of subintervals.

Definition 12.1 A **dissection** of a closed bounded interval $[a, b]$ is a finite subset $\mathscr{D}$ of $[a, b]$ containing both a and b. By convention, if $\mathscr{D}$ has $n + 1$ elements, we label these $a_0, a_1, \ldots, a_n$, so that

$$a = a_0 < a_1 < a_2 < \cdots < a_n = b,$$

and say that $\mathscr{D}$ is a dissection of **size** n. We say that $\mathscr{D}$ is a **regular** dissection if $a_j - a_{j-1} = (b-a)/n$ for all j, that is, if the points in the dissection are regularly spaced.

Recall that a function $f : D \to \mathbb{R}$ is **bounded** if its range $f(D) \subseteq \mathbb{R}$ is a bounded set, that is, if there exist $M, m \in \mathbb{R}$ such that, for all $x \in D$, $m \leq f(x) \leq M$.

Definition 12.2 Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $\mathscr{D}$ be a dissection of size n of $[a, b]$. For each $j \in \{1, 2, \ldots, n\}$, let

$$\begin{aligned} m_j &= \inf\{f(x) \,:\, a_{j-1} \leq x \leq a_j\}, \\ \text{and} \quad M_j &= \sup\{f(x) \,:\, a_{j-1} \leq x \leq a_j\}. \end{aligned}$$

Note that these numbers exist, since f is bounded. The **lower Riemann sum** of f with respect to $\mathscr{D}$ is

$$l_{\mathscr{D}}(f) = \sum_{j=1}^{n} m_j(a_j - a_{j-1}),$$

and the **upper Riemann sum** of f with respect to $\mathscr{D}$ is

$$u_{\mathscr{D}}(f) = \sum_{j=1}^{n} M_j(a_j - a_{j-1}).$$

The idea is that a dissection of size n divides $[a, b]$ into n subintervals, $[a_{j-1}, a_j]$ for $j = 1, 2, \ldots, n$. If $f(x) \geq 0$ for all x, then the lower Riemann sum can be visualized as the total area of the tallest rectangles with bases $[a_{j-1}, a_j]$ which fit under the graph $y = f(x)$ between $x = a$ and $x = b$. So $l_{\mathscr{D}}(f)$ is an underestimate of the area under the graph. Similarly, the upper Riemann sum can be visualized as the total area of the shortest rectangles with bases $[a_{j-1}, a_j]$ which the graph $y = f(x)$ between $x = a$ and $x = b$ fits under and is thus an overestimate of the area under the graph. This is illustrated in Figure 9.

Note that $m_j \leq M_j$ for all $j \in \{1, 2, \ldots, n\}$, so it follows immediately that, for all bounded functions $f : [a, b] \to \mathbb{R}$ and all dissections $\mathscr{D}$ of $[a, b]$,

$$l_{\mathscr{D}}(f) \leq u_{\mathscr{D}}(f).$$

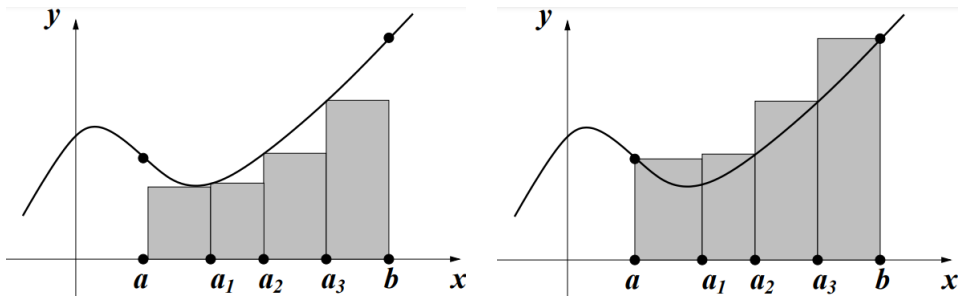


Figure 9: Lower and upper Riemann sums of a function with respect to a dissection of size 4.

Example 12.3 $\mathscr{D}_1 = \{0,1\}$, $\mathscr{D}_2 = \{0, \frac{1}{2}, 1\}$, and $\mathscr{D}_3 = \{0, \frac{1}{2}, \frac{3}{4}, 1\}$ are dissections of $[0,1]$. The function $f : [0,1] \to \mathbb{R}$, $f(x) = x^2$ is bounded, and its lower and upper Riemann sums with respect to these dissections are:

$$\begin{aligned}
l_{\mathscr{D}_1}(f) &= 0(1-0) = 0 \\
u_{\mathscr{D}_1}(f) &= 1(1-0) = 1 \\
l_{\mathscr{D}_2}(f) &= 0(\frac{1}{2} - 0) + \frac{1}{4}(1 - \frac{1}{2}) = \frac{1}{8} \\
u_{\mathscr{D}_2}(f) &= \frac{1}{4}(\frac{1}{2} - 0) + 1(1 - \frac{1}{2}) = \frac{5}{8} \\
l_{\mathscr{D}_3}(f) &= 0(\frac{1}{2} - 0) + \frac{1}{4}(\frac{3}{4} - \frac{1}{2}) + \frac{9}{16}(1 - \frac{3}{4}) = \frac{13}{64} \\
u_{\mathscr{D}_3}(f) &= \frac{1}{4}(\frac{1}{2} - 0) + \frac{9}{16}(\frac{3}{4} - \frac{1}{2}) + 1(1 - \frac{3}{4}) = \frac{33}{64}.
\end{aligned}$$

Note that

$$l_{\mathscr{D}_1} \le l_{\mathscr{D}_2} \le l_{\mathscr{D}_3} \le u_{\mathscr{D}_3} \le u_{\mathscr{D}_2} \le u_{\mathscr{D}_1}.$$

We will see shortly that this is no accident.

The set of all dissections of a fixed interval $[a,b]$ is a very large and complicated set. To tame this beast, we need to equip it with some extra structure. The crucial idea turns out to be rather simple: that of **refinement**.

Definition 12.4 Given dissections $\mathscr{D}, \mathscr{D}'$ of $[a,b]$, we say that $\mathscr{D}'$ is a **refinement** of $\mathscr{D}$ if $\mathscr{D} \subseteq \mathscr{D}'$. If $\mathscr{D}' \backslash \mathscr{D}$ contains k points, we say that $\mathscr{D}'$ is a **k-point refinement** of $\mathscr{D}$.

Note that:

- Given a pair $\mathscr{D}_1$, $\mathscr{D}_2$ of dissections, there's no reason why either should be a refinement of the other. For example $\mathscr{D}_1 = \{0, 1/2, 1\}$ and $\mathscr{D}_2 = \{0, 1/3, 2/3, 1\}$ are both dissections of $[0,1]$, but $\mathscr{D}_2$ is not a refinement of $\mathscr{D}_1$ (since $\mathscr{D}_1 \not\subseteq \mathscr{D}_2$) and nor is $\mathscr{D}_1$ a refinement of $\mathscr{D}_2$ (since $\mathscr{D}_2 \not\subseteq \mathscr{D}_1$).
- Every dissection $\mathscr{D}$ of $[a,b]$ is a refinement of the trivial dissection $\mathscr{D}_0 = \{a,b\}$.
- Every dissection is the unique 0-point refinement of itself.

The idea is that a refinement $\mathscr{D}'$ of $\mathscr{D}$ contains all the points in $\mathscr{D}$ and (unless $\mathscr{D}' = \mathscr{D}$) some more points too, so it splits $[a,b]$ up into more subintervals, at least some of which are narrower. Intuitively, one expects that passing from $\mathscr{D}$ to a refinement of $\mathscr{D}$ can only improve, that is, increase, the underestimate $l_{\mathscr{D}}(f)$. Similarly, passing to a refinement of $\mathscr{D}$, one expects, can only reduce the overestimate $u_{\mathscr{D}}(f)$. This expectation turns out to be essentially correct and is fundamental.

Lemma 12.5 (Refinement Lemma) *Let $f : [a, b] \to \mathbb{R}$ be bounded, $\mathscr{D}, \mathscr{D}'$ be dissections of $[a, b]$, and $\mathscr{D}'$ be a refinement of $\mathscr{D}$. Then*

$$l_{\mathscr{D}}(f) \leq l_{\mathscr{D}'}(f) \leq u_{\mathscr{D}'}(f) \leq u_{\mathscr{D}}(f).$$

Proof: We first prove this in the special case where $\mathscr{D}'$ is a 1-point refinement of $\mathscr{D}$. So, let $\mathscr{D} = \{a = a_0, a_1, \ldots, a_n = b\}$ and $\mathscr{D}' = \mathscr{D} \cup \{z\}$ where $z \in [a, b] \backslash \mathscr{D}$. Then there exists $k \in \{1, \ldots, n\}$ such that $z \in (a_{k-1}, a_k)$. Let

$$\begin{aligned}
m_j &= \inf\{f(x) \,:\, a_j \leq x \leq a_{j-1}\} \\
m' &= \inf\{f(x) \,:\, a_{k-1} \leq x \leq z\} \\
m'' &= \inf\{f(x) \,:\, z \leq x \leq a_k\} \\
M_j &= \sup\{f(x) \,:\, a_j \leq x \leq a_{j-1}\} \\
M' &= \sup\{f(x) \,:\, a_{k-1} \leq x \leq z\} \\
M'' &= \sup\{f(x) \,:\, z \leq x \leq a_k\}
\end{aligned}$$

and note that, since $[a_{k-1}, z]$ and $[z, a_k]$ are subsets of $[a_{k-1}, a_k]$, we know immediately that $m', m'' \geq m_k$, and $M', M'' \leq M_k$ (why?). Now

$$\begin{aligned}
l_{\mathscr{D}'}(f) &= \sum_{j \in \{1,2,\ldots,n\} \backslash \{k\}} m_j(a_j - a_{j-1}) + m'(z - a_{k-1}) + m''(a_k - z) \\
&= l_{\mathscr{D}}(f) - m_k(a_k - a_{k-1}) + m'(z - a_{k-1}) + m''(a_k - z) \\
&\geq l_{\mathscr{D}}(f) - m_k(a_k - a_{k-1}) + m_k(z - a_{k-1}) + m_k(a_k - z) \\
&= l_{\mathscr{D}}(f),
\end{aligned}$$

and

$$\begin{aligned}
u_{\mathscr{D}'}(f) &= \sum_{j \in \{1,2,\ldots,n\} \backslash \{k\}} M_j(a_j - a_{j-1}) + M'(z - a_{k-1}) + M''(a_k - z) \\
&= u_{\mathscr{D}}(f) - M_k(a_k - a_{k-1}) + M'(z - a_{k-1}) + M''(a_k - z) \\
&\leq u_{\mathscr{D}}(f) - M_k(a_k - a_{k-1}) + M_k(z - a_{k-1}) + M_k(a_k - z) \\
&= u_{\mathscr{D}}(f).
\end{aligned}$$

We have already observed that $l_{\mathscr{D}'}(f) \leq u_{\mathscr{D}'}(f)$, so we conclude that

$$l_{\mathscr{D}}(f) \leq l_{\mathscr{D}'}(f) \leq u_{\mathscr{D}'}(f) \leq u_{\mathscr{D}}(f),$$

that is, the Refinement Lemma holds for every 1-point refinement of every dissection.

Consider now the case where $\mathscr{D}' = \mathscr{D} \cup \{z_1, z_2, \ldots, z_p\}$ is a p-point refinement of $\mathscr{D}$. Then we can define a chain of dissections $\mathscr{D}_i$, $i = 0, \ldots, p$, by $\mathscr{D}_0 = \mathscr{D}$ and $\mathscr{D}_i = \mathscr{D}_{i-1} \cup \{z_i\}$ for each $i = 1, \ldots, p$. Note that $\mathscr{D}_p = \mathscr{D}'$, and that each $\mathscr{D}_i$ is a 1-point refinement of $\mathscr{D}_{i-1}$. Hence, by the result just proved,

$$l_{\mathscr{D}}(f) = l_{\mathscr{D}_0}(f) \leq l_{\mathscr{D}_1}(f) \leq l_{\mathscr{D}_2(f)} \leq \cdots \leq l_{\mathscr{D}_p}(f) = l_{\mathscr{D}'}(f)$$

and

$$u_{\mathscr{D}}(f) = u_{\mathscr{D}_0}(f) \geq u_{\mathscr{D}_1}(f) \geq u_{\mathscr{D}_2(f)} \geq \cdots \geq u_{\mathscr{D}_p}(f) = u_{\mathscr{D}'}(f).$$

Clearly $l_{\mathscr{D}'}(f) \leq u_{\mathscr{D}'}(f)$, and so

$$l_{\mathscr{D}}(f) \leq l_{\mathscr{D}'}(f) \leq u_{\mathscr{D}'}(f) \leq u_{\mathscr{D}}(f),$$

as was to be proved. □

Looking back at Example 12.3, we see that

$$\mathscr{D}_1 \subset \mathscr{D}_2 \subset \mathscr{D}_3$$

so $\mathscr{D}_3$ is a refinement of $\mathscr{D}_2$ which is a refinement of $\mathscr{D}_1$. Hence, the Refinement Lemma tells us that

$$l_{\mathscr{D}_1} \leq l_{\mathscr{D}_2} \leq l_{\mathscr{D}_3} \leq u_{\mathscr{D}_3} \leq u_{\mathscr{D}_2} \leq u_{\mathscr{D}_1},$$

as we discovered by explicit calculation.

It follows immediately from the Refinement Lemma that *every* upper Riemann sum is at least as large as *every* lower Riemann sum, whatever (possibly different) dissections we use to compute them.

Lemma 12.6 *Let $\mathscr{D}, \mathscr{D}'$ be two dissections of $[a,b]$ and $f : [a,b] \to \mathbb{R}$ be bounded. Then $l_{\mathscr{D}}(f) \leq u_{\mathscr{D}'}(f)$.*

Proof: $\mathscr{D}'' = \mathscr{D} \cup \mathscr{D}'$ is a refinement of both $\mathscr{D}$ and $\mathscr{D}'$, so by the Refinement Lemma,

$$l_{\mathscr{D}}(f) \leq l_{\mathscr{D}''}(f) \leq u_{\mathscr{D}''}(f) \leq u_{\mathscr{D}'}(f).$$

□

Note that we did *not* assume that either of $\mathscr{D}, \mathscr{D}'$ was a refinement of the other. The result holds completely generally!

Example 12.7 Let $f : [0,1] \to \mathbb{R}$, $f(x) = x^2$, and for each $a \in (0,1)$ let $\mathscr{D}_a = \{0, a, 1\}$. Then

$$\begin{aligned} l_{\mathscr{D}_a}(f) &= 0 \times a + a^2 \times (1-a) = a^2(1-a) \\ u_{\mathscr{D}_a}(f) &= a^2 \times a + 1 \times (1-a) = 1 - a + a^3. \end{aligned}$$

According to Lemma 12.6 for any pair $a, b \in (0,1)$ we should have $l_{\mathscr{D}_a}(f) \leq u_{\mathscr{D}_b}(f)$, since every upper sum must be at least as big as every lower sum. That is, for all $a, b \in (0,1)$,

$$a^2 - a^3 \leq 1 - b + b^3.$$

Is this obvious? Can you prove it directly, without using Lemma 12.6?

12.2 Definition of the Riemann integral

To define the Riemann integral we will use *sequences* of dissections.

Definition 12.8 A bounded function $f : [a,b] \to \mathbb{R}$ is **Riemann integrable** if there exists a sequence $(\mathscr{D}_n)$ of dissections of $[a,b]$ and a real number L such that

$$u_{\mathscr{D}_n}(f) \to L \quad \text{and} \quad l_{\mathscr{D}_n}(f) \to L.$$

In this case, we call L the **Riemann integral** of f over the interval $[a,b]$, and denote it

$$\int_a^b f.$$

Example 12.9 Let $f : [0,2] \to \mathbb{R}$ be the "step" function

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1, \\ 2 & \text{if } 1 \leq x \leq 2. \end{cases}$$

Show that f is Riemann integrable on $[0,2]$ and compute $\int_0^2 f$.

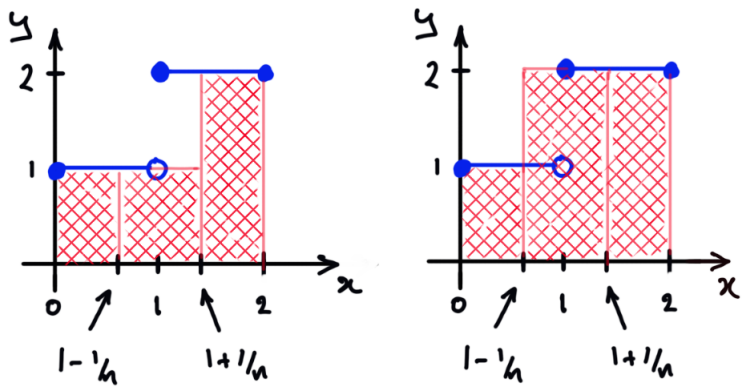


Figure 10: The sequence of dissections, $\mathscr{D}_n$, $n = 2, 3, \ldots$, used in Example 12.9, and their associated Riemann sums.

Solution: For each $n \in \{2, 3, \ldots\}$ define the dissection $\mathscr{D}_n = \{0, 1 - 1/n, 1 + 1/n, 2\}$ of $0, 1]$ (see Figure 10). Then the lower and upper Riemann sums of f with respect to this dissection are

$$\begin{aligned} l_{\mathscr{D}_n}(f) &= 1(1 - \frac{1}{n} - 0) + 1((1 + \frac{1}{n}) - (1 - \frac{1}{n})) + 2(2 - (1 + \frac{1}{n})) \\ &= 3 - \frac{1}{n} \\ u_{\mathscr{D}_n}(f) &= 1(1 - \frac{1}{n} - 0) + 2((1 + \frac{1}{n}) - (1 - \frac{1}{n})) + 2(2 - (1 + \frac{1}{n})) \\ &= 3 + \frac{1}{n}. \end{aligned}$$

Hence, $l_{\mathscr{D}_n}(f) \to 3$ and $u_{\mathscr{D}_n}(f) \to 3$, so f is Riemann integrable, and

$$\int_0^2 f = 3.$$

□

Example 12.10 Let $f : [0, 1] \to \mathbb{R}$ be the function $f(x) = x^2$. Show that f is Riemann integrable on $[0, 1]$ and compute $\int_0^1 f$.

Solution: For each integer $n \geq 1$ let $\mathscr{D}_n$ be the *regular* dissection of $[0, 1]$ of size n, that is, the dissection that divides $[0, 1]$ into n subintervals of equal width,

$$\mathscr{D}_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1\right\}.$$

Each subinterval $[a_{j-1}, a_j] = [(j-1)/n, j/n]$ has width $1/n$, and, since the function f is increasing,

$$\begin{aligned} m_j &= \inf\left\{f(x) : \frac{j-1}{n} \leq x \leq \frac{j}{n}\right\} = f\left(\frac{j-1}{n}\right) = \frac{(j-1)^2}{n^2}, \\ M_j &= \sup\left\{f(x) : \frac{j-1}{n} \leq x \leq \frac{j}{n}\right\} = f\left(\frac{j}{n}\right) = \frac{j^2}{n^2}, \end{aligned}$$

so the lower and upper Riemann sums with respect to $\mathscr{D}_n$ are

$$\begin{aligned} l_{\mathscr{D}_n}(f) &= \frac{1}{n}\sum_{j=1}^{n}\frac{(j-1)^2}{n^2} = \frac{1}{n^3}\sum_{k=1}^{n-1}k^2, \\ u_{\mathscr{D}_n}(f) &= \frac{1}{n}\sum_{j=1}^{n}\frac{j^2}{n^2} = \frac{1}{n^3}\sum_{j=1}^{n}j^2. \end{aligned}$$

I claim that, for all $n \in \mathbb{Z}^+$,

$$\sum_{j=1}^{n} j^2 = \frac{1}{6}n(n+1)(2n+1),$$

and leave the proof of this as an exercise (hint: use induction!). Hence, for each $n \in \mathbb{Z}^+$,

$$\begin{aligned} l_{\mathscr{D}_n}(f) &= \frac{1}{6}\left(1-\frac{1}{n}\right)\left(2-\frac{1}{n}\right) \to \frac{1}{3}, \\ u_{\mathscr{D}_n}(f) &= \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right) \to \frac{1}{3}. \end{aligned}$$

Hence, f is Riemann integrable and

$$\int_0^1 f = \frac{1}{3}.$$

□

There is something a bit troubling about our definition of $\int_a^b f$. In Example 12.10 we constructed a sequence $(\mathscr{D}_n)$ of dissections of $[0,1]$ such that $l_{\mathscr{D}_n}(f) \to \frac{1}{3}$ and $u_{\mathscr{D}_n}(f) \to \frac{1}{3}$, and concluded that

$$\int_0^1 f = \frac{1}{3}.$$

But how do we know that there isn't some other sequence of dissections, $(\mathscr{D}'_n)$ say, such that $l_{\mathscr{D}'_n}(f) \to \frac{1}{2}$ and $u_{\mathscr{D}'_n}(f) \to \frac{1}{2}$? If this were true, $\int_0^1 f$ would be both $\frac{1}{3}$ and $\frac{1}{2}$. "The" Riemann integral of f wouldn't be unique! Luckily, this disaster can never befall us:

Proposition 12.11 (Uniqueness of the Riemann Integral) *If $f : [a,b] \to \mathbb{R}$ is Riemann integrable then its integral $\int_a^b f$ is unique.*

Proof: Assume, towards a contradiction, that there exist sequences $(\mathscr{D}_n)$ and $(\mathscr{D}'_n)$ of dissections of $[a,b]$ and $L_1 \neq L_2$ such that

$$\begin{aligned} l_{\mathscr{D}_n}(f) &\to L_1 \qquad \text{and} \qquad u_{\mathscr{D}_n}(f) \to L_1, \\ l_{\mathscr{D}'_n}(f) &\to L_2 \qquad \text{and} \qquad u_{\mathscr{D}'_n}(f) \to L_2, \end{aligned}$$

meaning that $\int_a^b f$ equals both L_1 and L_2. Without loss of generality, we may assume $L_1 < L_2$. Then, by the Algebra of Limits,

$$u_{\mathscr{D}_n}(f) - l_{\mathscr{D}'_n}(f) \to L_1 - L_2 < 0.$$

But by Lemma 12.6

$$u_{\mathscr{D}_n}(f) - l_{\mathscr{D}'_n}(f) \geq 0,$$

which contradicts Proposition 2.14. □

We have now seen two examples of functions which are Riemann integrable. Can we cook up a function which is *not* Riemann integrable? This turns out to be surprisingly difficult, but is possible.

Example 12.12 (A nonintegrable function) Let $f : [0,1] \to \mathbb{R}$ such that $f(x) = 0$ if $x \in \mathbb{Q}$ and $f(x) = 1$ if $x \notin \mathbb{Q}$. I claim that f is not Riemann integrable.

Proof: Let $\mathscr{D} = \{a_0, \ldots, a_n\}$ be any dissection of $[0,1]$. Then, by the Density Theorems of $\mathbb{Q}$ in $\mathbb{R}$ and $\mathbb{R}\backslash\mathbb{Q}$ in $\mathbb{R}$ (Theorems 3.15 and 3.16), every subinterval $[a_{j-1}, a_j]$ contains both rational and irrational members, so $m_j = 0$ and $M_j = 1$ for all j. Hence

$$\begin{aligned} l_{\mathscr{D}}(f) &= \sum_{j=1}^{n} 0(a_j - a_{j-1}) = 0 \\ u_{\mathscr{D}}(f) &= \sum_{j=1}^{n} 1(a_j - a_{j-1}) = a_n - a_0 = 1. \end{aligned}$$

Since this is true for *all* dissections of $[0,1]$, given any sequence $\mathscr{D}_n$ of dissections,

$$u_{\mathscr{D}_n}(f) \to 1, \qquad l_{\mathscr{D}_n}(f) \to 0 \neq 1,$$

so f is not Riemann integrable. □

It is convenient to extend the definition of dissection to include the case where the interval is $[a,a] = \{a\}$. The only dissection of $[a,a]$ is the singleton set $\mathscr{D} = \{a\}$ (a dissection of size 0). Every function $f : [a,a] \to \mathbb{R}$ is bounded, above and below by $f(a)$, so the one and only lower Riemann sum is

$$l_{\{a\}}(f) = f(a)(a-a) = 0,$$

and the one and only upper Riemann sum is

$$u_{\{a\}}(f) = f(a)(a-a) = 0.$$

It follows that every function is integrable on $[a,a]$, and

$$\int_a^a f = 0.$$

Given a Riemann integrable function f on $[a,b]$ where $a \leq b$, it is also convenient to define

$$\int_b^a f = -\int_a^b f.$$

I leave it to the reader to verify that all the results we prove about $\int_a^b f$ trivially extend to the case $a \geq b$ with these conventions.

Remark It is very common to denote the integral of a function $f : [a,b] \to \mathbb{R}$

$$\int_a^b f(x)\,dx \tag{12.12}$$

rather than $\int_a^b f$. This has (at least) one advantage: one can define the function f (often called the "integrand") at the same time as one defines the integral to be computed. For example, it is clear that

$$\int_0^1 \sqrt{x^2+1}\,dx \tag{12.13}$$

means the Riemann integral (assuming this exists!) of the function

$$f : [0,1] \to \mathbb{R}, \qquad f(x) = \sqrt{x^2+1},$$

and we have conveyed this information without going to the trouble of formally defining f.

Nonetheless I will persist in preferring the more stripped down notation $\int_a^b f$, for two reasons. First, this is a course in Real Analysis: going to the trouble of formally defining the mathematical objects we study is the very heart of our endeavour. Second, there's something a bit naff about an expression like (12.12): the choice of symbol x is entirely arbitrary and without significance. That is

$$\int_a^b f(x)\,dx = \int_a^b f(y)\,dy = \int_a^b f(t)\,dt = \int_a^b f(\widetilde{\Upsilon}_{\sigma'}^{\mu_0})\,d\widetilde{\Upsilon}_{\sigma'}^{\mu_0}.$$

So this notation forces you to focus on something irrevelevant (the name of the symbol x) instead of thinking about what the function f really is.

Incidentally, do we know that the integral (12.13) exists? The next section will answer this question with a resounding "yes!"

12.3 Continuous functions are integrable

We have seen that all differentiable functions are continuous (Theorem 9.8), but not all continuous functions are differentiable. We next prove that every continuous function (on a closed bounded interval) is Riemann integrable. Once again, the converse is false: we've already seen an example of an integrable function that is not continuous (the step function Example 12.9).

Theorem 12.13 *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f is Riemann integrable.*

Proof: First note that, since f is continuous, it is certainly bounded (by the Extreme Value Theorem), so $l_{\mathscr{D}}(f)$ and $u_{\mathscr{D}}(f)$ exist for any dissection $\mathscr{D}$ of $[a, b]$.

Consider the sequence $(\mathscr{D}_n)$ of regular dissections of $[a, b]$ of size 2^n. Then $\mathscr{D}_n \subset \mathscr{D}_{n+1}$ for all n, so by the Refinement Lemma, $l_{\mathscr{D}_n}(f)$ is increasing (and bounded above) and $u_{\mathscr{D}_n}(f)$ is decreasing (and bounded below). Hence, by the Monotone Convergence Theorem (Theorem 3.20),

$$u_{\mathscr{D}_n}(f) \to L, \qquad l_{\mathscr{D}_n}(f) \to K$$

for some $L, K \in \mathbb{R}$ and, by Proposition 2.14 and Lemma 12.6, $K \leq L$. It remains to show that $L - K = 0$.

Assume, towards a contradiction, that $L - K > 0$. For a given fixed $n \in \mathbb{Z}^+$, define as usual

$$\begin{aligned} m_j &= \inf\{f(x) \,:\, a_{j-1} \leq x \leq a_j\} \\ \text{and} \quad M_j &= \sup\{f(x) \,:\, a_{j-1} \leq x \leq a_j\}. \end{aligned}$$

Then

$$u_{\mathscr{D}_n}(f) - l_{\mathscr{D}_n}(f) = \frac{b-a}{2^n}\sum_{j=1}^{2^n}(M_j - m_j) \geq L - K.$$

This sum consists of 2^n terms, each non-negative, so at least one term must be greater than or equal to $(L - K)/2^n$. That is, there must exist $j \in \{1, 2, \ldots, 2^n\}$ such that

$$M_j - m_j \geq \frac{L-K}{b-a}.$$

But $f : [a_{j-1}, a_j] \to \mathbb{R}$ is continuous so, by the Extreme Value Theorem, f attains both a maximum and minimum value on $[a_{j-1}, a_j]$, that is, there are points, x_n and y_n say, in $[a_{j-1}, a_j]$ such that $f(x_n) = M_j$ and $f(y_n) = m_j$. Hence, for each $n \in \mathbb{Z}^+$ there exist $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| \leq \frac{b-a}{2^n}, \tag{12.14}$$

$$f(x_n) - f(y_n) \geq \frac{L-K}{b-a}. \tag{12.15}$$

Consider the sequence (x_n). It is bounded, so, by the Bolzano–Weierstrass Theorem, it has a convergent subsequence $x_{n_k} \to c \in [a,b]$. By (12.14),

$$x_{n_k} - \frac{b-a}{2^{n_k}} \le y_{n_k} \le x_{n_k} + \frac{b-a}{2^{n_k}}$$

so $y_{n_k} \to c$ also, by the Squeeze Rule. Now f is continuous, so $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$, and hence $f(x_{n_k}) - f(y_{n_k}) \to f(c) - f(c) = 0$. But this contradicts (12.15) and Proposition 2.14. □

The function $f : [0,1] \to \mathbb{R}$, $f(x) = \sqrt{x^2+1}$ is a composition of continuous functions, so is continuous, so we know that the integral (12.13) exists by Theorem 12.13.

12.4 Elementary properties of the Riemann integral

The Riemann integral has many convenient properties.

Theorem 12.14 (The Join Rule) *Let f be Riemann integrable on $[a,b]$ and on $[b,c]$. Then f is Riemann integrable on $[a,c]$ and*

$$\int_a^c f = \int_a^b f + \int_b^c f.$$

Proof: By assumption, there exist sequences of dissections $(\mathscr{D}'_n)$ of $[a,b]$ and $(\mathscr{D}''_n)$ of $[b,c]$ such that $l_{\mathscr{D}'_n}(f) \to \int_a^b f$, $u_{\mathscr{D}'_n}(f) \to \int_a^b f$, $l_{\mathscr{D}''_n}(f) \to \int_b^c f$, and $u_{\mathscr{D}''_n}(f) \to \int_b^c f$. Let $\mathscr{D}_n = \mathscr{D}'_n \cup \mathscr{D}''_n$. Then $\mathscr{D}_n$ is a dissection of $[a,c]$ and it follows directly from Definition 12.2 that $l_{\mathscr{D}_n}(f) = l_{\mathscr{D}'_n}(f) + l_{\mathscr{D}''_n}(f)$ and $u_{\mathscr{D}_n}(f) = u_{\mathscr{D}'_n}(f) + u_{\mathscr{D}''_n}(f)$. Hence, by the Algebra of Limits, $l_{\mathscr{D}_n}(f) \to \int_a^b f + \int_b^c f$ and $u_{\mathscr{D}_n}(f) \to \int_a^b f + \int_b^c f$, so the claim follows. □

We next want to prove that the Riemann integral is *linear*, that is, $\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$, for any constants α, β. To do this, we need the following:

Lemma 12.15 *Let $f, g : [a,b] \to \mathbb{R}$ be bounded functions and $\mathscr{D}$ be a dissection of $[a,b]$. Then*

$$l_{\mathscr{D}}(f+g) \ge l_{\mathscr{D}}(f) + l_{\mathscr{D}}(g) \quad \textit{and} \quad u_{\mathscr{D}}(f+g) \le u_{\mathscr{D}}(f) + u_{\mathscr{D}}(g).$$

Proof: Let $\mathscr{D} = \{a_0, a_1, \ldots, a_n\}$ and $I_j = [a_{j-1}, a_j]$. Let

$$\begin{aligned} m_j(f) &= \inf\{f(x) : x \in I_j\}, \\ M_j(f) &= \sup\{f(x) : x \in I_j\}, \end{aligned}$$

and $m_j(g), M_j(g), m_j(g+f)$, and $M_j(g+f)$ be defined similarly. Then, for all $x \in I_j$, $f(x)+g(x) \ge m_j(f) + m_j(g)$, so $m_j(f) + m_j(g)$ is certainly a lower bound on $\{f(x)+g(x) : x \in I_j\}$. Since $m_j(f+g)$ is the *greatest* lower bound on this set, it follows that $m_j(f+g) \ge m_j(f) + m_j(g)$. Hence

$$\begin{aligned} l_{\mathscr{D}}(f+g) &= \sum_{j=1}^n m_j(f+g)(a_j - a_{j-1}) \\ &\ge \sum_{j=1}^n (m_j(f) + m_j(g))(a_j - a_{j-1}) = l_{\mathscr{D}}(f) + l_{\mathscr{D}}(g). \end{aligned}$$

Similarly, for all $x \in I_j$, $f(x)+g(x) \le M_j(f) + M_j(g)$, so $M_j(f) + M_j(g)$ is certainly an upper bound on $\{f(x)+g(x) : x \in I_j\}$. Since $M_j(f+g)$ is the *least* upper bound on this set, it follows that $M_j(f+g) \le M_j(f) + M_j(g)$. Hence

$$\begin{aligned} u_{\mathscr{D}}(f+g) &= \sum_{j=1}^n M_j(f+g)(a_j - a_{j-1}) \\ &\le \sum_{j=1}^n (M_j(f) + M_j(g))(a_j - a_{j-1}) = u_{\mathscr{D}}(f) + u_{\mathscr{D}}(g). \end{aligned}$$

□

Theorem 12.16 (Linearity of the Riemann Integral) *Let f, g be Riemann integrable on $[a, b]$, and $\alpha \in \mathbb{R}$ be a constant. Then*

(i) αf is Riemann integrable on $[a, b]$, and $\int_a^b \alpha f = \alpha \int_a^b f$,

(ii) $f + g$ is Riemann integrable on $[a, b]$, and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$.

Proof: (i) By assumption, there is a sequence of dissections $(\mathscr{D}_n)$ such that $l_{\mathscr{D}_n}(f) \to \int_a^b f$ and $u_{\mathscr{D}_n}(f) \to \int_a^b f$. It follows directly from Definition 12.2 that $l_{\mathscr{D}_n}(\alpha f) = \alpha l_{\mathscr{D}_n}(f)$ and $u_{\mathscr{D}_n}(\alpha f) = \alpha u_{\mathscr{D}_n}(f)$ if $\alpha \geq 0$, and $l_{\mathscr{D}_n}(\alpha f) = \alpha u_{\mathscr{D}_n}(f)$ and $u_{\mathscr{D}_n}(\alpha f) = \alpha l_{\mathscr{D}_n}(f)$ if $\alpha < 0$. In either case, $l_{\mathscr{D}_n}(\alpha f) \to \alpha \int_a^b f$ and $u_{\mathscr{D}_n}(\alpha f) \to \alpha \int_a^b f$ by the Algebra of Limits, and the claim follows.
(ii) Let $(\mathscr{D}_n)$ be as defined above. By assumption, there is also a sequence of dissections $(\mathscr{D}'_n)$ such that $l_{\mathscr{D}'_n}(g) \to \int_a^b g$ and $u_{\mathscr{D}'_n}(g) \to \int_a^b g$. Let $\mathscr{D}''_n = \mathscr{D}_n \cup \mathscr{D}'_n$, and note that this is a refinement of both $\mathscr{D}_n$ and $\mathscr{D}'_n$. Now, by Lemma 12.15 and the Refinement Lemma

$$\begin{aligned} l_{\mathscr{D}_n}(f) + l_{\mathscr{D}'_n}(g) \leq l_{\mathscr{D}''_n}(f) + l_{\mathscr{D}''_n}(g) &\leq l_{\mathscr{D}''_n}(f + g) \\ &\leq u_{\mathscr{D}''_n}(f + g) \leq u_{\mathscr{D}''_n}(f) + u_{\mathscr{D}''_n}(g) \leq u_{\mathscr{D}_n}(f) + u_{\mathscr{D}'_n}(g). \end{aligned} \tag{12.16}$$

Now

$$l_{\mathscr{D}_n}(f) + l_{\mathscr{D}'_n}(g) \to \int_a^b f + \int_a^b g, \quad u_{\mathscr{D}_n}(f) + u_{\mathscr{D}'_n}(g) \to \int_a^b f + \int_a^b g,$$

by the Algebra of Limits, so

$$u_{\mathscr{D}''_n}(f + g) \to \int_a^b f + \int_a^b g, \quad l_{\mathscr{D}''_n}(f + g) \to \int_a^b f + \int_a^b g$$

by the Squeeze Rule. Hence, $f + g$ is Riemann integrable and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g$$

□

We next prove that Riemann integration preserves inequalities.

Proposition 12.17 *Let $f, g : [a, b] \to \mathbb{R}$ be Riemann integrable and assume $f(x) \leq g(x)$ for all $x \in [a, b]$. Then*

$$\int_a^b f \leq \int_a^b g.$$

Proof: By assumption there exist sequences of dissections $\mathscr{D}'_n$, $\mathscr{D}''_n$ such that $l_{\mathscr{D}'_n}(f) \to \int_a^b f$ and $l_{\mathscr{D}''_n} \to \int_a^b g$. Let $\mathscr{D}_n = \mathscr{D}'_n \cup \mathscr{D}''_n$. Then, by the Refinement Lemma (and the Squeeze Rule) $l_{\mathscr{D}_n}(f) \to \int_a^b f$ and $l_{\mathscr{D}_n} \to \int_a^b g$ also. But, for each n, $l_{\mathscr{D}_n}(g) \geq l_{\mathscr{D}_n}(f)$ since $\inf g \geq \inf f$ on any subset of $[a, b]$. Hence, $\int_a^b g - \int_a^b f$, being the limit of a convergent non-negative sequence, is ≥ 0 (Proposition 2.14). □

Recall that the triangle inequality asserts that, for any pair of real numbers x_1, x_2,

$$|x_1 + x_2| \leq |x_1| + |x_2|.$$

It is but the work of a moment to prove (by induction) that this extends to arbitrary finite sets of real numbers:

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|.$$

Actually, with a bit more work one can show that the same claim holds for infinite sums, that is, series. That is, if a series $\sum x_n$ converges absolutely, then

$$\left| \sum_{n=1}^{\infty} x_n \right| \leq \sum_{n=1}^{\infty} |x_n|.$$

(Prove it!) Roughly speaking, we can think of an integral as being a kind of "continuously indexed infinite sum." So it's natural to ask whether a Triangle Inequality holds for integrals too.

Theorem 12.18 (The Triangle Inequality for Riemann Integrals) *If $f : [a, b] \to \mathbb{R}$ is continuous, then*

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Proof: For all $x \in [a, b]$, $-|f(x)| \leq f(x) \leq |f(x)|$ so, by Proposition 12.17,

$$-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|,$$

whence the claim immediately follows. □

12.5 The Fundamental Theorem of the Calculus

So far we have developed some powerful theoretical tools to show that a given function is integrable and to relate the integrals of related integrable functions, but we don't have any really convenient techniques for actually computing Riemann integrals. To compute

$$\int_0^1 x^2 \, dx,$$

for example, we had to resort to exhibiting a sequence of dissections whose upper and lower sums converge to a common limit, in this case, $\frac{1}{3}$ (see Example 12.10). In principle, we can always compute Riemann integrals like this, but in practice this method of computing integrals is very onerous and, unless the integrand (the function to be integrated) is fairly simple, is likely to be intractable. Consider, for example, attempting to compute

$$\int_0^{\pi} \sin x \, dx$$

using this method. We know that this integral exists (Theorem 12.13) because we know that sin is continuous, but we have no hope of computing it directly using Riemann sums. In this section, we establish a fundamental connexion between Riemann integration and differentiation which will give us a convenient means of computing $\int_a^b f$ whenever we can dream up a function whose *derivative* equals f. Once we have done this we will rarely have to resort to computing sequences of Riemann sums to compute integrals.

Theorem 12.19 (Fundamental Theorem of the Calculus version 1) *Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ be continuous and $a \in I$. Define $F : I \to \mathbb{R}$ by*

$$F(x) = \int_a^x f.$$

Then F is differentiable, and $F' = f$.

Proof: First note that F is well-defined (that is, the integral defining $F(x)$ exists for all $x \in I$) by Theorem 12.13 since f is continuous on $[a, x] \subseteq I$ if $x \geq a$, and continuous on $[x, a] \subseteq I$ if $x < a$. We wish to compute

$$\lim_{y \to x} \frac{F(y) - F(x)}{y - x}.$$

So, let (y_n) be any sequence in $I \backslash \{x\}$ such that $y_n \to x$, and

$$s_n = \frac{F(y_n) - F(x)}{y_n - x}.$$

We must show that $s_n \to f(x)$.

For each n, either $y_n > x$ or $y_n < x$. If $y_n > x$ then, by Theorem 12.14,

$$s_n = \frac{1}{y_n - x} \int_x^{y_n} f,$$

whereas if $y_n < x$,

$$s_n = \frac{1}{x - y_n} \int_{y_n}^{x} f.$$

In either case, by the Extreme Value Theorem, there exist w_n and z_n between x and y_n such that $f(w_n)$ is the minimum value of f on the closed interval with endpoints x and y_n, and $f(z_n)$ is the maximum value of f on this interval. Hence, by Proposition 12.17,

$$\begin{aligned} & \frac{1}{y_n - x} f(w_n)(y_n - x) \leq \quad s_n \quad \leq \frac{1}{y_n - x} f(z_n)(y_n - x) \qquad \text{if } y_n > x, \\ \text{and} \quad & \frac{1}{x - y_n} f(w_n)(x - y_n) \leq \quad s_n \quad \leq \frac{1}{x - y_n} f(z_n)(x - y_n) \qquad \text{if } y_n < x. \end{aligned}$$

In either case, we conclude that

$$f(w_n) \leq s_n \leq f(z_n).$$

Now, $y_n \to x$ so $w_n \to x$ and $z_n \to x$ also, by the Squeeze Rule, and f is continuous, so $f(w_n) \to f(x)$ and $f(z_n) \to f(x)$. Hence, by the Squeeze Rule again, $s_n \to f(x)$, which completes the proof. □

As a by-product of this theorem we see that any function which is continuous on an interval is the derivative of some differentiable function on that interval, an interesting and far from obvious fact. Even better, Theorem 12.19 immediately implies a second theorem which renders the job of computing many Riemann integrals almost trivial:

Theorem 12.20 (Fundamental Theorem of the Calculus version 2) *Let $f : [a, b] \to \mathbb{R}$ be continuous and $F : [a, b] \to \mathbb{R}$ be any differentiable function such that $F' = f$. Then*

$$\int_a^b f = F(b) - F(a).$$

Proof: Define the function $g : [a, b] \to \mathbb{R}$,

$$g(x) = \left(\int_a^x f \right) - F(x).$$

Then, by Theorem 12.19 and the definition of F,

$$g'(x) = f(x) - f(x) = 0$$

for all $x \in [a, b]$. Hence, by Proposition 10.9, g is constant, so $g(b) = g(a)$, that is

$$\left(\int_a^b f \right) - F(b) = \left(\int_a^a f \right) - F(a) = 0 - F(a),$$

and the result immediately follows. $\square$

Example 12.21 Compute the Riemann integrals

(i) $\displaystyle\int_{-1}^{1} x^2\,dx$,

(ii) $\displaystyle\int_{0}^{\pi} \sin x\,dx$,

Solution:

(i) Let $f(x) = x^2$ and $F(x) = x^3/3$. Then f is continuous and $F' = f$ on $[-1, 1]$ so, by Theorem 12.20,

$$\int_{-1}^{1} f = F(1) - F(-1) = \frac{2}{3}.$$

(ii) Let $f(x) = \sin x$ and $F(x) = -\cos x$. Then f is continuous and $F' = f$ on $[0, \pi]$ so, by Theorem 12.20,

$$\int_{0}^{\pi} f = F(\pi) - F(0) = -\cos\pi + \cos 0 = 2.$$

$\square$

Exercise 12.22 Let $f : \mathbb{R} \to \mathbb{R}$,

$$f(x) = \int_{x}^{x^2+1} \frac{t}{t^2+1}\,dt.$$

Compute $f'(x)$. Check your answer by watching this [VIDEO].

Having established Theorem 12.20, we see that we can explicitly compute

$$\int_a^b f(x)\,dx$$

if we can think of an **antiderivative** of f, that is, any function whose *derivative* is $f(x)$. This trick is so pervasive in integral calculus that it leads the unwary to identify *integration* of f (that is, the process of computing a Riemann integral $\int_a^b f$) with the process of writing down an antiderivative of f. Indeed, it is common practice to call an antiderivative of f an *indefinite integral* of f and to denote any such function by the symbol

$$\int f(x)\,dx.$$

This notation has some practical advantages, but it also, unfortunately, generates a huge amount of confusion. Note that

$$\int_a^b f(x)\,dx$$

is a single number and that its *definition* has nothing whatsoever to do with antiderivatives of f: it is the unique real number which is no greater than any upper Riemann sum of f on $[a, b]$ and no less than any lower Riemann sum of f on $[a, b]$. In particular, it is *not* a function of the "variable" x. Indeed, we could equally well have written it $\int_a^b f(y)\,dy$ or $\int_a^b f(\Gamma_{\widehat{\aleph}})\,d\Gamma_{\widehat{\aleph}}$, which is one reason to prefer the simpler notation $\int_a^b f$. It *is* a function of a and b, however.

By contrast, $\int f(x)\,dx$ is just a (slightly ambiguous) symbol denoting any function whose derivative (at x) is $f(x)$. It *is* a function of x, not a single number, and its *definition* has nothing to do with Riemann sums. That these two things turn out to be closely related is (as its name suggests) a very

important theorem: the Fundamental Theorem of the Calculus. To understand calculus properly it is important to maintain a clear conceptual distinction between the Riemann integral $\int_a^b f$ and any antiderivative that one might use to compute it.

We have now rigorously developed all the fundamental concepts of differential and integral calculus. It would be a straightforward matter to combine the Fundamental Theorem of the Calculus with the Product Rule of differentiation to develop the technique of integration by parts, or with the Chain Rule to develop the technique of integration by substitution. Such development requires no new analytic insight and so is better placed in a course on the methods of calculus, rather than a course on real analysis such as this.

Summary

- A **dissection** of $[a, b]$ is a finite subset $\mathscr{D} = \{a_0, a_1, \ldots, a_n\}$ of $[a, b]$ such that

$$a = a_0 < a_1 < a_2 < \cdots < a_n = b.$$

- Given a bounded function $f : [a, b] \to \mathbb{R}$, and a dissection $\mathscr{D}$, the **lower Riemann sum** is

$$l_{\mathscr{D}}(f) = \sum_{j=1}^{n} \inf\{f(x) \,:\, a_{j-1} \leq x \leq a_j\}(a_j - a_{j-1}),$$

  and the **upper Riemann sum** is

$$u_{\mathscr{D}}(f) = \sum_{j=1}^{n} \sup\{f(x) \,:\, a_{j-1} \leq x \leq a_j\}(a_j - a_{j-1}).$$

- f is **Riemann integrable** if there exists a sequence $(\mathscr{D}_n)$ of dissections of $[a, b]$ such that $u_{\mathscr{D}_n}(f)$ and $l_{\mathscr{D}_n}(f)$ converge to a common limit. In this case, we denote this limit by

$$\int_a^b f$$

  and call it the **Riemann integral** of f (on, or over, $[a, b]$).

- It follows from this definition that all **continuous** functions are Riemann integrable.

- We also proved

$$\text{The } \textbf{Join Rule}: \quad \int_a^b f + \int_b^c f = \int_a^c f,$$

$$\text{The } \textbf{Linearity Property}: \quad \int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$$

  where α, β are constants.

- There is an important link between Riemann integrals and derivatives, given by the **Fundamental Theorem of the Calculus**:
  - Version 1: if f is continuous and $F(x) = \int_a^x f$, then $F' = f$.
  - Version 2: if f is continuous and F is some antiderivative of f (that is, a function satisfying $F' = f$), then $\int_a^b f = F(b) - F(a)$.

- Version 2 of the Fundamental Theorem of the Calculus provides a flexible and convenient method for computing many Riemann integrals.